Chapter 1

Introduction

1.1 From a point to a curve

Classical Koszul duality is a phenomenon of graded algebras over a point: a quadratic algebra A determines a dual coalgebra $A^!$, and the bar construction mediates between them. On an algebraic curve X, the generators of the algebra become sections of a $\mathcal{D}_X$-module, the quadratic relations become operator product expansions with meromorphic singularities, and the bar differential becomes an integral transform whose kernel is the logarithmic propagator $d\log(z_i - z_j)$ on Fulton–MacPherson compactifications of configuration spaces. Already at genus 0, chiral Koszul duality differs from its classical counterpart. Vertex algebras live on the formal disk D, not on a bare point; the passage from D to a point discards the thickening data (completion, growth conditions) that the $\mathcal{D}$-module structure depends on. The affine line $\mathbb{A}^1$ deformation-retracts onto a point, but this retraction is additional data: relating the bar construction on $\mathbb{A}^1$ to the classical bar over a point requires specifying the retraction and its attendant homotopy transfer. Already on $\mathbb{A}^1$, the configuration spaces $\mathrm{Conf}_n(\mathbb{A}^1)$ carry Fulton–MacPherson compactifications and the Arnol'd algebra $H^*(\mathrm{Conf}_n(\mathbb{A}^1))$, generated by the forms $\omega_{ij} = d\log(z_i - z_j)$ subject to the Arnold relation: these structures are entirely absent over a point. On the projective line $\mathbb{P}^1 = \mathbb{A}^1 \cup \mathbb{A}^1$, compactness further changes the homotopy type of the configuration spaces. The classical theory embeds into this picture via the formal-disk restriction, but the embedding is not an equivalence: even at genus 0, the ordered-versus-unordered bar distinction (E_1 versus E_∞) and the configuration-space geometry are genuinely new. At genus $g \geq 1$, the topology of the curve forces genuinely new phenomena (central extensions, conformal anomalies, curved A_∞ structures) that have no analogue in the classical setting and whose geometric origin is the curvature of the Hodge bundle over $\overline{\mathcal{M}}_g$.

The classical theory goes as follows. Let V be a finite-dimensional graded vector space and $R \subset V^{\otimes 2}$ a subspace of quadratic relations. The quadratic algebra $A = T(V)/(R)$ is the free tensor algebra modulo R; its Koszul dual coalgebra is $A^{¡} = T^c(sV^*)$, the cofree conilpotent coalgebra on the suspended linear dual, with coproduct truncated by the orthogonal complement $R^\perp \subset (V^*)^{\otimes 2}$. The bar construction $B(A) = (T^c(s^{-1}\bar{A}), d_B)$ carries a differential d_B induced by the multiplication of A: on cogenerators $d_B(s^{-1}a) = \sum s^{-1}a' \otimes s^{-1}a''$ records the splitting of a into pairs of generators. The cobar construction $\Omega(C)$ reverses the process: starting from a conilpotent coalgebra C, one builds the free algebra $T(s^{-1}\bar{C})$ with differential induced by the comultiplication. The fundamental comparison map

$$\varepsilon\colon \Omega(B(A)) \longrightarrow A$$

is always a morphism of dg algebras; when it is a quasi-isomorphism, the algebra A is called *Koszul*, and $B(A)$ has cohomology concentrated in bar degree 1, equal to the Koszul dual coalgebra $A^{!\vee}$. The Koszul dual algebra is $A^! = (A^{!\vee})^\vee = T(V^*)/R^\perp$. Koszulness means that A admits a linear resolution by $A^!$-modules, and that the derived category of A-modules is equivalent to the derived category of $A^!$-modules (up to a shift and twist). This is the content of Priddy [?Priddy70], Beilinson–Ginzburg–Soergel [?BGS96], and Loday–Vallette [?LV12].

Let X be a smooth projective curve over $\mathbb{C}$. A *chiral algebra* $\mathcal{A}$ on X (Beilinson–Drinfeld [?BD04]) is a $\mathcal{D}_X$-module equipped with a chiral bracket

$$\mu^{\mathrm{ch}}\colon\ j_*j^*(\mathcal{A}\boxtimes\mathcal{A}) \longrightarrow \Delta_!(\mathcal{A}),$$

where $j\colon X^2\setminus\Delta \hookrightarrow X^2$ is the complement of the diagonal and $\Delta_!\colon D(\mathcal{D}_X)\to D(\mathcal{D}_{X^2})$ is the pushforward. The chiral bracket encodes the operator product expansion: locally, if $a, b \in \mathcal{A}$ are sections and z, w are local coordinates, then

$$a(z)b(w)\ \sim\ \sum_{n\geq 0}\frac{(a_{(n)}b)(w)}{(z-w)^{n+1}},$$

where the modes $a_{(n)}b$ are the Laurent coefficients of the OPE, and the bracket μ^{ch} records the entire singular part simultaneously. The Borcherds identity, the defining axiom of a vertex algebra, is the chiral-algebraic incarnation of the Jacobi identity, coupling all pole orders in a single statement.

Two reference algebras organize the theory (see §1.23 for the formal definitions). The **Heisenberg** algebra is the *CG opening*: its OPE has only double poles, so collisions are symmetric and unordered (E_∞-chiral); it is the commutative base case in which every clause of the thesis collapses to the single scalar k, but it is *not* an E_1 atom. The **Yangian** is the genuine E_1 *atom*: simple poles give collisions a canonical ordering (E_1-chiral, genuinely nonlocal), and it is the unique nonlocal generator of the E_1-chiral Koszul programme. Every standard family (Kac–Moody, Virasoro, $\mathcal{W}$-algebras, lattice vertex algebras) sits between the CG opening and the E_1 atom, and the bar complex on higher-genus curves classifies them. At genus $g\geq 1$, the bar differential inherits a curvature term $\kappa(\mathcal{A})\cdot\omega_g$ as the coalgebraic shadow of the cyclic trace composed with clutching: the Hodge class $\omega_g = c_1(\mathbb{E})$ on $\overline{\mathcal{M}}_g$ enters through the annular trace on the open sector, and the *modular characteristic* $\kappa(\mathcal{A})$ is the universal scalar measuring this obstruction (Theorem **??**). The bar construction on curves defines a pronilpotent dg Lie algebra $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ (the modular convolution algebra), whose Maurer–Cartan element $\Theta_{\mathcal{A}}$ encodes the full genus expansion. The finite-order projections $\Theta^{\leq r}_{\mathcal{A}}$ (the shadow obstruction tower) extract successively finer invariants: κ at degree 2, a cubic shadow C at degree 3, a quartic resonance class Q at degree 4. The shadow depth $r_{\max}(\mathcal{A})$ is the largest degree at which the shadow is nonzero; it equals the L_∞ formality level of the genus-0 part of $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ (Theorem **??**). The Heisenberg atom has $r_{\max}=2$ (formal, $\Delta=0$); affine Kac–Moody algebras have $r_{\max}=3$; the $\beta\gamma$ system has $r_{\max}=4$; Virasoro and $\mathcal{W}$-algebras have $r_{\max}=\infty$. These four cases exhaust the possibilities: the *critical discriminant* $\Delta = 8\kappa S_4$, where S_4 is the quartic shadow coefficient, forces $r_{\max}$ to lie in $\{2,3,\infty\}$ on any one-dimensional primary slice (Theorem **??**), with the fourth class $r_{\max}=4$ arising from a stratum separation phenomenon.

Three structure theorems govern this tower. All three are projections of the single Maurer–Cartan element $\Theta_{\mathcal{A}}\in \mathrm{MC}(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}})$ (Theorem **??**): algebraicity describes its restriction to a primary line, the formality identification describes its genus-0 component, and complementarity describes its Verdier dual. *Algebraicity* (Theorem **??**): on each primary line L, the shadow generating function satisfies $H(t)^2 =$

$t^4 Q_L(t)$ with Q_L a computable quadratic polynomial in three invariants (κ, α, S_4), so the tower is the Taylor expansion of $\sqrt{Q_L}$, algebraic of degree 2. The discriminant Δ governs the fundamental dichotomy: $\Delta = 0$ terminates the tower; $\Delta \neq 0$ makes it infinite. *Formality identification* (Theorem **??**): the genus-0 shadow obstruction tower coincides with the L_∞ formality obstruction tower of the convolution algebra at all degrees; genus $g \geq 1$ corrections form a separate layer. The genus and degree directions are orthogonal (Remark **??**). The *visibility formula* (Corollary **??**) controls which shadow coefficients contribute at each genus:

$$g_{\min}(S_r) = \lfloor r/2 \rfloor + 1 \qquad (r \geq 3).$$

Genus 1 sees only κ (degree 2). Genus 2 first sees S_3 (degree 3). Genus 3 first sees S_4 and S_5 (degrees 4, 5). In general, two new shadow coefficients enter at each genus. *Complementarity* (Theorem **??**): $Q_g(\mathcal{A}) \oplus Q_g(\mathcal{A}^!) \simeq H^*(\overline{\mathcal{M}}_g, Z(\mathcal{A}))$, a Lagrangian decomposition that constrains the Koszul dual pair. Classical Koszul duality over a point embeds into the genus 0, degree 2, $\Delta = 0$ stratum of this picture (the formal, Gaussian case where the tower terminates immediately), but the embedding requires specifying the retraction data of §1.1; even at genus 0, the E_1/E_∞ bar distinction and the Fulton–MacPherson geometry are absent over a bare point. Everything beyond this stratum is new.

1.2 The logarithmic seed

The preface constructs the bar complex from the logarithmic 1-form η_{ij}; the formal treatment follows.

Consider two distinct points z_1, z_2 on a smooth algebraic curve X, and the logarithmic 1-form

$$\eta_{12} = d\log(z_1 - z_2) = \frac{dz_1 - dz_2}{z_1 - z_2}$$

on the configuration space $C_2(X) = \{(z_1, z_2) \in X^2 : z_1 \neq z_2\}$. It has a simple pole along the diagonal with residue 1, and it is conformally invariant. For three distinct points, the Arnold relation

$$\eta_{12} \wedge \eta_{23} + \eta_{23} \wedge \eta_{31} + \eta_{31} \wedge \eta_{12} = 0 \tag{1.2.1}$$

forces the vanishing of triple residues along pairwise collision divisors of $\overline{C}_3(X)$ (see §1.11.1 for the precise definitions of $\overline{C}_n(X)$ and the Arnold relation).

1.2.1 From OPE to bar differential

Let $\mathcal{A}$ be a chiral algebra on X. Place n elements $a_1, \ldots, a_n$ at positions $z_1, \ldots, z_n$ and form

$$a_1(z_1) \otimes \cdots \otimes a_n(z_n) \otimes \bigwedge_{(i,j)\in E(T)} \eta_{ij} \in \mathcal{A}^{\boxtimes n} \otimes \Omega^{n-1}_{\log}(\overline{C}_n(X)),$$

where T is a spanning tree of the complete graph K_n on $\{1, \ldots, n\}$ (selecting which pairs of log forms to wedge), and $\Omega^{n-1}_{\log}(\overline{C}_n(X))$ denotes $(n-1)$-forms on the FM compactification with at worst simple poles along the boundary divisors $D_{\{i,j\}}$. A bar element of degree n is a sum of such terms over all spanning trees, modulo the relations from the Arnold identity. Define a differential d by extracting residues along collision divisors: $d(a_1 \otimes \cdots \otimes a_n) = \sum_{i<j} \mathrm{Res}_{D_{ij}}[\eta_{ij} \cdot \mathrm{OPE}(a_i, a_j)] \otimes$ (remaining factors). The Arnold relation (1.2.1) makes the triple residue vanish, so $d_0^2 = 0$ at genus 0, where d_0 denotes the genus-0 bar differential. The resulting cochain complex is the *bar complex* $\bar{B}_X(\mathcal{A})$.

As a graded object, the bar complex is

$$\bar{B}_X(\mathcal{A}) \;=\; \bigoplus_{n\geq 1}(s^{-1}\bar{\mathcal{A}})^{\otimes n}_{\Sigma_n}\otimes\Gamma(\overline{C}_n(X),\Omega^{n-1}_{\log}),$$

where s^{-1} denotes desuspension ($|s^{-1}a| = |a| - 1$; cohomological convention $|d| = +1$), and $\bar{\mathcal{A}} = \ker(\varepsilon\colon \mathcal{A}\to\omega_X)$ is the *augmentation ideal*, the kernel of the vacuum map (§1.11.3). The subscript Σ_n indicates coinvariants under the symmetric group action $\sigma\cdot(a_1\otimes\cdots\otimes a_n\otimes\omega) = a_{\sigma^{-1}(1)}\otimes\cdots\otimes a_{\sigma^{-1}(n)}\otimes\sigma^*\omega$ that permutes both the algebra factors and the marked points simultaneously. The bar degree (also called tensor degree or degree) is n; the internal degree comes from the conformal grading of $\mathcal{A}$. The total differential decomposes as

$$d_{\bar{B}} \;=\; d_{\text{internal}} + d_{\text{residue}} + d_{\text{dR}},$$

where d_{internal} acts on each tensor factor by the internal differential of $\mathcal{A}$; d_{residue} extracts OPE residues at collision divisors weighted by η_{ij} (the component encoding the bracket); and d_{dR} is the de Rham differential on configuration-space forms. Since the generators η_{ij} are closed, $d_{\text{dR}} = 0$ on the logarithmic subcomplex.

1.2.2 Arnold's identity and $d^2 = 0$

The identity $d^2_{\bar{B}} = 0$ reduces to $d^2_{\text{residue}} = 0$. Indeed, $d^2_{\text{internal}} = 0$ and $d^2_{\text{dR}} = 0$ since each is individually a differential, and the mixed anticommutators vanish by the residue exact sequence on $\overline{C}_n(X)$.

The square $d^2_{\text{residue}} = \sum_{(i,j)}\sum_{(k,\ell)}\operatorname{Res}_{D_{k\ell}}\circ\operatorname{Res}_{D_{ij}}$ splits by index overlap into three cases:

(i) *Disjoint indices* ($\{i,j\}\cap\{k,\ell\} = \emptyset$): residue operations commute up to the Koszul sign and cancel pairwise.

(ii) *One shared index* (say $j = k$): three boundary divisors meet at the codimension-2 stratum $D_{ij\ell}$ of triple collision, producing a cyclic sum

$$\sum_{\text{cyclic}}\operatorname{Res}_{D_{j\ell}}\circ\operatorname{Res}_{D_{ij}}\big[\mu(\mu(a_i,a_j),a_\ell)\otimes\eta_{ij}\wedge\eta_{j\ell}\big]$$

that vanishes if and only if the Arnold relation (1.2.1) kills the form factor and the Borcherds identity kills the algebraic factor.

(iii) *Same pair* ($\{i,j\} = \{k,\ell\}$): $\operatorname{Res}^2_{D_{ij}} = 0$ automatically.

Cases (i) and (iii) impose no condition; case (ii) imposes exactly the Borcherds identity. Thus $d^2_{\bar{B}} = 0$ if and only if the OPE defines a chiral algebra (§1.3).

1.2.3 The bar complex as categorical logarithm

The adjunction $\bar{B}_X \dashv \Omega_X$ converts the multiplicative structure of the OPE into the conilpotent structure of the bar differential. The *Koszul dual* of a chiral algebra $\mathcal{A}$ is constructed in three steps (§1.11.3):

(i) *Bar*: form the bar coalgebra $\bar{B}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$, a conilpotent dg coalgebra.

(ii) *Cohomology*: take bar cohomology $\mathcal{A}^{!`} := H^*(\bar{B}(\mathcal{A}))$, the *Koszul dual coalgebra*.

(iii) *Dualize*: take the graded linear dual $\mathcal{A}^! := (\mathcal{A}^{!`})^\vee$, the *Koszul dual algebra*.

When $\mathcal{A}$ is Koszul, $\bar{B}(\mathcal{A})$ has cohomology concentrated in bar degree 1, and the classical Koszul formula gives $\mathcal{A}^! = T(V^*)/R^\perp$.

Verdier duality $\mathbb{D}_{\mathrm{Ran}}$ on the Ran space (1.11.2) interchanges $\mathcal{A}$ and $\mathcal{A}^!$ (Theorem **??**):

$$\int_X \mathcal{A} \;\simeq\; \mathbb{D}\Big(\int_{-X} \mathcal{A}^!\Big),$$

where $\int_X \mathcal{A}$ denotes factorization homology (§1.11.4) and $\int_{-X}$ denotes factorization cohomology on the orientation-reversed curve.

The counit $\varepsilon\colon \Omega_X(\bar{B}_X(\mathcal{A})) \to \mathcal{A}$ is a quasi-isomorphism on the Koszul locus, where the bar spectral sequence collapses at E_2. Off the Koszul locus, the bar complex may have vanishing ordinary cohomology while carrying nontrivial coderived data (§1.11.4; Positselski [?Positselski11]).

1.2.4 The E_1 story as primitive: averaging and the modular shadow

The bar complex $\bar{B}(\mathcal{A})$ presented in §1.2.1 is the symmetric (commutative) bar: it lives on $(s^{-1}\bar{\mathcal{A}})^{\otimes n}_{\Sigma_n}$ and forgets the order in which collisions occur. This presentation is convenient for formulating the five main theorems and connecting to the modular operad on $\overline{\mathcal{M}}_{g,n}$, but it is not the natural primitive of the theory. The natural primitive is the *ordered* bar

$$\bar{B}^{\mathrm{ord}}(\mathcal{A}) \;=\; T^c(s^{-1}\bar{\mathcal{A}}) \;=\; \bigoplus_{n\geq 0} (s^{-1}\bar{\mathcal{A}})^{\otimes n}$$

with deconcatenation coproduct, which by Stasheff's theorem [?Sta63] is the universal cofree conilpotent coalgebra on $s^{-1}\bar{\mathcal{A}}$. Every A_∞-chiral algebra admits a canonical ordered bar; the symmetric bar arises only after applying Σ_n-coinvariants externally on each tensor factor. The unsymmetrized object retains strictly more information (§1.2.4).

1.2.4.0.1 The ordered convolution algebra and its MC element.

The ordered convolution dg Lie algebra

$$\mathfrak{g}^{\mathrm{mod}\,E_1}_{\mathcal{A}} \;:=\; \prod_{g,n} \mathrm{Hom}\big(F\mathrm{Ass}(g,n),\ \mathrm{End}_{\mathcal{A}}(n)\big),$$

built on the ribbon (associative) modular operad $F\mathrm{Ass}$ rather than the commutative cooperad Sym^c, houses the bar-intrinsic Maurer–Cartan element

$$\Theta^{E_1}_{\mathcal{A}} \;:=\; D^{E_1}_{\mathcal{A}} - d^{(0)}_{\mathcal{A}} \;\in\; \mathrm{MC}\big(\mathfrak{g}^{\mathrm{mod}\,E_1}_{\mathcal{A}}\big).$$

The degree-2 component of $\Theta^{E_1}_{\mathcal{A}}$ at genus 0 is the *classical r-matrix* $r(z) \in \mathrm{End}(V^{\otimes 2}) \otimes \mathcal{O}(*\Delta)$, the meromorphic function whose residue at $z = 0$ controls the spectral scattering of $\mathcal{A}$. The degree-3 component is the *KZ associator* $r_3(z_1, z_2)$, and the degree-≥ 4 components are the higher Yangian/braided coherences that satisfy the boundary equations of Stasheff associahedra: classical Yang–Baxter for $r(z)$, the pentagon for r_3, the quartic R-matrix identity for r_4, and so on (Proposition **??**). The components of $\Theta^{E_1}_{\mathcal{A}}$ are the full data of the line-operator algebra: an R-matrix, a braided category, a Drinfeld–Jimbo Yangian.

1.2.4.0.2 Averaging as the sole information loss. The modular convolution algebra $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ is the Σ_n-coinvariant of $\mathfrak{g}^{\mathrm{mod}\,E_1}_{\mathcal{A}}$ via the *averaging map* (Definition ??, equation (??)):

$$\mathrm{av}\colon \mathfrak{g}^{\mathrm{mod}\,E_1}_{\mathcal{A}} \twoheadrightarrow \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}, \qquad \mathrm{av}(\phi)(g,n) \;=\; \frac{1}{n!}\sum_{\sigma\in\Sigma_n} \sigma\cdot\bigl(\phi(g,n)\circ\iota^{\mathrm{rib}}_{g,n}\bigr). \tag{1.2.2}$$

The map av is a surjective dg Lie morphism (Theorem ??) sending the ordered MC element to the symmetric one (Theorem ??):

$$\mathrm{av}\bigl(\Theta^{E_1}_{\mathcal{A}}\bigr) \;=\; \Theta_{\mathcal{A}}. \tag{1.2.3}$$

At degree 2, this projection records the Σ_2-coinvariant scalar shadow of the residue of $r(z)$ at $z=0$:

$$\kappa_{\mathrm{cl}}(\mathcal{A}) \;=\; \mathrm{av}\bigl(r(z)\bigr) \;=\; \tfrac{1}{2}\,\mathrm{tr}\bigl[\mathrm{Res}_{z=0}\, r(z)\bigr],$$

where the trace is the cyclic trace on $\mathrm{End}(V^{\otimes 2})$. For abelian and Virasoro-type families, $\kappa(\mathcal{A}) = \kappa_{\mathrm{cl}}(\mathcal{A})$; for non-abelian affine Kac–Moody in the trace-form convention, $\kappa(\widehat{\mathfrak{g}}_k) = \kappa_{\mathrm{cl}}(\widehat{\mathfrak{g}}_k) + \dim(\mathfrak{g})/2$. The shadow obstruction tower $(\kappa, \mathfrak{C}, \mathfrak{Q}, \dots)$ is the degree-by-degree image of $(r(z), r_3, r_4, \dots)$ under av (Proposition ??). The E_1 side contains the matrix-valued meromorphic function; the E_∞ side contains its Σ_n-symmetrized scalar.

Theorem 1.2.1 (E_1 primacy;). *Let $\mathcal{A}$ be a cyclic chiral algebra on X.*

- (i) (Surjectivity.) *The averaging map* $\mathrm{av}\colon \mathfrak{g}^{\mathrm{mod}\,E_1}_{\mathcal{A}} \twoheadrightarrow \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ ((1.2.2)) *is a surjective dg Lie morphism.*
- (ii) (MC projection.) av *sends* $\Theta^{E_1}_{\mathcal{A}} \mapsto \Theta_{\mathcal{A}}$ ((1.2.3)). *In particular, the shadow obstruction tower $(\kappa, \mathfrak{C}, \mathfrak{Q}, \dots)$ is the degree-by-degree Σ_n-coinvariant image of the R-matrix tower $(r(z), r_3, r_4, \dots)$.*
- (iii) (Bracket preservation.) av *preserves the dg Lie bracket by Σ_n-equivariance of the operad composition maps.*
- (iv) (Non-splitting.) *The kernel* $\ker(\mathrm{av})$ *is non-trivial: at degree* 3 *it contains the Drinfeld associator* $\Phi_{\mathrm{KZ}}(\mathcal{A})$*, which is annihilated by* av *but satisfies the pentagon equation in* $\mathfrak{g}^{\mathrm{mod}\,E_1}_{\mathcal{A}}$. *In particular, the short exact sequence* $0 \to \ker(\mathrm{av}) \to \mathfrak{g}^{\mathrm{mod}\,E_1}_{\mathcal{A}} \xrightarrow{\mathrm{av}} \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}} \to 0$ *does not split as dg Lie algebras.*

Proof sketch. Parts (i)–(iii) are proved in Chapter ??. The map av is the Reynolds operator $\phi \mapsto \frac{1}{n!}\sum_\sigma \sigma\cdot\phi$; surjectivity follows because av restricts to the identity on Σ_n-invariant elements, which span the target $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$. The bracket is preserved because operad composition in $F\mathrm{Ass}$ is Σ_n-equivariant: coinvariant projection converts T^c-convolution to Sym^c-convolution (Theorem ??). The MC projection $\mathrm{av}(\Theta^{E_1}_{\mathcal{A}}) = \Theta_{\mathcal{A}}$ follows from $\mathrm{av}(D^{E_1}_{\mathcal{A}}) = D_{\mathcal{A}}$ (the genus-completed bar differential descends under coinvariants) and $\mathrm{av}(d_0) = d_0$ (Theorem ??).

For (iv), the Drinfeld associator $\Phi_{\mathrm{KZ}}(\mathcal{A}) \in \ker(\mathrm{av})_{0,3}$ satisfies the pentagon equation in $\mathfrak{g}^{\mathrm{mod}\,E_1}_{\mathcal{A}}$ (the genus-0, degree-3 MC equation; Theorem ??, equation (??)). Its Σ_3-symmetrization vanishes because the pentagon identity is antisymmetric under transposition of the two internal edges. Were av split, there would exist a dg Lie section $s\colon \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}} \hookrightarrow \mathfrak{g}^{\mathrm{mod}\,E_1}_{\mathcal{A}}$ with $\mathrm{av}\circ s = \mathrm{id}$. Such a section would lift the commutative MC element $\Theta_{\mathcal{A}}$ to an ordered element in the image of s; but the pentagon constraint on $\ker(\mathrm{av})$ forces the degree-3 component of any lift to involve Φ_{KZ}, which does not lie in the image of any linear section. $\square$

Remark 1.2.2 (Categorical E_1 primacy). Theorem 1.2.1 is a dg Lie statement. The categorical analogue amounts to DK-5: the braided monoidal category $\mathcal{C}_{\mathrm{line}}$ of line operators is the primitive categorical datum, with modular tensor categories derived via the Drinfeld center $Z(\mathcal{C}_{\mathrm{line}})$ (Conjecture ??). The proved content (MC3 on the evaluation-generated core for all simple types, Theorem ??) establishes the base case. The full categorical E_1 primacy theorem = DK-5.

Principle 1.2.3 (The E_1 primacy thesis). *Theorem* 1.2.1 *admits a concise restatement. The E_1/ordered structure is the natural primitive of the theory. The E_∞/symmetric/modular structure is its Σ_n-coinvariant shadow under* av. *The five main theorems A–H are the invariants that survive averaging. Information flows in only one direction: from the rich E_1 side, which contains the R-matrix, the KZ associator, the Drinfeld–Jimbo Yangian, the full braided category of line operators, and the entire spectral scattering data, to the coarser E_∞ side, which contains only the modular characteristic κ, the shadow obstruction tower, and the genus expansion of the closed sector.*

1.2.4.0.3 Five reasons for the inversion. Operadic primacy. In the hierarchy of E_n-operads, E_1 = Ass is the primitive associative operad and E_∞ = Com arises from it as the colimit Com = $\mathrm{colim}_n\, E_n$ or, equivalently, by Σ_n-coinvariants on each degree. At the level of cofree coalgebras, Stasheff's theorem identifies T^c (deconcatenation) as the universal cofree conilpotent coalgebra; the symmetric cooperad Sym^c is recovered from T^c by external Σ_n-coinvariants. The bar functor that respects this universality is $\bar{B}^{\mathrm{ord}} = T^c(s^{-1}\bar{\mathcal{A}})$, not the symmetric bar.

Information content. An equivariant $\mathrm{End}(V^{\otimes n})$-valued meromorphic function strictly contains its scalar trace. The averaging map av is surjective but not injective: the kernel records the non-symmetric components of the line-operator data. For Heisenberg, $r(z) = k/z$ is already scalar, so the symmetrization recovers $\kappa(\mathcal{H}_k) = k$ without losing matrix-valued data. For affine Kac–Moody, $r(z) = k\,\Omega/z$ contains the full Yangian double; the symmetrization recovers only $\kappa_{\mathrm{dp}}(\widehat{\mathfrak{g}}_k) = k \dim \mathfrak{g}/(2h^\vee)$, and the full $\kappa(\widehat{\mathfrak{g}}_k) = (k + h^\vee)\dim \mathfrak{g}/(2h^\vee)$ appears only after adding the Sugawara shift.

Physical primacy. The natural physical objects are ordered. Three-dimensional holomorphic-topological theories have line operators along the topological direction $\mathbb{R}_t$ (Costello–Gaiotto [?CostelloGaiotto2020]); these line operators form the ordered convolution algebra $\mathfrak{g}^{\mathrm{mod}^{E_1}}_{\mathcal{A}}$ on $\mathbb{C}_z \times \mathbb{R}_t$, and their fusion is the E_1-multiplication. Open-string field theory and factorization on bordered curves carry intrinsic boundary orderings. Integrable lattice models have R-matrices that satisfy the (ordered) Yang–Baxter equation, not its symmetric projection. In every physical setting where the bar complex appears, the line/boundary direction is ordered before any Σ_n-symmetrization is taken.

Categorical primacy. The categorical objects of representation theory are ordered. Quantum groups, Yangians, and braided tensor categories are the natural framework for chiral algebras coupled to a spectral parameter; their structure constants are matrix-valued and obey ordered identities (CYBE, pentagon, hexagon, mixed Yang–Baxter). Symmetric monoidal categories are obtained from braided ones by the Σ_n-coinvariant procedure on the braid groupoid; this is exactly the categorical analogue of the averaging map (1.2.2).

Operadic primacy of Swiss-cheese. The bar complex $B(\mathcal{A})$ is an E_1 chiral coassociative coalgebra over $(\mathrm{ChirAss})^!$; the two-coloured Swiss-cheese operad $\mathrm{SC}^{\mathrm{ch,top}}$ organizes the open/closed derived center pair $(C^\bullet_{\mathrm{ch}}(\mathcal{A},\mathcal{A}),\ \mathcal{A})$, not the bar complex itself (Theorem ??). The operad takes E_1-data as its input colour: the open colour is associative, the closed colour appears only as a retract. The closed sector of any Swiss-cheese algebra is recovered from the open sector by passing to Σ_n-coinvariants along the open-to-closed boundary operator. The closed sector is intrinsically a quotient of the open sector, never an enrichment of it.

1.2.4.0.4 Historical context. The manuscript developed from the Beilinson–Drinfeld theory of factorization algebras on the Ran space [?BD04], which is unordered: a factorization algebra assigns sections to finite subsets of points, with no ordering. That framing made the symmetric bar complex the default starting point and treated the ordered bar as a refinement. The reframing recorded here reverses this: the ordered bar is the universal object, and the symmetric bar is its Σ_n-coinvariant. All five-theorem statements and proofs are unchanged; only the reading order of the dependency DAG shifts. Concretely: Chapter **??** presents the ordered theory in full, and the present chapter and Chapter **??** onward present the av-images of the ordered constructions.

Remark 1.2.4 (Scope of the E_1 reframing). The proofs of Theorems A–D and H are unchanged. The conceptual reordering is as follows: the modular convolution algebra $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ is now read as $\mathrm{av}(\mathfrak{g}^{\mathrm{mod}\,E_1}_{\mathcal{A}})$, the universal MC element $\Theta_{\mathcal{A}}$ as $\mathrm{av}(\Theta^{E_1}_{\mathcal{A}})$, and the modular characteristic as the degree-2 scalar shadow of $r(z)$: for abelian and Virasoro-type families $\kappa(\mathcal{A}) = \mathrm{av}(r(z))$, while non-abelian affine Kac–Moody adds the Sugawara shift $\dim(\mathfrak{g})/2$. The ordered theory of Chapter **??** is the upstream object; the symmetric theory is its image.

The E_1 side is strictly richer: at degree 2 alone it distinguishes algebras that the symmetric side identifies. The Heisenberg E_1-shadow $r_{\mathcal{H}_k}(z) = k/z$ and the $\widehat{\mathfrak{sl}}_2$-shadow $r_{\widehat{\mathfrak{sl}}_2,k}(z) = k\,\Omega_{\mathfrak{sl}_2}/z$ have proportional κ-values when $k \dim \mathfrak{g} = k(k + h^\vee)$, but the underlying r-matrices distinguish them because the Heisenberg kernel is scalar while the affine one is matrix-valued. The ordered bar separates what the symmetric bar collapses.

1.2.5 The genus-g propagator and curvature

At genus 0, the logarithm η_{12} is single-valued on $C_2(\mathbb{P}^1)$, and $d_0^2 = 0$. At genus $g \geq 1$, the function $z_1 - z_2$ has no global meaning. The replacement (Definition **??**) is

$$\eta_{12}^{(g)} \;=\; d\log E(z_1, z_2) \;+\; \sum_{\alpha=1}^{g} (\mathrm{Im}\,\Omega)^{-1}_{\alpha\beta}\, \omega_\alpha(z_1)\, \overline{\omega_\beta(z_2)},$$

where $E(z_1, z_2)$ is the prime form (a section of $K^{-1/2} \boxtimes K^{-1/2}$), $\{\omega_\alpha\}$ are the normalized abelian differentials, and Ω is the period matrix. The second term couples the collision geometry to the complex structure of the curve via the Hodge bundle $\mathbb{E} \to \overline{\mathcal{M}}_g$.

The prime form $E(z_1, z_2)$ is the unique section of $K^{-1/2} \boxtimes K^{-1/2}$ (not $K^{+1/2}$, a common error in the literature) that vanishes to first order along the diagonal $z_1 = z_2$ with unit residue and has prescribed monodromy along the B-cycles determined by a choice of homology basis. Its logarithmic derivative $d\log E$ is the genus-g propagator, replacing $d\log(z_1 - z_2)$.

The Arnold relation acquires a defect:

$$d_{\mathrm{fib}}^2 = \kappa(\mathcal{A}) \cdot \omega_g, \tag{1.2.4}$$

where $\mathbb{E} = \pi_*\omega_\pi$ is the Hodge bundle on $\overline{\mathcal{M}}_g$, $\omega_g = c_1(\det \mathbb{E})$, and $\kappa(\mathcal{A})$ is a scalar determined by the leading OPE singularity (§1.11.1 for $\overline{\mathcal{M}}_g$).

The logarithm $d \log E(z_1, z_2)$ acquires monodromy around the B-cycles of Σ_g; the scalar $\kappa(\mathcal{A})$ is the infinitesimal generator of this monodromy. Period integrals along A-cycles restore nilpotence: $\mathrm{d}g g^2 = 0$ (Theorem ??). The passage from κ to the full monodromy is the chiral Riemann–Hilbert correspondence: the nilpotent datum determines the periodic datum via exponentiation.

As $[\Sigma_g]$ varies over $\overline{\mathcal{M}}_g$, the bar complex traces a family of curved cochain complexes. For uniform-weight algebras (single-generator, or multi-generator with all generators of the same conformal weight), the obstruction class $\mathrm{obs}_g = \kappa(\mathcal{A}) \cdot \lambda_g \in H^*(\overline{\mathcal{M}}_g)$, where $\lambda_g = c_g(\mathbb{E})$ is the top Chern class of the Hodge bundle, is a characteristic class of this family at all genera (Theorem ??). For multi-weight families such as $\mathcal{W}_N$, the weight-1 propagator controls the edge contribution, but the scalar formula $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}}$ *fails* at $g \geq 2$: a nonzero cross-channel correction from mixed-propagator graphs appears (Theorem ??, Open Problem ?? resolved negatively). The Koszul dual $\mathcal{A}^!$ traces a complementary family: their obstruction classes pair to give the full cohomology of moduli with coefficients in the center $Z(\mathcal{A})$.

1.3 The selection principle

The bar construction of §1.2 is a *geometric* machine: take a $\mathcal{D}_X$-module V with a meromorphic pairing $\mu\colon j_* j^*(V \boxtimes V) \to \Delta_!(V)$ (the OPE), place sections at points of $\overline{C}_n(X)$, and extract residues along boundary divisors. The output is a graded object $\bar{B}_X(V, \mu)$ with a candidate differential $d = d_{\mathrm{internal}} + d_{\mathrm{residue}} + d_{\mathrm{dR}}$.

The governing question:

For which algebraic inputs (V, μ) does the FM bar construction produce $d^2 = 0$?

Expanding $(d_1 + d_2 + d_3)^2$, the terms d_1^2, d_3^2, and all mixed anticommutators vanish for formal reasons (cochain complex, de Rham, residue exact sequence). The content is $d_2^2 = 0$: residue extraction must be nilpotent (Theorem ??). Three cases arise from index overlap in $d_2^2 = \sum_{i<j} \sum_{k<\ell} \mathrm{Res}_{D_{k\ell}} \circ \mathrm{Res}_{D_{ij}}$:

(i) *Disjoint indices* ($\{i, j\} \cap \{k, \ell\} = \emptyset$). Residue operations commute; cancellation is by the Koszul sign rule. No condition on μ.

(ii) *One shared index* (say $j = k$). Three boundary strata meet at the codimension-2 locus $D_{ij\ell}$ of triple collision. The cyclic sum of iterated residues

$$\sum_{\text{cyclic}} \mathrm{Res}_{D_{j\ell}} \circ \mathrm{Res}_{D_{ij}} \left[\mu(\mu(\phi_i, \phi_j), \phi_\ell) \otimes \omega \right] = 0$$

holds if and only if μ satisfies the Borcherds identity, the associativity of the chiral product, coupling all OPE pole orders simultaneously.

(iii) *Same pair* ($\{i, j\} = \{k, \ell\}$). $\mathrm{Res}^2_{D_{ij}} = 0$ automatically.

Cases (i) and (iii) impose no condition. Case (ii) imposes exactly the Borcherds identity (equation (??)), axiom (ii) of a chiral algebra (Definition ??). The FM bar construction is well-defined if and only if the OPE satisfies it. This is the *selection principle*: the geometry of configuration space compactifications determines the algebraic axioms.

Two witnesses from Chapter ??:

Heisenberg ($\alpha(z)\alpha(w) \sim k/(z - w)^2$, double pole only). No simple pole, so $d_{\mathrm{bracket}} = 0$. The full differential reduces to $d_{\mathrm{curvature}}$, which produces scalars: $d^2_{\mathrm{curvature}} = 0$ automatically. The Arnold relation handles all of case (ii). No algebraic input needed.

Kac–Moody ($h(z)\,e(w) \sim 2e(w)/(z-w) + \cdots$, simple *and* double poles). The bracket component d_{bracket} is now nonzero, and $d^2_{\text{bracket}} \neq 0$, computed explicitly for all 2048 sign conventions (§??, Remark ??). The obstruction is proportional to the level k: it is the double-pole data that d_{bracket} ignores. The Borcherds identity absorbs this obstruction:

$$d^2_{\text{bracket}} = -\{d_{\text{bracket}},\, d_{\text{curvature}}\}.$$

The curvature is the obstruction; the Borcherds identity is the integrability condition (Proposition ??).

Remark 1.3.1 (Classical vs. chiral selection). The bar construction asks the same question at every level of geometry:

Base	**Condition for $d^2 = 0$**	**Selected objects**
Point	Jacobi identity	Lie algebras
Curve X	Borcherds identity	Chiral algebras
X over $\overline{\mathcal{M}}_g$	+ period corrections	Modular Koszul algebras

Each row is forced by the geometry of the next: the Jacobi identity handles triple collisions when all poles are simple; on a curve, multiple pole orders mix at the codimension-2 strata of $\overline{C}_n(X)$, and the Borcherds identity is the unique condition that makes this mixing consistent; at higher genus, period corrections from the Hodge bundle restore $\mathrm{d}g g^{\,2} = 0$. Chiral algebras are what configuration space geometry accepts; modular Koszul algebras are what the genus tower accepts.

1.4 The arithmetic face

The genus-1 free energy $F_1^{\text{conn}}(\mathcal{A}; q) = -\log\det(1 - K_q)$ encodes connected sewing amplitudes $a_{\mathcal{A}}(N)$ as Fourier coefficients. Here K_q is the *sewing kernel*, the operator on $H^*(\bar{B}^{(1,1)}(\mathcal{A}))$ with eigenvalues q^n weighted by the conformal-weight-n dimensions ($q = e^{2\pi i\tau}$), and $\det(1-K_q)$ is its Fredholm determinant. The *Dirichlet–sewing lift*

$$S_{\mathcal{A}}(u) := \sum_{N=1}^{\infty} a_{\mathcal{A}}(N)\, N^{-u}$$

is a Dirichlet series whose analytic structure is controlled by the shadow obstruction tower. The sewing amplitudes define a reproducing kernel $K_{\mathcal{A}}(s,t) = S_{\mathcal{A}}(s{+}t)$, and the surface moment matrix $\mathscr{M}_{ij} = K_{\mathcal{A}}(\alpha/2{+}i,\ \alpha/2{+}j)$ is positive definite for all $\alpha \geq 2$ (Theorem ??).

Explicit values:

$$S_{\mathcal{H}}(u) = \zeta(u)\zeta(u{+}1), \quad S_{\mathrm{Vir}}(u) = \zeta(u{+}1)(\zeta(u)-1), \quad S_{\mathcal{W}_N}(u) = \zeta(u{+}1)\Big((N{-}1)\zeta(u) - \textstyle\sum_{j=1}^{N-1} H_j(u)\Big).$$

1.4.1 The three-tier Euler–Koszul classification

The Euler defect $D_{\mathcal{A}}(u) := S_{\mathcal{A}}(u)/(|W(\mathcal{A})|\,\zeta(u)\zeta(u{+}1))$, where $|W(\mathcal{A})|$ is the order of the Weyl group of the Lie algebra underlying $\mathcal{A}$ (or 1 for non-Lie-type algebras), classifies chiral algebras into three tiers:

Tier	Condition	Families
Exact Euler–Koszul	$D_{\mathcal{A}} \equiv 1$	$\mathcal{H}, V_k(\mathfrak{g}), \beta\gamma$
Finitely defective	$1 - D_{\mathcal{A}}$ is finite harmonic	Virasoro, $\mathcal{W}_N$
Non-Euler–Koszul	Hecke L-function decomposition	lattice with $h(D) \geq 2$

The Euler–Koszul class $\mathrm{ek}(\mathcal{A})$ and shadow depth $r_{\max}(\mathcal{A})$ are independent invariants (Theorem ??).

1.4.2 The two-variable L-object and shadow-moduli resolution

Rankin–Selberg unfolding produces

$$L_{\mathcal{A}}(s,u) := \int_{\mathcal{F}}^{\mathrm{reg}} F_{\mathcal{A}}^{\mathrm{conn}}(\tau)\, E^*(\tau,s)\, y^{u-2}\, dx\, dy,$$

collapsing to $L_{\mathcal{A}}(s,u) = \frac{\Gamma(v)}{(2\pi)^v} S_{\mathcal{A}}(v)$ with $v = s + u - 1$. The shadow-moduli map $t\colon \mathfrak{H} \to \mathbb{C}$ solves $\prod_j (1 - \lambda_j t(q))^{c_j} = \det(1 - K_q^{\mathrm{conn}})$, and the MC recursion determines shadow coefficients S_r from seed data (S_2, S_3, S_4) via a quadratic recurrence. Defining power sums $p_r := -r\, S_r$ and elementary symmetric polynomials e_k via the Newton relations yields a formal spectral polynomial whose roots (the *spectral atoms*) encode the shadow obstruction tower in spectral coordinates (Proposition ??).

For even unimodular lattice VOAs, the Hecke–Newton closure (Theorem ??) gives the chain: MC equation $\Rightarrow$ Hecke decomposition $\Rightarrow$ CPS automorphy $\Rightarrow$ Sym^{r-1} identification $\Rightarrow$ Ramanujan bound (Corollary ??). The MC framework supplies the Hecke decomposition and CPS input; the bound itself relies on Deligne's theorem [?Deligne1974] as external input.

1.4.3 The Weil analogy

Weil proof for $C/\mathbb{F}_q$	MC equation for $\mathcal{A}$
Frobenius Frob_q	MC element $\Theta_{\mathcal{A}} = D_{\mathcal{A}} - d_0$
$\mathrm{Frob}^2 = q$	$D_{\mathcal{A}}^2 = 0$
Étale cohomology	cyclic deformation complex
Weight filtration $W_\bullet$	degree filtration $\Theta_{\mathcal{A}}^{\leq r}$
Eigenvalues $\lVert\alpha_i\rVert = q^{1/2}$	Ramanujan $\lvert a_{f_j}(p)\rvert \leq 2p^{(k-1)/2}$

The descent from $\Theta_{\mathcal{A}}$ to arithmetic is a fan, not a chain: three independent projections (categorical, spectral, modular) share a source but do not compose. The categorical projection (Hecke decomposition, Euler product, Ramanujan bound) passes through the converse theorem and Langlands functoriality. The spectral projection (shadow-metric Epstein zeta, L-function factorization) passes through the discriminant Δ and the shadow field K_L. The modular projection (clutching law, residue clutching defect, Conjecture ??) passes through the boundary stratification of $\overline{\mathcal{M}}_g$. Each projection extracts distinct arithmetic content from the single MC element.

1.4.4 The Koszul self-duality principle

Koszul self-duality $\mathcal{A} \simeq \mathcal{A}^!$ forces L-function factorisation. The Verdier involution $\sigma\colon \bar{B}(\mathcal{A}) \to \bar{B}(\mathcal{A}^!) = \bar{B}(\mathcal{A})$ commutes with Hecke operators (Theorem ??); combined with strong multiplicity one for newforms, $Z_{\mathcal{A}}$ decomposes into Hecke eigenforms, each giving a standard L-function with Euler product (Theorem ??). Self-dual lattice VOAs ($h(D) = 1$) produce factored Epstein zeta functions whose zeros lie on the critical line; non-self-dual lattices ($h(D) > 1$) escape this constraint and exhibit the Davenport–Heilbronn off-line zeros (Remark ??). For rational VOAs, the theta decomposition bridge (Proposition ??) gives Dirichlet L-functions modulo the auxiliary level $2pq$, resolving the non-lattice gap and showing that the scalar non-multiplicativity of $\sum |\chi_i|^2$ is a projection artefact: each theta summand carries its own Euler product.

1.5 The tautological programme

The MC element $\Theta_{\mathcal{A}}$ determines shadow tautological classes $\tau_{g,n}(\mathcal{A}) \in R^*(\overline{\mathcal{M}}_{g,n+1})$ forming a cohomological field theory (Theorem ??). The MC equation at (g, n) yields a tautological relation (Theorem ??). At genus 0, degree 3: WDVV (Proposition ??). At degree 2, genus $g \geq 2$: Mumford's relation $\lambda_g^2 = 0$ (Proposition ??).

The complementarity propagator $P_{\mathcal{A}}$ is the Givental R-matrix; when the shadow CohFT has a flat unit (which holds for all standard families with the vacuum in V), Teleman reconstruction recovers $\{\tau_{g,n}\}$ from genus-0 data (Theorem ??). Eynard–Orantin topological recursion is the recursive structure of the MC equation (Corollary ??).

1.6 The primitive kernel

By the Milnor–Moore theorem for modular operads (a cofree conilpotent modular coalgebra is determined by its primitive part), the MC element $\Theta_{\mathcal{A}}$ is determined by its restriction to primitive corollas, the *primitive logarithmic modular kernel* $\mathfrak{K}_{\mathcal{A}}$. The *modular homological vector field* $Q^{\mathrm{mod}}_{\mathcal{A}}$ is the coderivation on the modular bar coalgebra $\bar{B}^{\mathrm{mod}}(\mathcal{A})$ induced by the full differential $D_{\mathcal{A}}$; its square $(Q^{\mathrm{mod}}_{\mathcal{A}})^2 = 0$ encodes $D_{\mathcal{A}}^2 = 0$. The fundamental equivalence (Theorem ??) rewrites the global equation in terms of primitives:

$$(Q^{\mathrm{mod}}_{\mathcal{A}})^2 = 0 \qquad \Longleftrightarrow \qquad d\,\mathfrak{K}_{\mathcal{A}} + \mathfrak{K}_{\mathcal{A}} \star \mathfrak{K}_{\mathcal{A}} + \hbar\,\Delta_{\mathrm{ns}}(\mathfrak{K}_{\mathcal{A}}) = 0,$$

where the three terms are:

(i) $d\,\mathfrak{K}_{\mathcal{A}}$: the internal differential;

(ii) $\mathfrak{K}_{\mathcal{A}} \star \mathfrak{K}_{\mathcal{A}}$: the *primitive pre-Lie product*, defined by composing two primitive corollas at a shared leg via the log-FM residue map ($\star$ is pre-Lie: $f \star g - g \star f$ is the Lie bracket);

(iii) $\hbar\,\Delta_{\mathrm{ns}}(\mathfrak{K}_{\mathcal{A}})$: the *non-separating loop operator*, contracting two legs of a single corolla with the invariant pairing and raising genus by 1 (the BV operator restricted to primitives).

The branch BV action on the finite-rank quotient $V^{\mathrm{br}}_{\mathcal{A}}$ (the branch-active vector space, spanned by the spectral generators of the Koszul dual Hilbert series) satisfies the quantum master equation (QME), and its *metaplectic half-density* $\delta_{\mathcal{A}} \in 1 + x\,\Bbbk[\![x]\!]$ satisfies $\delta_{\mathcal{A}}^2 = \Delta_{\mathcal{A}}$, where $\Delta_{\mathcal{A}}$ is the *spectral discriminant* (the BV Laplacian on the branch quotient, §??).

The primitive kernel determines the four archetype families completely. The components $K_{g,n}$ are the restrictions of $\mathfrak{K}_{\mathcal{A}}$ to the corolla of type (g, n) (genus g, n legs); R_{pf} denotes the planted-forest (FM boundary) corrections. The "active channels" column records which terms in the fundamental equivalence are nonzero:

Family	Primitive kernel $\mathfrak{K}_{\mathcal{A}}$	Active channels
Heisenberg	$K_{0,2} + K_{1,1}$	Δ_{ns} only
Affine $\widehat{\mathfrak{g}}_k$	$K_{0,2} + K_{0,3} + K_{1,1}$	Δ_{ns}, $[-,-]$
$\beta\gamma$	$K_{0,2} + K_{0,4} + K_{1,1} + R_{\mathrm{pf}}$	Δ_{ns}, Rig (rigidity)
Virasoro/$\mathcal{W}_N$	$K_{0,2} + K_{0,3} + K_{0,4} + K_{1,1} + R_{\mathrm{pf}} + \cdots$	all three

1.7 The flat connection and its affine conformal-block comparison

The MC equation $D\Theta_{\mathcal{A}} + \frac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = 0$ defines a flat connection on the primitive deformation complex. The modular tangent complex (Construction ??) carries the twisted differential

$$d_{\Theta_{\mathcal{A}}}(x) \;=\; \sum_{g\geq 0}\sum_{n\geq 0} \frac{\hbar^g}{n!}\, \ell_{n+1}^{(g)}(\Theta_{\mathcal{A}}^{\otimes n},\, x),$$

where $\ell_{n+1}^{(g)}$ are the Taylor coefficients of the quantum L_∞ structure on $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$: the operation $\ell_k^{(g)}$ is a k-linear bracket of genus g, with $\ell_1^{(0)} = D$ (the differential), $\ell_2^{(0)} = [-,-]$ (the Lie bracket), and $\ell_k^{(g)}$ for $k \geq 3$ or $g \geq 1$ the higher homotopy brackets obtained by Feynman graph summation over stable graphs of type (g,k) (Theorem ??). The strict chart is $\nabla_{\mathcal{A}}^{\mathrm{mod}} = d + [\Theta_{\mathcal{A}}, -]$ on the modular convolution algebra $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$. The MC equation is equivalent to flatness: $(\nabla_{\mathcal{A}}^{\mathrm{mod}})^2 = 0$.

The chain from primitive kernel to the flat modular connection:

$$\mathfrak{K}_{\mathcal{A}} \xrightarrow{\mathrm{FT}^{\log}_{\mathrm{mod}}} \Theta_{\mathcal{A}} \xrightarrow{\text{linearise}} \nabla_{\mathcal{A}}^{\mathrm{mod}} \xrightarrow{\text{integrable affine } H^0 \text{ flat sections}} \mathrm{CB}_g(\widehat{\mathfrak{g}}_k).$$

The flat connection $\nabla_{\mathcal{A}}^{\mathrm{mod}}$ lives on the cohomology sheaf of the bar complex over $\overline{\mathcal{M}}_{g,n}$. For $\widehat{\mathfrak{g}}_k$ at integrable level, its H^0 flat sections recover the Tsuchiya–Ueno–Yamada conformal blocks, and its monodromy furnishes the bar-side comparison surface for the KZ/Hitchin package. More generally, the primitive kernel $\mathfrak{K}_{\mathcal{A}}$ controls the modular flat connection through the Feynman transform: the degree-2 corolla $K_{0,2}$ determines the projectively flat genus-0 structure, the degree-3 corolla $K_{0,3}$ determines, on the affine Kac–Moody comparison surface, the genus-0 KZ equations, and the higher corollas $K_{0,n}$ ($n \geq 4$) encode quantum corrections to that tree-level connection.

For the four archetypes: the Heisenberg connection $\nabla_{\mathcal{H}}^{\mathrm{mod}} = d$ is trivially flat; the affine genus-0 restriction $\nabla_{\widehat{\mathfrak{g}}}^{\mathrm{mod}}|_{g=0}$ is the KZ shadow connection, and on the integrable affine lane its monodromy gives the bar-side comparison surface for the KZ/Hitchin package; the $\beta\gamma$ connection carries a quartic correction from $K_{0,4}$; the Virasoro connection $\nabla_{\mathrm{Vir}}^{\mathrm{mod}} = d + [\Theta_{\mathrm{Vir}}, -]$ is the modular flat connection controlled by the infinite shadow tower, producing an infinite hierarchy of quantum corrections to the tree-level projectively flat structure.

1.8 The three-dimensional reading

The bar differential extracts residues in $\mathbb{C}$; the bar coproduct splits ordered sequences in $\mathbb{R}$. The bar complex $B(\mathcal{A})$ is an E_1 chiral coassociative coalgebra over $(\mathrm{ChirAss})^!$: its differential encodes holomorphic factorization on $\mathbb{C}$, its deconcatenation coproduct encodes topological factorization on $\mathbb{R}$. The two-coloured $\mathsf{SC}^{\mathrm{ch,top}}$ structure of Volume II emerges on the chiral derived center pair $(C^\bullet_{\mathrm{ch}}(\mathcal{A},\mathcal{A}),\ \mathcal{A})$, not on $B(\mathcal{A})$ itself. The *holomorphic-topological Swiss-cheese operad* $\mathsf{SC}^{\mathrm{ch,top}}$ has *closed-colour* operations parametrised by $\mathrm{FM}_k(\mathbb{C})$ (Fulton–MacPherson configurations of k points in $\mathbb{C}$), *open-colour* operations parametrised by $\mathrm{Conf}^<_k(\mathbb{R})$ (ordered configurations of k points in $\mathbb{R}$), and mixed operations encoding the bulk-boundary interaction. It is *homotopy-Koszul* (i.e. the bar-cobar resolution is a quasi-isomorphism: Kontsevich formality + classical Swiss-cheese Koszulity + homotopy transfer), and the bar-cobar adjunction on $\mathsf{SC}^{\mathrm{ch,top}}$-algebras is a Quillen equivalence.

The product $\mathrm{FM}_k(\mathbb{C}) \times \mathrm{Conf}^<_k(\mathbb{R})$ is the *chiral Steinberg correspondence*. The analogy: in geometric representation theory, the classical Steinberg variety $\widetilde{\mathcal{N}} \times_{\mathfrak{g}} \widetilde{\mathcal{N}}$ (the fibre product of two copies of the Springer resolution $\widetilde{\mathcal{N}} \to \mathfrak{g}$ over the Lie algebra) carries a convolution product whose endomorphism algebra is the Hecke algebra. Here, $\mathrm{FM}_k(\mathbb{C})$ plays the role of $\widetilde{\mathcal{N}}$ and $\mathrm{Conf}^<_k(\mathbb{R})$ plays the role of the fibre parameter. The *Swiss-cheese convolution algebra* $\mathfrak{g}^{\mathrm{SC}}_T := \mathrm{Hom}_\Sigma(\bar{B}(\mathsf{SC}^{\mathrm{ch,top}}), \mathrm{End}_{(B_\partial, \mathcal{C}_{\mathrm{line}})})$ (where T denotes a holomorphic-topological theory, B_∂ its boundary algebra, and $\mathcal{C}_{\mathrm{line}}$ its line-operator category) carries a single MC element $\alpha_T \in \mathrm{MC}(\mathfrak{g}^{\mathrm{SC}}_T)$ that extracts six structures: the A_∞ operations, the PVA λ-bracket, the spectral R-matrix and Yang–Baxter equation, the bulk-boundary-line triangle, the genus tower, and the line-operator category. For the affine lineage, the monodromy of the KZ connection is compared with the quantum-group R-matrix of $U_q(\mathfrak{g})$ by the affine Drinfeld–Kohno package; on the reduced evaluation comparison surface, the reduced line category is identified with the corresponding quantum-group side, and the colored Jones polynomial is recovered from the monodromy representation classified by the bar complex.

The genus expansion of the modular bar differential,

$$D = \sum_{g \geq 0} \hbar^g D^{(g)}, \qquad D^{(g)} = \sum_{\substack{\Gamma \in \mathsf{StGraph}^{\mathrm{conn}} \\ g(\Gamma)=g}} \frac{1}{|\mathrm{Aut}(\Gamma)|} D_\Gamma,$$

where $\mathsf{StGraph}^{\mathrm{conn}}$ is the set of connected stable graphs (§1.11.1) and D_Γ is the graph-amplitude operator (the composition of transferred operations at each vertex of Γ, contracted along internal edges by the propagator), is the Feynman transform of the modular operad $\{\overline{\mathcal{M}}_{g,n}\}$ (§1.11.2). The canonical MC element $\delta_{\mathcal{A}} = D - D^{(0)} = \Theta_{\mathcal{A}}$ (the positive-genus correction of the master MC element, defined in §1.12.2) computes the total positive-genus correction; the genus-by-genus identities $\sum_{p+q=g} D^{(p)} D^{(q)} = 0$ are the modular sewing relations. The primitive boundary kernel $\mathfrak{K}_{\mathcal{A}} = K_{0,2} + K_{0,3} + \cdots + K_{1,1} + R_{\mathrm{pf}} + \cdots$ reconstructs the full modular MC field via the logarithmic modular Feynman transform $\Theta_{\mathcal{A}} = \mathrm{FT}^{\log}_{\mathrm{mod}}(\mathfrak{K}_{\mathcal{A}})$.

The deformation-quantization bridge from PVA to vertex algebra passes through the half-space $\mathbb{C} \times \mathbb{R}_{\geq 0}$. The *doubling theorem* of Volume II embeds the half-space BV problem into the doubled whole-space $\mathbb{C} \times \mathbb{R}$ via the method of images: the reflected propagator $K^{\mathrm{refl}}(p_i, p_j) = K(p_i, p_j) + K(p_i, \sigma(p_j))$ (where σ is reflection across the boundary and K is the BV propagator) makes boundary contributions vanish by symmetry. Every logarithmic $\mathsf{SC}^{\mathrm{ch,top}}$-algebra has a boundary vertex algebra. The five-component datum (bulk factorization algebra, boundary vertex algebra, derived center, line-operator category, spectral R-matrix) assembles into a single holographic package: for any logarithmic $\mathsf{SC}^{\mathrm{ch,top}}$-algebra satisfying the standing physical axioms (H1)–(H4) of Volume II, all five components exist and are consistent (Volume II, Theorem 4.1). The standard landscape enters through two gates: the affine bridge (the affine PV sigma model satisfies the physics bridge hypotheses) and Drinfeld–Sokolov transport (DS reduction preserves the $\mathsf{SC}^{\mathrm{ch,top}}$ structure).

The bar complex $\bar{B}(\mathcal{A})$ classifies *twisting morphisms* (universal couplings between $\mathcal{A}$ and $\mathcal{A}^!$); the *chiral derived center* $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}) = H^*(C^\bullet_{\mathrm{ch}}(\mathcal{A},\mathcal{A}),\delta)$ (Definition ??) classifies *bulk operators* acting on the boundary. The annulus trace $\mathrm{Tr}_{\mathcal{A}} \simeq HH_*(\mathcal{A})$ (Theorem ??) is the first modular shadow of the open sector, connecting boundary topological Hochschild homology to the genus-1 open partition function. Both structures are packaged by the *open/closed MC element* $\Theta^{\mathrm{oc}}_{\mathcal{A}} = \Theta_{\mathcal{A}} + \sum \mu^{M_j}$ (Construction ??), which encodes the boundary MC class and the bulk module contributions in a single equation. The resulting *completed modular Koszul datum* $\Pi^{\mathrm{oc}}_X(\mathcal{A})$ (Definition ??) upgrades the six-fold modular Koszul datum to an eight-fold datum by adjoining the derived center, the annulus trace, and the nonlinear resonance tower in $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$. The primitive object is C_{op} (the open factorization dg-category), not the boundary algebra $A_b = \mathrm{End}(b)$. The bulk is the derived center $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$, not the bar complex. Modularity is trace plus clutching on the open sector.

1.9 The five theorems

The protagonist of this monograph is the universal Maurer–Cartan element $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}})$ (Theorem ??), defined by $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d^{(0)}_{\mathcal{A}}$: the difference between the full bar differential $D_{\mathcal{A}}$ and its genus-0 restriction. Since $D^2_{\mathcal{A}} = 0$, the element $\Theta_{\mathcal{A}}$ is automatically Maurer–Cartan. The five theorems characterize five structural properties of $\Theta_{\mathcal{A}}$: *existence* (Theorem A constructs the arena $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$), *faithfulness* (Theorem B shows $\Theta_{\mathcal{A}}$ determines $\mathcal{A}$), *decomposition* (Theorem C splits the genus-g projection into complementary Lagrangians), *leading coefficient* (Theorem D extracts the universal scalar $\kappa(\mathcal{A})$), and *coefficient ring* (Theorem H identifies $\mathrm{ChirHoch}^*(\mathcal{A})$ as the finite-dimensional coefficient space over which $\Theta_{\mathcal{A}}$ varies). The shadow obstruction tower $\Theta^{\leq 2}_{\mathcal{A}}, \Theta^{\leq 3}_{\mathcal{A}}, \Theta^{\leq 4}_{\mathcal{A}}, \ldots$ consists of finite-order projections of this single element; κ is the degree-2 projection, the cubic shadow C is the degree-3 projection, and the quartic resonance class Q is the degree-4 projection.

The element $\Theta_{\mathcal{A}}$ is itself the closed-sector projection of a larger structure. The *open/closed MC element* $\Theta^{\mathrm{oc}}_{\mathcal{A}} = \Theta_{\mathcal{A}} + \sum \mu^{M_j}$ (Construction ??, Theorem ??) packages both boundary data (the bar-intrinsic class $\Theta_{\mathcal{A}}$, governing twisting morphisms between $\mathcal{A}$ and $\mathcal{A}^!$) and bulk data (the chiral derived center $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$, governing operators in the interior). The five theorems characterize the closed projection $\Theta_{\mathcal{A}} = \pi_{\mathrm{cl}}(\Theta^{\mathrm{oc}}_{\mathcal{A}})$; the bulk projection is controlled by the $\mathrm{SC}^{\mathrm{ch,top}}$-algebra structure on the derived center pair $(\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}), \mathcal{A})$ (Theorem ??). Modularity (the genus expansion and its tautological descent to $\overline{\mathcal{M}}_g$) belongs to the trace and clutching operations on the open sector, not to the closed algebra alone.

1.9.1 The Heisenberg algebra

Take $\mathcal{A} = \mathcal{H}_k$ with OPE $J(z)J(w) \sim k/(z-w)^2$. One generator, one double pole, no simple pole.

Theorem A (bar-cobar adjunction). The bar complex is $\bar{B}(\mathcal{H}_k) = T^c(s^{-1}\mathbb{C}\cdot J)$, the cofree coalgebra on a single cogenerator. The bar differential on a degree-2 element is

$$d_{\bar{B}}[s^{-1}J \mid s^{-1}J] \;=\; \mathrm{Res}_{z_1=z_2}\left[\frac{k}{(z_1-z_2)^2}\cdot d\log(z_1-z_2)\right] \;=\; k.$$

At degree 3: the Arnold relation on $\overline{C}_3(\mathbb{C})$ gives $d^2_{\bar{B}} = 0$. The three collision divisors contribute $k\cdot(\eta_{12}\wedge\eta_{23} + \eta_{23}\wedge\eta_{13} + \eta_{13}\wedge\eta_{12}) = 0$.

Verdier duality on $\mathrm{Ran}(X)$ produces the homotopy Koszul dual: $\mathbb{D}_{\mathrm{Ran}}\,\bar{B}(\mathcal{H}_k) \simeq (\mathcal{H}^!_k)_\infty$, with $\kappa(\mathcal{H}^!_k) = -k$. The Heisenberg is not Koszul self-dual: $\mathcal{H}^!_k \neq \mathcal{H}_{-k}$ (§??).

Theorem B (bar-cobar inversion). The bar cohomology is $H^{-1}(\bar{B}(\mathcal{H}_k)) = \mathbb{C} \cdot s^{-1}J$ (concentrated in bar degree 1), so the Koszul dual coalgebra is $\mathcal{H}_k^{\text{i}} = \mathbb{C} \cdot s^{-1}J$. The Koszul dual algebra is $\mathcal{H}_k^! = (\mathcal{H}_k^{\text{i}})^\vee$: the symmetric chiral algebra $\mathrm{Sym}^{\mathrm{ch}}(V^*)$ with curvature $m_0 = -k\,\omega$ encoding the opposite level. By the Verdier intertwining of Theorem A, $\kappa(\mathcal{H}_k^!) = -k$, the opposite modular characteristic (the Heisenberg is *not* Koszul self-dual; see §??). The cobar of the bar is $\Omega(\bar{B}(\mathcal{H}_k)) = T(s\,\mathbb{C} \cdot s^{-1}J) = \mathbb{C}[J]$, which is $\mathcal{H}_k$ as a graded algebra. The counit $\varepsilon\colon \Omega(\bar{B}(\mathcal{H}_k)) \to \mathcal{H}_k$ is a quasi-isomorphism: the Heisenberg is Koszul.

Theorem C (complementarity). At genus $g \geq 1$: $Q_g(\mathcal{H}_k) \oplus Q_g(\mathcal{H}_k^!) \cong H^*(\overline{\mathcal{M}}_g, \mathcal{Z}_{\mathcal{H}_k})$. The modular characteristic is $\kappa(\mathcal{H}_k) = k$, and $\kappa(\mathcal{H}_k) + \kappa(\mathcal{H}_k^!) = k + (-k) = 0$. The complementarity sum of central charges is $c(\mathcal{H}_k) + c(\mathcal{H}_k^!) = 1 + 1 = 2$ (noting that the dual is curved, so $c(\mathcal{H}_k^!) = 1$ refers to the underlying graded algebra).

Theorem D (modular characteristic). At genus 1: the self-loop graph Γ_{irr} on $\overline{\mathcal{M}}_{1,1}$ contributes $d_{\mathrm{fib}}^2 = k \cdot \omega_1$. The genus-1 obstruction number is $F_1 = k/24$. The generating function:

$$\sum_{g\geq 1} F_g(\mathcal{H}_k)\, x^{2g} \;=\; k\!\left(\frac{x/2}{\sin(x/2)} - 1\right) \;=\; \frac{k}{24}\,x^2 + \frac{7k}{5760}\,x^4 + \frac{31k}{967680}\,x^6 + \cdots$$

The coefficients are $F_g = k \cdot \frac{2^{2g-1}-1}{2^{2g-1}} \cdot \frac{|B_{2g}|}{(2g)!}$, where B_{2g} is the $2g$-th Bernoulli number. (The chiral partition function $Z_{\mathrm{ch}} = 1/\eta(\tau)$ gives $F_1 = \kappa(\mathcal{H}_k)/24 = k/24$ as the genus-1 free energy on $\overline{\mathcal{M}}_{1,1}$.)

Theorem H (chiral Hochschild). $\mathrm{ChirHoch}^*(\mathcal{H}_k) = (\mathbb{C}, \mathbb{C}, \mathbb{C})$ concentrated in degrees $\{0, 1, 2\}$: the class in degree 0 is the center (scalars), the class in degree 1 is the outer chiral derivation $D(\alpha) = \mathbf{1}$ (non-inner because $\alpha_{(0)} = 0$ when the OPE has no simple pole), and the class in degree 2 is the level deformation (double-pole OPE datum). The Koszul resolution has length 2, so $\mathrm{ChirHoch}^n = 0$ for $n \geq 3$; Hilbert series $1 + t + t^2$, the rank-1 case of the polynomial-growth theorem.

The full computation appears in Chapter **??**.

1.9.2 The general theorems

Here $\mathrm{Fact}(X)$ denotes the symmetric monoidal category of factorization algebras on X (§1.11.4); $\mathrm{Alg}_{\mathrm{aug}}(\mathrm{Fact}(X))$ is the category of *augmented* factorization algebras (equipped with a map $\mathcal{A} \to \omega_X$ to the vacuum); $\mathrm{CoAlg}_{\mathrm{conil}}(\mathrm{Fact}(X))$ is the category of *conilpotent* factorization coalgebras (the iterated reduced coproduct eventually vanishes).

Each statement below is the av-image of a corresponding E_1 statement on the ordered convolution algebra $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}\,E_1}$ (see §1.2.4 for the averaging map $\mathrm{av}\colon \mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}\,E_1} \twoheadrightarrow \mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$ and Chapter **??** for the ordered counterparts A^{E_1}–H^{E_1}). The five theorems are the invariants that survive averaging.

- *Theorem A* (Geometric bar-cobar duality, Theorem **??**). The bar functor

$$\bar{B}_X\colon\ \mathrm{Alg}_{\mathrm{aug}}(\mathrm{Fact}(X)) \to \mathrm{CoAlg}_{\mathrm{conil}}(\mathrm{Fact}(X))$$

  is intertwined with Verdier duality (§1.2.3): $\mathbb{D}_{\mathrm{Ran}}\bar{B}_X(\mathcal{A}) \simeq \mathcal{A}_\infty^!$ (factorization *algebra*, not coalgebra), functorial in families over $\overline{\mathcal{M}}_{g,n}$. It decomposes into chiral twisting morphisms (A_0, Theorem **??**), bar concentration (A_1, Theorem **??**), and Verdier-geometric duality (A_2, Theorem **??**). On the proved genus-0 locus, the bar complex admits a BRST comparison; Verdier duality is the algebraic shadow of CPT.

- *Theorem B* (Bar-cobar inversion, Theorem ??). On the Koszul locus, the counit $\Omega_X \bar{B}_X(\mathcal{A}) \xrightarrow{\sim} \mathcal{A}$ is an equivalence; the spectral sequence collapses at E_2. Off the locus, the bar-cobar object persists but becomes curved; the failure is measured by $\Theta_{\mathcal{A}}$ (1.12.2). The BRST resolution is complete.
  More precisely, there is a spectral sequence with E_1 page $E_1^{p,q} = H^q(\bar{B}^p_X(\mathcal{A}))$ converging to $H^{p+q}(\Omega_X \bar{B}_X(\mathcal{A}))$. On the Koszul locus, $E_1^{p,q} = 0$ for $p \neq 1$, so $E_2 = E_\infty$, and the comparison map $\varepsilon\colon \Omega_X \bar{B}_X(\mathcal{A}) \to \mathcal{A}$ induces an isomorphism on cohomology. This is the categorical exponential: $\exp \circ \log = \mathrm{id}$.
- *Theorem C* (Deformation-obstruction complementarity, Theorem ??). The *center local system* $\mathcal{Z}_{\mathcal{A}}$ is the local system on $\overline{\mathcal{M}}_g$ whose fibre over a smooth curve $[\Sigma_g]$ is the center $Z(\mathcal{A}|_{\Sigma_g})$ (the commutant of $\mathcal{A}$ acting on itself via the chiral bracket). The *ambient complex* $\mathbf{C}_g(\mathcal{A}) := R\Gamma(\overline{\mathcal{M}}_g, \mathcal{Z}_{\mathcal{A}})$ (derived global sections of this local system) carries a shifted-symplectic pairing from Verdier duality. The Verdier involution decomposes it into *complementary Lagrangians* $\mathbf{Q}_g(\mathcal{A}) \oplus \mathbf{Q}_g(\mathcal{A}^!)$: the $+1$ and -1 eigenspaces of the involution (Definition ??). At the cohomological level,

$$Q_g(\mathcal{A}) \oplus Q_g(\mathcal{A}^!) \cong H^*(\overline{\mathcal{M}}_g, \mathcal{Z}_{\mathcal{A}}).$$

  The anomaly of one algebra is the ghost-number violation of its dual.
  The shifted-symplectic structure is a $(-(3g-3))$-shifted symplectic form on the derived moduli of genus-g bar families, in the sense of Pantev–Toën–Vaquié–Vezzosi [?PTVV13]. The complementary Lagrangians $Q_g(\mathcal{A})$ and $Q_g(\mathcal{A}^!)$ are the ± 1 eigenspaces of the Verdier involution $\sigma = \mathbb{D}_{\mathrm{Ran}}$ acting on $\mathbf{C}_g(\mathcal{A})$. Their intersection is the center $Z(\mathcal{A})$; the ambient complex is the direct sum.
- *Theorem D_{scal}* (Scalar modular characteristic, Theorem ??). A single scalar $\kappa(\mathcal{A})$, the genus-1 curvature coefficient (1.2.4), determines the entire *scalar modular package*: the collection $(\kappa, \{\mathrm{obs}_g\}_{g\geq 1}, \{F_g\}_{g\geq 1})$ consisting of the modular characteristic, its genus-g obstruction classes, and the free energies. For algebras whose strong generators have the same conformal weight (including all single-generator families), $\mathrm{obs}_g = \kappa \cdot \lambda_g \in H^*(\overline{\mathcal{M}}_g)$ and $F_g = \kappa \cdot \hat{A}_g$ at all genera. For multi-weight algebras (e.g. $\mathcal{W}_N$, $N \geq 3$), the new rigidity theorem forces the minimal-model MC element into the one-dimensional cyclic direction $\Theta^{\min} = \eta \otimes \Gamma_{\mathcal{A}}$; the stronger identification $\Gamma_{\mathcal{A}} = \kappa(\mathcal{A})\Lambda$, and hence the all-genera identities $\mathrm{obs}_g = \kappa \cdot \lambda_g$ and $F_g = \kappa \cdot \hat{A}_g$, hold on the uniform-weight lane but *fail* for multi-weight algebras: the genus-g free energy receives a cross-channel correction $\delta F_g^{\mathrm{cross}}$ from mixed-channel boundary graphs (Theorem ??, Open Problem ?? resolved negatively). The scalar κ is universal, additive under tensor product, and duality-constrained ($\kappa + \kappa' = 0$ for affine Kac–Moody and free fields, and a fixed nonzero constant for $\mathcal{W}$-algebras). On the uniform-weight lane, the generating function is the $\hat{A}$-genus (Theorem 1.10.1):

$$\sum_{g\geq 1} F_g(\mathcal{A}) \cdot \hbar^{2g} = \kappa(\mathcal{A}) \cdot \big(\hat{A}(i\hbar) - 1\big).$$

  At genus 1 this reduces to $F_1(\mathcal{A}) = \kappa(\mathcal{A})/24$ for every family.
  Theorem D_Δ (Spectral characteristic, Theorem ??): the spectral discriminant $\Delta_{\mathcal{A}}(x)$ is a separately proved non-scalar invariant.
  κ is the one-loop coefficient of the effective action; its vanishing removes the scalar genus-1 anomaly, while higher-genus multi-weight corrections are a separate issue. Together, D_{scal} and D_Δ are the characteristic classes of the factorization homology bundle $\int_{\Sigma_g} \mathcal{A}$ (Remark ??); the bundle itself (its fibers, flat connection, and modular functor structure) is the full modular homotopy invariant (Conjecture ??).

- *Theorem H* (Chiral Hochschild cohomology, Theorem ??). The *chiral Hochschild cohomology* $\mathrm{ChirHoch}^*(\mathcal{A})$ is the cohomology of the complex $\mathrm{CoDer}(\bar{B}(\mathcal{A}))$ (coderivations of the bar coalgebra, with differential induced by the bar differential). On the Koszul locus, $\mathrm{ChirHoch}^*(\mathcal{A})$ is polynomial and dual to $\mathrm{ChirHoch}^*(\mathcal{A}^!)$. Its brace and Gerstenhaber structures control the moduli of quantizations and the passage from Poisson to chiral (Chapter ??).

Principle 1.9.1 (Five facets of $\Theta_{\mathcal{A}}$). *Let $\mathcal{A}$ be a modular Koszul chiral algebra on a smooth projective curve X, and let $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d^{(0)}_{\mathcal{A}} \in \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ be the bar-intrinsic Maurer–Cartan element* (*Theorem* ??). *The five main theorems characterize five* facets *of $\Theta_{\mathcal{A}}$: existence, faithfulness, decomposition, leading coefficient, and coefficient ring.*

(A) Arena (*Theorem A*). $\Theta_{\mathcal{A}}$ ***exists****. The bar–cobar adjunction constructs the modular convolution algebra $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ in which $\Theta_{\mathcal{A}}$ lives; with it, $\Theta_{\mathcal{A}}$ arises automatically as $D_{\mathcal{A}} - d^{(0)}_{\mathcal{A}}$.*

(B) Faithfulness (*Theorem B*). $\Theta_{\mathcal{A}}$ ***determines*** $\mathcal{A}$. *On the Koszul locus, the counit $\Omega_X(\bar{B}_X(\mathcal{A})) \xrightarrow{\sim} \mathcal{A}$ is a quasi-isomorphism.*

(C) Decomposition (*Theorem C*). $\Theta_{\mathcal{A}}$ ***splits****. The genus-g projection decomposes under the Verdier involution into complementary Lagrangians $\mathbf{Q}_g(\mathcal{A}) \oplus \mathbf{Q}_g(\mathcal{A}^!) = H^*(\overline{\mathcal{M}}_g, Z(\mathcal{A}))$.*

(D) Leading coefficient (*Theorem D*). $\Theta_{\mathcal{A}}$ *has* ***universal scalar projection****. On the uniform-weight lane, the degree-2 genus-g component is* $\mathrm{obs}_g(\mathcal{A}) = \kappa(\mathcal{A})\,\lambda_g$ *at all genera, with $\kappa(\mathcal{A})$ universal and governed by the $\hat{A}$-genus. At genus* 1 *this holds unconditionally for all families.*

(E) Coefficient ring (*Theorem H*). $\Theta_{\mathcal{A}}$ *has* ***polynomial coefficients****.* $\mathrm{ChirHoch}^*(\mathcal{A})$ *is concentrated in degrees* $\{0, 1, 2\}$, *controlling the deformation ring over which $\Theta_{\mathcal{A}}$ varies.*

Remark 1.9.2 (E_1-chiral counterparts of the five theorems). Each of Theorems A–H has an E_1 counterpart (Theorems A^{E_1}–H^{E_1}, Chapter ??), in which the symmetric bar $\bar{B}^{\Sigma}(\mathcal{A})$ on unordered configuration spaces is replaced by the *ordered bar* $\bar{B}^{\mathrm{ord}}(\mathcal{A})$ on $\mathrm{Conf}^{\mathrm{ord}}_n(X)$, the chiral Koszul dual $\mathcal{A}^!$ is replaced by the *dg-shifted Yangian* $\mathcal{A}^!_{\mathrm{line}}$, and the commutative modular operad FCom is replaced by the ribbon modular operad FAss. The five facets transfer: $\Theta^{E_1}_{\mathcal{A}}$ exists in the ordered convolution algebra $\mathfrak{g}^{\mathrm{mod},E_1}_{\mathcal{A}}$ (Theorem A^{E_1}); the ordered cobar inverts on the ordered Koszul locus (Theorem B^{E_1}); the genus-g projection decomposes into complementary Lagrangians via Verdier duality on ribbon graphs (Theorem C^{E_1}); the leading coefficient is the same scalar $\kappa(\mathcal{A})$ (Theorem D^{E_1}); and the coefficient ring is the ordered chiral Hochschild cohomology (Theorem H^{E_1}). The Σ_n-coinvariant map $F\mathrm{Ass} \to \mathrm{FCom}$ sends $\Theta^{E_1}_{\mathcal{A}} \mapsto \Theta_{\mathcal{A}}$: the symmetric theory is the unordered projection of the ordered theory.

The two Koszul dualities are co-equal but geometrically distinct. The symmetric bar sees unordered collisions and produces the chiral Koszul dual $\mathcal{A}^!$, an E_∞-chiral algebra on the curve. The ordered bar sees ordered collisions and produces the dg-shifted Yangian $\mathcal{A}^!_{\mathrm{line}}$, an E_1-chiral algebra whose R-matrix is the Koszul-dual spectral parameter. The symmetric dual lives on X; the ordered dual lives on a line in the 3d bulk $\mathbb{C}_z \times \mathbb{R}_t$.

Theorem 1.9.3 (Central charge complementarity;). [Regime: curved-central on the Koszul locus (Convention 1.25.2).]

For Koszul dual pairs related by the Feigin–Frenkel involution $k \mapsto k' = -k - 2h^\vee$:

(a) Affine Kac–Moody. $c(\widehat{\mathfrak{g}}_k) + c(\widehat{\mathfrak{g}}_{k'}) = 2\dim\mathfrak{g}$.

(b) Principal $\mathcal{W}$-algebras: $c(\mathcal{W}^k(\mathfrak{g})) + c(\mathcal{W}^{k'}(\mathfrak{g})) = 2\,\mathrm{rank}(\mathfrak{g}) + 4h^\vee \dim\mathfrak{g}$.

In particular, $\mathfrak{g} = \mathfrak{sl}_2$ in part (b) *gives $c + c' = 26$.*

Proof. Part (a). The Sugawara central charge is $c(\widehat{\mathfrak{g}}_k) = k \dim \mathfrak{g}/(k + h^\vee)$. Since $k' + h^\vee = -(k + h^\vee)$:

$$c(k) + c(k') = \frac{k \cdot d}{k + h^\vee} + \frac{(k + 2h^\vee) \cdot d}{k + h^\vee} = \frac{2(k + h^\vee) \cdot d}{k + h^\vee} = 2d$$

where $d = \dim \mathfrak{g}$.

Part (b). The Drinfeld–Sokolov central charge is $c(\mathcal{W}^k(\mathfrak{g})) = r - 12|\rho|^2(t-1)^2/t$ where $r = \operatorname{rank}(\mathfrak{g})$ and $t = k + h^\vee$ (verified for $\mathfrak{sl}_2$ in Theorem **??** and for $\mathfrak{sl}_3$ in Theorem **??**). The involution sends $t \mapsto -t$:

$$c(k) + c(k') = 2r + 12|\rho|^2 \cdot \frac{(t+1)^2 - (t-1)^2}{t} = 2r + 48|\rho|^2.$$

The Freudenthal–de Vries identity $|\rho|^2 = h^\vee \dim \mathfrak{g}/12$ (under normalization $(\theta, \theta) = 2$; see [?Kac], Ch. 13) yields $c + c' = 2r + 4h^\vee d$.

Verification. $\mathfrak{sl}_2$: $2(1) + 4(2)(3) = 26$. $\mathfrak{sl}_3$: $2(2) + 4(3)(8) = 100$. $\mathfrak{sl}_N$: $2(N-1) + 4N(N^2-1) = 2(N-1)(2N^2+2N+1)$. □

Remark 1.9.4. The identity $c + c' = 26$ for $\mathfrak{sl}_2$ is the algebraic incarnation of $c_{\text{matter}} + c_{\text{ghost}} = 0$. At the critical point, the relative BRST complex of the bosonic string is quasi-isomorphic to the genus-0 bar complex (Theorem **??**). The identification of the ghost system with the cobar of the matter bar complex requires the heuristic BV/bar bridge of Chapter **??**; the bosonic string critical dimension follows from Koszul duality at the scalar level without this bridge.

The five theorems assemble into a single commutative square:

$$\begin{tikzcd}
\mathrm{Alg}_{\mathrm{aug}}(\mathrm{Fact}(X)) \arrow[r, "\bar{B}_X"] \arrow[d, "(\cdot)^!"'] & \mathrm{CoAlg}_{\mathrm{conil}}(\mathrm{Fact}(X)) \arrow[d, "\mathbb{D}_{\mathrm{Ran}}"] \\
\mathrm{Alg}_{\mathrm{aug}}(\mathrm{Fact}(X)) \arrow[r, "\bar{B}_X"'] & \mathrm{CoAlg}_{\mathrm{conil}}(\mathrm{Fact}(X))
\end{tikzcd}$$

Commutativity is Theorem A. The cobar functor Ω_X, left adjoint to $\bar{B}_X$, gives the counit of Theorem B. At genus $g \geq 1$, the universal Maurer–Cartan class $\Theta_{\mathcal{A}}$ measures the failure of strict commutativity. Vanishing of the scalar curvature term $\kappa(\mathcal{A})$ removes one obstruction, but the manuscript's reduction from the intrinsic coderived/contraderived comparison to ordinary derived categories is proved only on the flat finite-type completed-dual loci singled out later in the Bar Complex part.

Off the Koszul locus, the coderived category (Positselski [?Positselski11]) is the correct ambient category: the curved bar object can lose ordinary cohomological visibility while remaining nontrivial in the coderived framework. Ordinary derived categories are recovered only on the flat finite-type completed-dual loci.

Beyond the five theorems, the programme produces: (i) the Pixton ideal of the modular curve from the MC tautological relations (Theorem **??**, proved for semisimple shadow CohFTs; obstructed for non-semisimple by Proposition **??**); (ii) a complete resolution of the BV/BRST identification for free fields (Theorem **??**); (iii) the multi-weight genus tower through genus 4 with explicit closed forms and a growth theorem (Proposition **??**); (iv) non-principal Koszul duality for the Bershadsky–Polyakov algebra and nilpotent transport through $\mathfrak{sl}_8$ (Computation **??**, Proposition **??**).

1.9.3 The main theorem

Theorem 1.9.5 (Modular Koszul duality;). *Let $\mathcal{A}$ be a chiral algebra on a smooth projective curve X over $\mathbb{C}$, equipped with a non-degenerate invariant pairing. The genus-graded bar complex $\{\bar{B}^{(g)}_X(\mathcal{A})\}_{g\geq 0}$ is an algebra over the Feynman transform* FCom *of the commutative modular operad* (*Theorem* ??), *with genus-completed differential*

$$D_{\mathcal{A}} = \sum_{g\geq 0} \hbar^g\, d^{(g)}_{\mathcal{A}} \quad \textit{satisfying} \quad D^2_{\mathcal{A}} = 0.$$

(i) Bar–cobar duality. *The bar functor $\bar{B}_X$ and cobar functor Ω_X form an adjoint pair on augmented factorization algebras, with Verdier intertwining $\mathbb{D}_{\mathrm{Ran}}\, \bar{B}_X(\mathcal{A}) \simeq \bar{B}_X(\mathcal{A}^!)$* (*Theorem* ??). *On the Koszul locus, the counit $\Omega_X(\bar{B}_X(\mathcal{A})) \xrightarrow{\sim} \mathcal{A}$ is a quasi-isomorphism at every genus* (*Theorem* ??).

(ii) Universal Maurer–Cartan class. *The positive-genus correction*

$$\Theta_{\mathcal{A}} \;:=\; D_{\mathcal{A}} - d^{(0)}_{\mathcal{A}} \;=\; \sum_{g\geq 1} \hbar^g\, d^{(g)}_{\mathcal{A}}$$

is a Maurer–Cartan element: $\Theta_{\mathcal{A}} \in \mathrm{MC}\big(\mathrm{Def}_{\mathrm{cyc}}(\mathcal{A}) \mathbin{\widehat{\otimes}} \mathfrak{G}^{\mathrm{mod}}\big)$. The Maurer–Cartan equation holds because $D^2_{\mathcal{A}} = 0$; no construction is required beyond the bar differential itself.

(iii) Projections. *The scalar trace channel of $\Theta_{\mathcal{A}}$ is governed by the genus-1 coefficient $\kappa(\mathcal{A}) \in \mathbb{C}$. For uniform-weight modular Koszul algebras,* $\mathrm{obs}_g(\mathcal{A}) = \kappa(\mathcal{A})\,\lambda_g$ *for all $g \geq 1$; for arbitrary families the unconditional scalar statement is $F_1 = \kappa/24$* (*Remark* ??). *The coefficient $\kappa(\mathcal{A})$ is universal, additive under tensor product, duality-constrained ($\kappa + \kappa' = 0$ for KM/free fields, a fixed constant for $\mathcal{W}$-algebras), and has $\hat{A}$-genus generating function* (*Theorem* ??). *Evaluated against a conformal deformation $g_1 = e^{2\sigma} g_0$ of the worldsheet metric, $\kappa(\mathcal{A})$ yields the Polyakov formula with coefficient $\kappa/(6\pi)$ (Proposition* ??*): the modular characteristic is the algebraic form of the conformal anomaly. The eigenspace decomposition under the Verdier involution gives the Lagrangian polarization $\mathbf{Q}_g(\mathcal{A}) \oplus \mathbf{Q}_g(\mathcal{A}^!)$* (*Theorem* ??). *The coefficient ring is the polynomial chiral Hochschild cohomology* $\mathrm{ChirHoch}^*(\mathcal{A})$ (*Theorem* ??).

(iv) Shadow obstruction tower. *The finite-order projections $\Theta^{\leq 2}_{\mathcal{A}} \to \Theta^{\leq 3}_{\mathcal{A}} \to \Theta^{\leq 4}_{\mathcal{A}} \to \cdots$ give the shadow obstruction tower: $\kappa(\mathcal{A})$ at degree 2, the cubic shadow $\mathfrak{C}(\mathcal{A})$ at degree 3, the quartic resonance class $\mathfrak{Q}(\mathcal{A})$ at degree 4. The element $\Theta_{\mathcal{A}}$ satisfies clutching factorization under stable-curve degeneration and Verdier duality: $\mathbb{D}(\Theta_{\mathcal{A}}) = \Theta_{\mathcal{A}^!}$.*

(v) The chiral center theorem. *The bar complex $\bar{B}_X(\mathcal{A})$ is an E_1 chiral coassociative coalgebra over* $(\mathrm{ChirAss})^!$ (*deconcatenation coproduct = factorization along an ordered line; bar differential = holomorphic OPE residues*). *The $\mathsf{SC}^{\mathrm{ch,top}}$ structure emerges on the chiral derived center pair, not on $\bar{B}_X(\mathcal{A})$ itself. The operadic center of this structure is the chiral Hochschild cochain complex:*

$$Z_{\mathsf{SC}}(\mathcal{A}) \;\simeq\; C^\bullet_{\mathrm{ch}}(\mathcal{A},\mathcal{A}),$$

and the pair $(C^\bullet_{\mathrm{ch}}(\mathcal{A},\mathcal{A}),\ \mathcal{A})$ is the terminal local chiral Swiss-cheese pair over $\mathcal{A}$ (*Theorem* ??, *Theorem* ??). *This is the chiral analogue of the Deligne–Tamarkin theorem: the bulk of any holomorphic boundary theory is forced by locality to be the chiral Hochschild cochains.*

Remark 1.9.6. The genus tower is the Taylor expansion of the monodromy $\kappa(\mathcal{A}) \cdot \omega_g$. The nonlinear characteristic layer computes the higher Taylor jets of this expansion: the quartic jets of the complementarity potential (Chapter ??), the spectral cumulants of the branch-line quotient (Appendix ??), and the canonical homotopy degree in the non-principal $\mathcal{W}$-algebra sector.

Remark 1.9.7 (Two proofs of the center theorem). Item (v) is proved by two independent routes: an *operadic* proof via the recognition theorem and homotopy-Koszulity of $\mathsf{SC}^{\mathrm{ch,top}}$ (§??), and a *direct* proof constructing the unique brace morphism $\Phi\colon B \to C^\bullet_{\mathrm{ch}}(\mathcal{A}, \mathcal{A})$ with explicit Koszul signs (§??). The key new difficulty versus the topological Deligne conjecture is the passage from smooth to meromorphic propagators: Arnold relations replace Stokes, and genus $g \geq 1$ introduces curvature $\kappa(\mathcal{A}) \cdot \omega_g$ into the center (curved E_2-algebra).

1.9.4 Shadow separation, tautological classes, and analytic realization

Corollary 1.9.8 (Shadow separation; ; Theorem ??). (i) Completeness. $[\Theta^{\leq r}_{\mathcal{A}}] = [\Theta^{\leq r}_{\mathcal{B}}]$ *for all* $r \geq 2$ *implies* $[\Theta_{\mathcal{A}}] = [\Theta_{\mathcal{B}}]$ *in* $\mathrm{MC}(\widehat{\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}})/\!\sim$.

(ii) Strict refinement. $\kappa(\mathcal{H}_1) = \kappa(\mathrm{Vir}_2) = 1$ *but* $\mathfrak{C}_{\mathcal{H}} = 0 \neq 2x^3 = \mathfrak{C}_{\mathrm{Vir}}$*; on the weight-changing line,* $\mathfrak{Q}_{\mathcal{H}} = 0 \neq \mathrm{cyc}(m_3) = \mathfrak{Q}_{\beta\gamma}$.

(iii) Koszul invariance. $r_{\max}(\mathcal{A}^!) = r_{\max}(\mathcal{A})$ (*Corollary* ??).

The cyclic trace of the graph-sum formula defines shadow tautological classes $\tau_{g,n}(\mathcal{A}) \in R^*(\overline{\mathcal{M}}_{g,n+1})$ (Definition ??, Theorem ??): $\tau_{g,2}(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \pi^*\lambda_g$ at degree 2, with new classes at higher degrees whose vertex weights are the cyclic traces of the transferred operations $\ell_n^{(g)}(\mathcal{A})$. For every algebra in the standard landscape, the algebraic shadow data extends to the *sewing envelope* $\mathcal{A}^{\mathrm{sew}}$, the Hausdorff completion of $\mathcal{A}$, with respect to the seminorms defined by sewing amplitudes (Corollary ??).

1.10 The modular characteristic and the spectral discriminant

The genus-1 curvature $\kappa(\mathcal{A})$ is the scalar component of the MC element at degree 2: it controls the genus-1 obstruction $d^2 = \kappa\, \omega_1$, the leading term of the $\hat{A}$-genus generating function, and the Hodge-bundle eigenvalue of the spectral measure.

1.10.1 The explicit formula for κ

Let $\mathcal{A}$ be a chiral algebra on a smooth projective curve X with OPE $a(z)\, b(w) \sim \sum_{n\geq 0} (a_{(n)}b)(w)/(z-w)^{n+1}$. The modular characteristic $\kappa(\mathcal{A})$ is the genus-1 curvature coefficient: $\kappa(\mathcal{H}_k) = k$ for the Heisenberg algebra, $\kappa(\widehat{\mathfrak{g}}_k) = (k + h^\vee)\dim\mathfrak{g}/(2h^\vee)$ for affine Kac–Moody, and $\kappa(\mathrm{Vir}_c) = c/2$ for Virasoro (Theorem ??). For uniform-weight algebras (Definition ??), the obstruction class at genus g is universally determined:

$$\mathrm{obs}_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g \in H^{2g}(\overline{\mathcal{M}}_g, \mathbb{Q}),$$

where $\lambda_g = c_g(\mathbb{E})$ is the top Chern class of the Hodge bundle. This holds at all genera; for multi-weight families such as $\mathcal{W}_N$, the scalar formula fails at $g \geq 2$: a cross-channel correction from mixed-propagator boundary graphs is generically nonzero (Theorem ??). The free-energy formula $F_g = \kappa \cdot \frac{2^{2g-1}-1}{2^{2g-1}} \cdot \frac{|B_{2g}|}{(2g)!}$

holds for uniform-weight algebras (Theorem ??). The first explicit multi-channel computation is the $\mathcal{W}_3$ genus-2 cross-channel correction (Remark ??):

$$F_2(\mathcal{W}_3) = \underbrace{\frac{7c}{6912}}_{\kappa\cdot\lambda_2^{\mathrm{FP}}} + \underbrace{\frac{c+204}{16c}}_{\delta F_2},$$

verified by graph-by-graph recomputation over all seven genus-2 stable graphs and 2^e channel assignments per graph (Theorem ??(vi), Computation ??). The cross-channel correction $\delta F_2 = (c{+}204)/(16c)$ is the multi-generator signal at genus 2; it tends to 1/16 as $c \to \infty$ and diverges as $c \to 0$. This resolves Open Problem ?? negatively: the scalar formula fails for multi-weight algebras at $g \geq 2$. The gravitational cross-channel correction has a universal closed form for all principal $\mathcal{W}_N$ algebras (Proposition ??): $\delta F_2^{\mathrm{grav}}(\mathcal{W}_N, c) = (N{-}2)(N{+}3)/96 + (N{-}2)(3N^3{+}14N^2{+}22N{+}33)/(24c)$, vanishing if and only if $N = 2$ (Virasoro). The formula is exact for $\mathcal{W}_3$ and a lower bound for $N \geq 4$.

1.10.2 Universality, additivity, anti-symmetry

Three properties make κ a topological characteristic class:

(i) **Universality.** The genus-1 identity $\mathrm{obs}_1(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_1$ is unconditional for every chirally Koszul algebra. On the uniform-weight lane, the all-genera extension $\mathrm{obs}_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g$ and the Faber–Pandharipande free-energy formula hold via clutching morphisms on $\partial\overline{\mathcal{M}}_g$ (an independent GRR derivation provides an alternative path); see Theorem ??. For multi-weight algebras (e.g. $\mathcal{W}_N$, $N \geq 3$), the scalar formula fails at $g \geq 2$: a nonzero cross-channel correction appears (Theorem ??).

(ii) **Additivity.** For a tensor product $\mathcal{A} \otimes \mathcal{B}$ of chiral algebras,

$$\kappa(\mathcal{A} \otimes \mathcal{B}) = \kappa(\mathcal{A}) + \kappa(\mathcal{B}).$$

This follows from the factorization of the bar complex: $\bar{B}_X(\mathcal{A} \otimes \mathcal{B}) \simeq \bar{B}_X(\mathcal{A}) \otimes \bar{B}_X(\mathcal{B})$.

(iii) **Duality constraint.** Under Koszul duality, $\kappa(\mathcal{A}) + \kappa(\mathcal{A}^!) = 0$ for affine Kac–Moody algebras (where $\mathcal{A}^! = \widehat{\mathfrak{g}}_{k'}$ with $k' = -k - 2h^\vee$) and for free-field algebras. For $\mathcal{W}$-algebras from Drinfeld–Sokolov reduction, $\kappa + \kappa'$ is a fixed nonzero constant depending on $\mathfrak{g}$: $\kappa(\mathrm{Vir}_c) + \kappa(\mathrm{Vir}_{26-c}) = 13$ for Virasoro (Theorem ??(iii)).

1.10.3 The $\hat{A}$-genus generating function

A single function from index theory controls the genus expansion of every uniform-weight modular Koszul algebra. Grothendieck–Riemann–Roch applied to the universal curve $\pi\colon \overline{C}_g \to \overline{\mathcal{M}}_g$ yields $\mathrm{ch}(R\pi_*\bar{B}_g(\mathcal{A})) = \kappa(\mathcal{A}) \cdot \mathrm{td}(\mathbb{E}^\vee)$, and the Todd class of the dual Hodge bundle produces the $\hat{A}$-series.

Theorem 1.10.1 ($\hat{A}$-genus universality;). *Let $\mathcal{A}$ be a modular Koszul chiral algebra whose strong generators have uniform conformal weight. Then the free energies $F_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g^{\mathrm{FP}}$ at all genera $g \geq 1$, where*

λ_g^{FP} *are the Faber–Pandharipande evaluations of the Hodge class* ($\lambda_1^{\mathrm{FP}} = 1/24$, $\lambda_2^{\mathrm{FP}} = 7/5760$, $\lambda_3^{\mathrm{FP}} = 31/967680, \ldots$). *The generating function is the $\hat{A}$-genus:*

$$\sum_{g\geq 1} F_g(\mathcal{A}) \cdot \hbar^{2g} = \kappa(\mathcal{A}) \cdot \left(\hat{A}(i\hbar) - 1\right) = \kappa(\mathcal{A}) \cdot \left(\frac{\hbar/2}{\sin(\hbar/2)} - 1\right),$$

with radius of convergence $|\hbar| = 2\pi$, independent of $\mathcal{A}$. For arbitrary modular Koszul algebras (*including multi-weight families*)*, the genus-1 coefficient $F_1(\mathcal{A}) = \kappa(\mathcal{A})/24$ holds unconditionally.*

The function $(x/2)/\sin(x/2)$ is the Wick rotation of the $\hat{A}$-genus $(x/2)/\sinh(x/2)$ from the Atiyah–Singer index theorem; the positivity $\lambda_g^{\mathrm{FP}} > 0$ reflects the Hodge index theorem on $\overline{\mathcal{M}}_g$. For a free fermion, the formula yields the spin partition function; for the Virasoro algebra, the one-loop string amplitude (Theorem ??; equation (??)).

1.10.4 The spectral discriminant

The spectral discriminant $\Delta_{\mathcal{A}}(x) := \mathrm{disc}(P_{\mathcal{A}^!}(x))$, the discriminant of the Koszul dual Hilbert series, is the first non-scalar invariant, invariant under both Koszul duality ($\Delta_{\mathcal{A}} = \Delta_{\mathcal{A}^!}$, Proposition ??) and Drinfeld–Sokolov reduction (Theorem ??).

The Koszul dual Hilbert series $P_{\mathcal{A}^!}(x) = \sum_{n\geq 0} \dim H^n(\bar{B}(\mathcal{A})) \cdot x^n$ satisfies $P_{\mathcal{A}^!}(x) = 1/P_{\mathcal{A}}(-x)$ for Koszul algebras. Its discriminant detects spectral structure invisible to κ.

The shared discriminant $\Delta(x) = (1-3x)(1+x) = 1-2x-3x^2$ across $\widehat{\mathfrak{sl}}_2$, Virasoro, and $\beta\gamma$ is not coincidence: these algebras lie on the same spectral sheet of the Koszul dual parameter space. The zeros of Δ at $x = 1/3$ and $x = -1$ mark the transitions between bar-acyclic and bar-active sectors.

1.11 Algebraic prerequisites

The definitions below are standard; full treatments appear in the cited chapters.

1.11.1 Configuration spaces and their combinatorics

Fulton–MacPherson compactification. For a smooth projective curve X, the *open configuration space* $C_n(X) = \{(z_1, \ldots, z_n) \in X^n : z_i \neq z_j\}$ parametrises n distinct labelled points. The *Fulton–MacPherson compactification* $\overline{C}_n(X)$ is the iterated real-oriented blowup of X^n along all diagonals, starting from the smallest (pairwise) and proceeding to larger ones (Definition ??). The result is a smooth manifold with corners whose boundary strata D_S, one for each subset $S \subseteq \{1, \ldots, n\}$ with $|S| \geq 2$, parametrise configurations in which the points labelled by S collide. The *logarithmic one-forms* $\eta_{ij} = d\log(z_i - z_j) \in \Omega^1_{\log}(\overline{C}_n(X))$ have simple poles along $D_{\{i,j\}}$ and no other poles.

Arnold relation. On $\overline{C}_3(X)$, three logarithmic one-forms satisfy the *Arnold relation*:

$$\eta_{12} \wedge \eta_{23} \ + \ \eta_{23} \wedge \eta_{31} \ + \ \eta_{31} \wedge \eta_{12} \ = \ 0 \quad \text{as logarithmic differential forms on } \overline{C}_3(X). \qquad (1.11.1)$$

This is the geometric content behind $d^2 = 0$ at genus 0: the bar differential, built from residues along collision divisors, squares to zero because any two codimension-one boundaries meet transversally and the three-term relation (1.11.1) makes the codimension-two contributions cancel in pairs.

Stable graphs. A *stable graph* Γ of type (g, n) consists of

(i) a connected graph with vertex set $V(\Gamma)$ and edge set $E(\Gamma)$;
(ii) a *genus function* $g_\bullet : V(\Gamma) \to \mathbb{Z}_{\geq 0}$;
(iii) n labelled *tails* (half-edges) attached to vertices;
(iv) *stability*: $2g(v) - 2 + \mathrm{val}(v) > 0$ for every vertex v, where $\mathrm{val}(v)$ counts all half-edges (edges and tails) at v.

The total genus is $g(\Gamma) = \sum_v g(v) + h^1(\Gamma)$, where $h^1(\Gamma) = |E(\Gamma)| - |V(\Gamma)| + 1$ is the first Betti number. Every boundary stratum of the Deligne–Mumford compactification $\overline{\mathcal{M}}_{g,n}$ is indexed by a stable graph: $\overline{\mathcal{M}}_\Gamma = \prod_{v \in V(\Gamma)} \overline{\mathcal{M}}_{g(v),\mathrm{val}(v)}$ (Definition ??).

Ran space. The *Ran space* of X is the colimit

$$\mathrm{Ran}(X) = \varinjlim_{I \in \mathrm{fSet}^{\mathrm{surj}}} X^I, \tag{1.11.2}$$

taken over the category of non-empty finite sets with surjections. A surjection $I \twoheadrightarrow J$ induces a diagonal embedding $X^J \hookrightarrow X^I$, so $\mathrm{Ran}(X)$ is the space of finite non-empty subsets of X; points are allowed to collide (Definition ??).

1.11.2 Operadic algebra

Symmetric sequences. A *symmetric sequence* (or $\mathbb{S}$-module) is a collection $V = \{V(n)\}_{n\geq 0}$ of chain complexes, each equipped with a left Σ_n-action. A morphism $f : V \to W$ of symmetric sequences consists of Σ_n-equivariant chain maps $f(n) : V(n) \to W(n)$ for each n. The *equivariant hom* $\mathrm{Hom}_{\Sigma_n}(V(n), W(n))$ denotes the chain complex of Σ_n-equivariant maps.

Endomorphism operad. For a chain complex A, the *endomorphism operad* End_A is the symmetric sequence with

$$\mathrm{End}_A(n) = \mathrm{Hom}(A^{\otimes n}, A), \tag{1.11.3}$$

where Σ_n acts by permuting the tensor factors of $A^{\otimes n}$. Composition is given by substitution: for $f \in \mathrm{End}_A(k)$ and $g_1, \dots, g_k \in \mathrm{End}_A(n_1), \dots, \mathrm{End}_A(n_k)$,

$$(f \circ (g_1, \dots, g_k))(a_1 \otimes \cdots \otimes a_N) = f\big(g_1(a_1 \otimes \cdots \otimes a_{N_1}) \otimes \cdots \otimes g_k(a_{N_{k-1}+1} \otimes \cdots \otimes a_{N_k})\big),$$

where $N_j = n_1 + \cdots + n_j$, $N_0 = 0$, and $N = N_k$. An *algebra over an operad $\mathcal{P}$* is a morphism of operads $\mathcal{P} \to \mathrm{End}_A$.

Modular operad. A *modular operad* (Getzler–Kapranov [?GK98]) consists of a symmetric sequence $\mathcal{O} = \{\mathcal{O}(g, n)\}$, indexed by pairs (g, n) satisfying the stability condition $2g - 2 + n > 0$, equipped with two families of structure maps (Definition ??):

(i) *Separating gluing*: for each decomposition $(g, n) = (g_1, n_1 + 1) \sqcup (g_2, n_2 + 1)$ with $g = g_1 + g_2$ and $n = n_1 + n_2$,

$$\xi_{\mathrm{sep}} : \mathcal{O}(g_1, n_1 + 1) \otimes \mathcal{O}(g_2, n_2 + 1) \longrightarrow \mathcal{O}(g, n),$$

gluing two marked points (one from each factor) into a separating node.

(ii) *Non-separating gluing*: for each $(g - 1, n + 2) \to (g, n)$,

$$\xi_{\mathrm{ns}} : \mathcal{O}(g - 1, n + 2) \longrightarrow \mathcal{O}(g, n),$$

gluing two marked points on the same component into a non-separating node, raising genus by 1.

These satisfy equivariance under Σ_n, associativity (iterated gluings are independent of order), and commutativity (separating gluing is symmetric).

The canonical example is the *modular operad of stable curves*: $\overline{\mathcal{M}}(g,n) = C_*(\overline{\mathcal{M}}_{g,n})$, with ξ_{sep} and ξ_{ns} induced by the clutching maps of Knudsen and Mumford.

Feynman transform. Let $\mathcal{O}$ be a modular operad. Its *Feynman transform $F\mathcal{O}$* is the modular cooperad whose underlying graded vector space is (Definition **??**)

$$F\mathcal{O}(g,n) \;=\; \bigoplus_{\Gamma\in\mathcal{G}_{g,n}} \Bigg(\bigotimes_{v\in V(\Gamma)} \mathcal{O}\big(g(v),\mathrm{val}(v)\big)\Bigg) \otimes \det\big(E(\Gamma)\big),$$

summed over isomorphism classes of stable graphs Γ of type (g,n), modulo $\mathrm{Aut}(\Gamma)$. The *orientation line* $\det(E(\Gamma)) = \bigwedge^{\mathrm{top}} \Bbbk\langle E(\Gamma)\rangle$ is the one-dimensional vector space spanned by orderings of the edge set; it controls signs. The differential contracts edges:

$$d_{F\mathcal{O}}(\gamma) = \sum_{e\in E(\Gamma)} \pm\, \xi_e(\gamma),$$

where ξ_e applies the appropriate gluing map of $\mathcal{O}$ at edge e and contracts e from Γ, with sign determined by the position of e in the orientation line. The identity $d^2 = 0$ follows from the associativity axiom of $\mathcal{O}$.

The *commutative modular operad* Com has $\mathrm{Com}(g,n) = \Bbbk$ (with trivial Σ_n-action) for every stable (g,n), and all structure maps are the identity $\Bbbk\otimes\Bbbk \xrightarrow{\sim} \Bbbk$. Its Feynman transform $\mathrm{FCom} := F(\mathrm{Com})$ has components $\mathrm{FCom}(g,n) = \bigoplus_\Gamma \det(E(\Gamma))$, the direct sum of orientation lines over all stable graphs of type (g,n).

Cofree conilpotent coalgebra. For a chain complex V, the *cofree conilpotent coalgebra* is

$$T^c(V) = \bigoplus_{n\geq 0} V^{\otimes n}$$

with *deconcatenation coproduct* $\Delta(v_1\cdots v_n) = \sum_{i=0}^n (v_1\cdots v_i)\otimes(v_{i+1}\cdots v_n)$. The coalgebra is *conilpotent*: $\bar\Delta^{(k)}$ vanishes on $V^{\otimes n}$ for $k>n$. The bar complex of a chiral algebra $\mathcal{A}$ has underlying coalgebra $\bar B(\mathcal{A}) = \bar T^c(s^{-1}\bar{\mathcal{A}})$ (the reduced version, omitting the $n=0$ summand), where $\bar{\mathcal{A}} = \ker(\varepsilon)$ is the augmentation ideal (the kernel of the vacuum map $\varepsilon\colon \mathcal{A}\to\omega_X$) and s^{-1} is the desuspension shifting cohomological degree by -1.

1.11.3 Twisting and convolution

Convolution dg Lie algebra. Let (C,Δ,d_C) be a conilpotent dg coalgebra and (A,μ,d_A) a dg algebra. The *convolution dg Lie algebra* $\mathrm{Conv}(C,A) = \mathrm{Hom}(C,A)$ carries (Definition **??**):

(i) *Convolution product*: $f\star g := \mu\circ(f\otimes g)\circ\Delta$.

(ii) *Differential*: $\partial(f) := d_A\circ f - (-1)^{|f|} f\circ d_C$.

(iii) *Lie bracket*: $[f,g] := f\star g - (-1)^{|f||g|}\, g\star f$.

The triple $(\mathrm{Hom}(C,A),\ [-,-],\ \partial)$ is a dg Lie algebra.

Twisting morphism. A *twisting morphism* $\tau\colon C \to A$ between a conilpotent dg coalgebra C and a dg algebra A is a degree +1 linear map satisfying the Maurer–Cartan equation in the convolution algebra (Definition **??**):

$$\partial(\tau) + \tau \star \tau = 0, \qquad \text{i.e., } d_A \circ \tau + \tau \circ d_C + \mu \circ (\tau \otimes \tau) \circ \Delta = 0.$$

The set of twisting morphisms $\mathrm{Tw}(C, A) = \mathrm{MC}(\mathrm{Conv}(C, A))$ is the Maurer–Cartan locus of the convolution algebra. The set $\mathrm{Tw}(C, A)$ is in natural bijection with bar-coalgebra maps $C \to \bar{B}(A)$ and with cobar-algebra maps $\Omega(C) \to A$; these three bijections constitute bar-cobar adjunction (Theorem **??**).

Scalar saturation symbols η and Λ. In the simplest form of the universal MC element $\Theta_{\mathcal{A}}^{\leq 2} = \kappa(\mathcal{A}) \cdot \eta \otimes \Lambda$ (Theorem **??**), two geometric objects appear:

- $\eta = d\log(z_1 - z_2) \in \Omega^1_{\log}(\overline{C}_2(X))$: the universal logarithmic propagator, the same form that defines the bar differential at genus 0.
- $\Lambda = \lambda_1 \in H^2(\overline{\mathcal{M}}_{1,1}, \mathbb{C}) \cong \mathbb{C}$: the first Chern class of the Hodge line bundle on $\overline{\mathcal{M}}_{1,1}$, i.e. the generator of $H^*(\overline{\mathcal{M}}_{1,1})$. At genus g, Λ extends to the Hodge class $\lambda_g = c_g(\mathbb{E})$ of the Hodge bundle $\mathbb{E} = \pi_*\omega_\pi$ (pushforward of the relative dualizing sheaf along the universal curve $\pi\colon \mathcal{C}_g \to \overline{\mathcal{M}}_g$).

The product $\eta \otimes \Lambda$ lives in $\Omega^1_{\log}(\overline{C}_2(X)) \otimes H^*(\overline{\mathcal{M}}_{1,1})$, the degree-2, genus-1 component of the modular convolution algebra.

1.11.4 Homotopy transfer and derived categories

Strong deformation retract. A *strong deformation retract* (SDR) of a chain complex M onto its cohomology $H = H^*(M)$ is a quintuple (M, H, π, ι, h) satisfying (Definition **??**):

$$\begin{aligned} &\pi\colon M \to H, \quad \iota\colon H \to M, \qquad\qquad h\colon M \to M[-1], \\ &\pi \circ \iota = \mathrm{id}_H, \quad\ \ \iota \circ \pi - \mathrm{id}_M = dh + hd, \end{aligned}$$

with side conditions $h^2 = 0$, $h \circ \iota = 0$, $\pi \circ h = 0$. The *harmonic projection* $P := \iota \circ \pi$ projects onto the image of cohomology; the *contracting homotopy* h (the propagator) inverts d up to chain homotopy. Homological perturbation theory transfers algebraic structures (A_∞, L_∞, brackets, products) from M to H along the SDR: transferred operations are graph sums whose vertices are original operations and whose internal edges are h.

Factorization algebra. A *factorization algebra* $\mathcal{F}$ on a curve X assigns to each finite subset $S \subset X$ a chain complex $\mathcal{F}_S$ with *factorization isomorphisms* (Definition **??**):

$$\mu_{S_1,S_2}\colon \mathcal{F}_{S_1} \otimes \mathcal{F}_{S_2} \xrightarrow{\ \sim\ } \mathcal{F}_{S_1 \sqcup S_2}$$

for disjoint finite subsets $S_1, S_2 \subset X$, satisfying associativity, commutativity, and unitality. *Factorization homology* $\int_X \mathcal{A}$ is the global sections of the associated constructible cosheaf on $\mathrm{Ran}(X)$: the derived tensor product of all fibres of $\mathcal{F}$ as points collide.

Coderived category. For curved dg coalgebras ($d^2 = m_0 \neq 0$), ordinary cohomology can vanish while nontrivial coderived invariants survive. The *coderived category* $D^{\mathrm{co}}(C)$ (Positselski [?Positselski11]) replaces acyclic complexes by the smaller class of *coacyclic* complexes (totalizations of short exact sequences of curved dg comodules), preserving curved objects that ordinary cohomology destroys. The bar-cobar adjunction $\bar{B} \dashv \Omega$ descends to an equivalence on the coderived category (Definition ??). At genus $g \geq 1$, curvature $m_0 = \kappa(\mathcal{A}) \cdot \omega_g \neq 0$ forces passage to D^{co} rather than D^b.

1.12 The master object

1.12.1 The modular convolution dg Lie algebra

The genus-graded bar complex $\{\bar{B}^{(g)}(\mathcal{A})\}_{g\geq 0}$ is an algebra over the Feynman transform FCom of the commutative modular operad (Theorem ??). The *modular convolution dg Lie algebra*

$$\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}} := \prod_{g,n} \mathrm{Hom}_{\Sigma_n}\big(C_*(\overline{\mathcal{M}}_{g,n}),\ \mathrm{End}_{\mathcal{A}}(n)\big) \tag{1.12.1}$$

is the equivariant internal hom in $\mathbb{S}$-modules (§1.11.2; Definition ??), where $C_*(\overline{\mathcal{M}}_{g,n})$ denotes the singular chain complex of the Deligne–Mumford moduli space (rational coefficients) and $\mathrm{End}_{\mathcal{A}}(n) = \mathrm{Hom}(\mathcal{A}^{\otimes n}, \mathcal{A})$ the endomorphism operad (1.11.3). An element $\alpha \in \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ assigns to each stable graph Γ of type (g, n) (§1.11.1) an $\mathrm{Aut}(\Gamma)$-equivariant operation $\alpha_\Gamma\colon\ \bar{B}(\mathcal{A})^{\otimes n} \to \bar{B}(\mathcal{A})$.

Differential. The differential on $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ has five components, one per boundary stratum of $\overline{\mathcal{M}}_{g,n}$ (Theorem ??):

$$D = d_{\mathrm{int}} + [\tau, -] + d_{\mathrm{sew}} + d_{\mathrm{pf}} + \hbar\Delta.$$

(i) $d_{\mathrm{int}}(\alpha)_\Gamma = d_{\mathcal{A}} \circ \alpha_\Gamma - (-1)^{|\alpha|}\, \alpha_\Gamma \circ d_{\mathcal{A}}$: the internal differential of $\mathcal{A}$, applied vertexwise.

(ii) $[\tau, -](\alpha)_\Gamma = \sum_{e\in E(\Gamma)} \pm (\tau \circ_e \alpha)_{\Gamma/e}$: twisting by the universal propagator $\tau = \eta_{12} = d\log(z_1 - z_2) \in \Omega^1_{\log}(\overline{C}_2(X))$. Each edge e of Γ inserts τ and contracts.

(iii) $d_{\mathrm{sew}}(\alpha)_\Gamma = \sum \alpha_{\Gamma_1} \circ_i \alpha_{\Gamma_2}$, summed over separating-edge decompositions $\Gamma = \Gamma_1 \cup_e \Gamma_2$ (clutching $\xi_{\mathrm{sep}}\colon \overline{\mathcal{M}}_{g_1,n_1+1} \times \overline{\mathcal{M}}_{g_2,n_2+1} \to \overline{\mathcal{M}}_{g,n}$).

(iv) $d_{\mathrm{pf}}(\alpha)$: the planted-forest correction from Fulton–MacPherson codimension-2 boundary, encoding the failure of Arnold-type relations at higher genus.

(v) $\hbar\Delta(\alpha)_\Gamma$: the non-separating (BV) operator, arising from the clutching map $\xi_{\mathrm{ns}}\colon \overline{\mathcal{M}}_{g,n+2} \to \overline{\mathcal{M}}_{g+1,n}$ that glues two marked points on the same component and contracts with the invariant pairing $\langle -, - \rangle_{\mathcal{A}}$.

Lie bracket. The Lie bracket is the convolution bracket of §1.11.3, lifted to the modular operad:

$$[\alpha, \beta]_\Gamma = \sum_{\Gamma=\Gamma_1\cup_e\Gamma_2} \alpha_{\Gamma_1} \circ_e \beta_{\Gamma_2} - (-1)^{|\alpha||\beta|}\, \beta_{\Gamma_1} \circ_e \alpha_{\Gamma_2},$$

summed over all ways to decompose Γ into two subgraphs joined by a separating edge e; the operation $\circ_e$ composes the outputs and inputs at the half-edges of e.

Why $D^2 = 0$. The identity $D^2 = 0$ is the codimension-2 boundary cancellation $\partial^2 = 0$ on $\overline{\mathcal{M}}_{g,n}$, transported through the Hom functor (Lemma ??). At the ambient relative log-FM level, Theorem ?? is conditional on Mok's relative geometry and the signed residue-pushforward construction.

1.12.2 The universal Maurer–Cartan element

The genus-completed bar differential decomposes as $D_{\mathcal{A}} = d_{\mathcal{A}}^{(0)} + \sum_{g\geq 1} \hbar^g\, d_{\mathcal{A}}^{(g)}$. Define

$$\Theta_{\mathcal{A}} \;:=\; D_{\mathcal{A}} - d_{\mathcal{A}}^{(0)} \;=\; \sum_{g\geq 1} \hbar^g\, d_{\mathcal{A}}^{(g)} \;\in\; \mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}. \tag{1.12.2}$$

Since $D_{\mathcal{A}}^2 = 0$ and $(d_{\mathcal{A}}^{(0)})^2 = 0$ (Arnold relation at genus 0), expanding $D_{\mathcal{A}}^2 = (d_{\mathcal{A}}^{(0)} + \Theta_{\mathcal{A}})^2 = 0$ gives

$$d_{\mathcal{A}}^{(0)}\,\Theta_{\mathcal{A}} + \Theta_{\mathcal{A}}\,d_{\mathcal{A}}^{(0)} + \Theta_{\mathcal{A}}^2 = 0,$$

which is the Maurer–Cartan equation

$$D\,\Theta_{\mathcal{A}} + \tfrac{1}{2}[\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = 0$$

in $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$ (Theorem **??**). No construction is required: the positive-genus part of the bar differential *is* a Maurer–Cartan element.

1.12.3 The five theorems and the MC element

The five main theorems serve three logically distinct roles with respect to $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}})$:

Theorem 1.12.1 (Completeness of the MC element;). *Let $\mathcal{A}$ be a modular Koszul chiral algebra on a smooth projective curve X.*

(i) Existence. *The MC element $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_{\mathcal{A}}^{(0)}$ exists unconditionally: it is MC because $D_{\mathcal{A}}^2 = 0$ (Theorem* **??**). *No construction beyond the bar differential is required.*

(ii) Completeness. *On the Koszul locus, $\Theta_{\mathcal{A}}$ is a complete invariant of $\mathcal{A}$: the bar-cobar counit $\Omega_X(\bar{B}_X(\mathcal{A})) \xrightarrow{\sim} \mathcal{A}$ is a quasi-isomorphism (Theorem B, Theorem* **??**)*, so $\mathcal{A}$ is recoverable from $\bar{B}_X(\mathcal{A})$ and hence from $\Theta_{\mathcal{A}}$.*

(iii) Structure. *The structural content of $\Theta_{\mathcal{A}}$ decomposes into three proved projections: the Verdier splitting into complementary Lagrangians $Q_g(\mathcal{A}) \oplus Q_g(\mathcal{A}^!)$ (Theorem C); the scalar modular characteristic $\kappa(\mathcal{A})$ governing* $\mathrm{obs}_g = \kappa\lambda_g$ *on the uniform-weight lane and the $\hat{A}$-genus generating function (Theorem D); and the polynomial chiral Hochschild ring* $\mathrm{ChirHoch}^*(\mathcal{A})$ *(Theorem H).*

Theorem A provides the categorical framework: the bar-cobar adjunction $\bar{B}_X \dashv \Omega_X$ and Verdier intertwining $\mathbb{D}_{\mathrm{Ran}}\bar{B}(\mathcal{A}) \simeq \bar{B}(\mathcal{A}^!)$ are the ambient structure in which both $\Theta_{\mathcal{A}}$ and its projections live (Theorem **??**).

Proof. Item (i) is Theorem **??**: $D_{\mathcal{A}}^2 = 0$ implies the MC equation for $\Theta_{\mathcal{A}} = D_{\mathcal{A}} - d_{\mathcal{A}}^{(0)}$ by direct expansion. Item (ii) is Theorem **??**: the PBW spectral sequence collapse and Barr–Beck–Lurie comparison prove the counit is a quasi-isomorphism on the Koszul locus, so $\bar{B}_X(\mathcal{A})$, and therefore the MC element $\Theta_{\mathcal{A}}$ extracted from its differential, faithfully determines $\mathcal{A}$. Item (iii) is the conjunction of Theorem **??** (Theorem C, complementarity), Theorem **??** (Theorem D, modular characteristic), and Theorem **??** (Theorem H, chiral Hochschild polynomial growth), each extracting a different component of $\Theta_{\mathcal{A}}$. □

Remark 1.12.2. The three roles in Theorem 1.12.1 are logically independent. Existence (i) requires only $D_{\mathcal{A}}^2 = 0$; it uses none of the five lettered theorems. Completeness (ii) is a faithfulness statement, not a consequence of the MC equation: the MC element exists for *every* modular Koszul algebra, but completeness (no information lost in passing from $\mathcal{A}$ to $\Theta_{\mathcal{A}}$) depends on bar-cobar inversion (Theorem B), whose proof uses PBW concentration, spectral sequence collapse, and Barr–Beck–Lurie comparison. Structure (iii) consists of genuine projections: Theorems C, D, and H each extract a specific algebraic invariant from the single element $\Theta_{\mathcal{A}}$.

The following table records the role of each theorem:

Theorem	**Role**	**Mechanism**	**Content**
A (Adjunction)	Framework	Categorical	$\bar{B}_\kappa \dashv \Omega_\kappa$; $\mathbb{D}_{\mathrm{Ran}}\bar{B}(\mathcal{A}) \simeq \bar{B}(\mathcal{A}^!)$
B (Inversion)	Completeness	PBW + Barr–Beck–Lurie	$\Omega(\bar{B}(\mathcal{A})) \xrightarrow{\sim} \mathcal{A}$ on Koszul locus
C (Complementarity)	Projection	Verdier eigenspace	$Q_g(\mathcal{A}) \oplus Q_g(\mathcal{A}^!) = H^*(\overline{\mathcal{M}}_g, Z(\mathcal{A}))$
D (Modular char.)	Projection	Scalar trace at g=1	$\kappa(\mathcal{A})$; $\mathrm{obs}_g = \kappa\lambda_g$ (uniform-weight); $\hat{A}$-genus GF
H (chiral Hochschild)	Projection	Coderivation sector	$\mathrm{ChirHoch}^*(\mathcal{A})$ polynomial, Koszul-functorial

The shadow tautological classes $\tau_{g,n}(\mathcal{A}) \in R^*(\overline{\mathcal{M}}_{g,n+1})$ form a CohFT (equivariance and splitting unconditional; flat identity conditional on the vacuum lying in the generator space V), with the factorization axiom descending from the clutching law for $\Theta_{\mathcal{A}}$ (Theorem ??). The MC equation projects to tautological relations in $R^*(\overline{\mathcal{M}}_{g,n})$ at each (g, n) (Theorem ??): at genus 0 the degree-4 projection recovers WDVV; at degree 2 the genus-g projection recovers Mumford's relation $\lambda_g^2 = 0$. The complementarity propagator $P_{\mathcal{A}}\colon H^*(\mathcal{A}) \otimes H^*(\mathcal{A}^!) \to \mathbb{C}$ is the Givental R-matrix, and Teleman reconstruction recovers the full system $\{\tau_{g,n}\}$ from genus-0 data and $P_{\mathcal{A}}$ (Theorem ??).

1.12.4 The modular cyclic deformation complex and coefficient algebras

The Maurer–Cartan element $\Theta_{\mathcal{A}}$ lives in the *modular cyclic deformation complex* (Definition ??):

$$\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}(\mathcal{A}) \;:=\; \prod_{\substack{g,n \\ 2g-2+n>0}} \mathrm{CoDer}^{\mathrm{cyc}}\big(\bar{B}^{(g,n)}(\mathcal{A})\big)[1],$$

where $\mathrm{CoDer}^{\mathrm{cyc}}$ denotes *cyclic coderivations*: coderivations of $\bar{B}^{(g,n)}(\mathcal{A})$ that respect the pairing $\langle -, - \rangle_{\mathcal{A}}$ on $\mathcal{A}$ (i.e. the coderivation ξ satisfies $\langle \xi(a), b\rangle = \pm\langle a, \xi(b)\rangle$). The shift [1] is cohomological, making the Lie bracket degree 0. The genus-0 truncation recovers the ordinary cyclic deformation complex: $\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}(\mathcal{A})|_{g=0} = \mathrm{Def}_{\mathrm{cyc}}(\mathcal{A})$.

The MC element takes values in a completed tensor product with a *coefficient algebra*: $\Theta_{\mathcal{A}} \in \mathrm{MC}\big(\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}(\mathcal{A}) \mathbin{\widehat{\otimes}} \mathfrak{G}^{\mathrm{mod}}\big)$, where $\widehat{\otimes}$ is completion with respect to the genus filtration $F^{\geq g}$.

Two coefficient algebras control the combinatorics, both completed commutative dg algebras.

Stable-graph coefficient algebra. The *stable-graph coefficient algebra* (Definition ??) is the completed dg algebra

$$\mathfrak{G}^{\mathrm{mod}}_{\mathrm{st}} \;:=\; \prod_{\Gamma \in \mathsf{Gr}_{\mathrm{st}}} \mathbb{Q} \cdot e_\Gamma \Big/ \mathrm{Aut}(\Gamma),$$

with *generators*: basis vectors e_Γ, one for each isomorphism class of connected stable graph Γ; *product*: disjoint union of graphs, $e_{\Gamma_1} \cdot e_{\Gamma_2} = e_{\Gamma_1 \sqcup \Gamma_2}$; and *differential*: edge contraction, $d(e_\Gamma) = \sum_{e \in E(\Gamma)} e_{\Gamma/e}$, where Γ/e is the graph obtained by contracting edge e. This models the boundary stratification of $\overline{\mathcal{M}}_{g,n}$:

boundary strata are indexed by stable graphs, and the boundary operator on $C_*(\overline{\mathcal{M}}_{g,n})$ induces edge contraction.

Planted-forest coefficient algebra. The *planted-forest coefficient algebra* (Definition ??) is the completed dg algebra

$$\mathfrak{G}_{\mathrm{pf}}^{\mathrm{mod}} := \prod_{F \in \mathsf{For}_{\mathrm{plant}}} \mathbb{Q} \cdot e_F \Big/ \mathrm{Aut}(F),$$

with *generators*: planted forests F (forests of rooted trees where each vertex carries a non-negative weight); *product*: disjoint union of forests; and *differential*: internal edge contraction with Koszul sign, $d_{\mathrm{pf}}(e_F) = \sum_{e \in E(F)} \pm e_{F/e}$. Each planted forest indexes a boundary stratum of the Fulton–MacPherson compactification $\overline{C}_n(X)$: roots correspond to first-order collision points, interior vertices to higher-order nested collisions. By the tropicalization theorem (Theorem ??), $\mathfrak{G}_{\mathrm{pf}}^{\mathrm{mod}} = \mathrm{Trop}(\mathrm{FM}_n(\mathbb{C}|D))$: planted forests are dual graphs of tropical curves under Mok's logarithmic Fulton–MacPherson compactification [?Mok25].

Differential decomposition. The differential on $\mathrm{Def}_{\mathrm{cyc}}^{\mathrm{mod}}(\mathcal{A})$ decomposes as $D_{\mathrm{cyc}} = D_{\mathrm{edge}} + D_{\mathrm{vertex}}$, where D_{edge} contracts edges (applying the gluing maps of $\overline{\mathcal{M}}_{g,n}$) and D_{vertex} splits vertices (applying the degenerations of the modular operad). The identity $D_{\mathrm{cyc}}^2 = 0$ follows from $\partial^2 = 0$ on $\overline{\mathcal{M}}_{g,n}$.

1.12.5 The shadow algebra and the shadow obstruction tower

The cohomology

$$\mathcal{A}^{\mathrm{sh}} := H_\bullet\big(\mathrm{Def}_{\mathrm{cyc}}^{\mathrm{mod}}(\mathcal{A})\big)$$

is the *shadow algebra* (Definition ??): a bigraded Lie algebra ($\mathcal{A}^{\mathrm{sh}}_{r,g}$ at degree r and genus g) for finite-order modular invariants of $\mathcal{A}$, whose Lie bracket descends from the convolution bracket on $\mathrm{Def}_{\mathrm{cyc}}^{\mathrm{mod}}(\mathcal{A})$ and drives the Maurer–Cartan recursion. It factorizes under tensor product ($\mathcal{A}^{\mathrm{sh}} \cong \mathcal{A}_1^{\mathrm{sh}} \otimes \mathcal{A}_2^{\mathrm{sh}}$) and collapses to $\mathbb{C}$ when $\mathcal{A}$ has a mass gap.

The finite-order projections of $\Theta_{\mathcal{A}}$ form the *shadow obstruction tower*

$$\Theta_{\mathcal{A}}^{\leq 2} \longrightarrow \Theta_{\mathcal{A}}^{\leq 3} \longrightarrow \Theta_{\mathcal{A}}^{\leq 4} \longrightarrow \cdots \longrightarrow \Theta_{\mathcal{A}} = \varprojlim \Theta_{\mathcal{A}}^{\leq r}$$

(Theorem ??). Each truncation $\Theta_{\mathcal{A}}^{\leq r}$ is a graph sum through degree r, with obstruction class $o_{r+1}(\mathcal{A}) \in H^*(\mathrm{Def}_{\mathrm{cyc}}^{\mathrm{mod},(r+1)}(\mathcal{A}))$ measuring failure of extension. The base is $\Theta_{\mathcal{A}}^{\leq 2} = \kappa(\mathcal{A}) \cdot \eta \otimes \Lambda$; at degree $r \geq 3$, extension solves $D_{\mathrm{cyc}}(\Theta_{\mathcal{A}}^{(r)}) + \frac{1}{2}[\Theta_{\mathcal{A}}^{\leq r-1}, \Theta_{\mathcal{A}}^{\leq r-1}]^{(r)} = 0$, and o_{r+1} is the residual obstruction.

The degree-r projections extract named invariants:

Degree	Invariant	Status
$r = 2$	$\kappa(\mathcal{A})$ (scalar modular characteristic)	proved, all families
$r = 3$	$\mathfrak{C}(\mathcal{A})$ (cubic shadow)	proved, computed for standard landscape
$r = 4$	$\mathfrak{Q}(\mathcal{A})$ (quartic resonance class)	proved, with clutching law
$r = \infty$	$\Theta_{\mathcal{A}}$ (full universal class)	proved (bar-intrinsic, Thm ??)

The shadow obstruction tower classifies chiral algebras into four *shadow archetypes*, according to the termination degree $r_{\max}(\mathcal{A}) := \min\{r : o_{r+1}(\mathcal{A}) = 0\}$:

Class	$r_{\max}$	Archetype	Families
G	2	Gaussian	Heisenberg, free fermion, lattice (abelian)
L	3	Lie/tree	Affine Kac–Moody $\widehat{\mathfrak{g}}_k$, $V_k(\mathfrak{g})$
C	4	Contact/quartic	$\beta\gamma$, bc, symplectic bosons
M	∞	Mixed	Virasoro, $\mathcal{W}_N$, non-principal $\mathcal{W}$-algebras

Shadow depth classifies the complexity of Koszul algebras, not Koszulness itself: all four classes contain Koszul algebras. The depth is a modular-homotopy-type invariant (Corollary ??: $r_{\max}(\mathcal{A}^!) = r_{\max}(\mathcal{A})$). The class assignment depends on *generator count*, not statistics: both $\beta\gamma$ (bosonic) and bc (fermionic) are class C because each has two generators; the single-generator free fermion is class G because fermionic antisymmetry on one generator kills all higher bar cohomology.

The four-class partition has interpretations across mathematics: G/L/C/M classifies the integrable hierarchy type (trivial, Gelfand–Dickey, KdV + contact, full N-th GD), the Calogero–Moser interaction (free, tree, contact, full), the CSFT vertex structure (quadratic, cubic, quartic, infinite), and the celestial OPE truncation order (1, 2, 3, infinite soft factors).

Remark 1.12.3 (Shadow depth as functional-integral complexity). The effective worldsheet action of a chirally Koszul algebra $\mathcal{A}$ decomposes by shadow depth:

$$S_{\text{eff}}(\mathcal{A}) \;=\; \underbrace{\kappa \cdot S_{\text{Polyakov}}}_{r=2} \;+\; \underbrace{[\mathfrak{C}] \cdot S_{\text{cubic}}}_{r=3} \;+\; \underbrace{[\mathfrak{Q}^{\text{contact}}] \cdot S_{\text{quartic}}}_{r=4} \;+\; \cdots$$

Class **G** truncates at $r = 2$: the Polyakov action $S_{\text{Polyakov}} = (1/(4\pi\alpha')) \int \sqrt{g}\, g^{ab}\partial_a X\, \partial_b X$ captures the full content, and $F_g = \kappa \cdot \lambda_g^{\text{FP}}$ (uniform-weight) determines the genus expansion completely. Each subsequent class adds one order of nonlinear OPE interaction: tree-level Lie bracket for **L**, one-loop contact for **C**, all orders for **M**.

The quartic resonance class satisfies a clutching law under stable-curve degeneration (Theorem ??). For Virasoro:

$$Q^{\text{contact}}_{\text{Vir}} = \frac{10}{c(5c+22)}, \qquad \delta^{(1)}_{H,\text{Vir}} = \frac{120}{c^2(5c+22)} \cdot x^2$$

(Theorem ??).

The shadow obstruction tower, a priori an infinite sequence of obstruction classes, is algebraic of degree 2 on each *primary line* (a one-dimensional subspace of $H^2_{\text{cyc}}(\mathcal{A}, \mathcal{A})$ spanned by a single quasi-primary; Theorem ??). A single quadratic form Q_L, the *shadow metric* (Definition ??), controls the entire tower; its Gaussian decomposition

$$Q_L(t) = (2\kappa + 3\alpha\, t)^2 + 2\Delta\, t^2$$

separates the free-field contribution from the interaction correction, where $\alpha := S_3$ is the *cubic shadow coefficient* (the degree-3 projection of $\Theta_{\mathcal{A}}$ evaluated on the primary line) and $\Delta := 8\kappa S_4$ is the *critical discriminant* (S_4 being the degree-4 shadow coefficient). The signature of Q_L classifies the shadow depth: $\Delta = 0$ gives a perfect square and a terminating tower (classes **G** and **L**); $\Delta \neq 0$ gives an irreducible quadratic and an infinite but algebraic tower (class **M**). The contact class **C** ($r_{\max} = 4$) is realized by the standard conformal-weight family $\beta\gamma_\lambda$ (equivalently bc_λ), for which $S_3 = 0$, $S_4 = -5/12$, and $S_r = 0$ for $r \geq 5$ (Theorem ??). The shadow metric defines a logarithmic connection ∇^{sh} with residue $\frac{1}{2}$ at the branch points and monodromy -1 (Theorem ??). Under Koszul duality, the critical discriminants satisfy a complementarity relation with constant numerator:

$$\Delta(\mathcal{A}) \;+\; \Delta(\mathcal{A}^!) \;=\; \frac{6960}{(5c+22)(152-5c)}.$$

For multi-channel algebras (algebras with more than one primary generator), the connection acquires curvature equal to the *propagator variance*

$$\delta_{\mathrm{mix}} = \sum_i \frac{f_i^2}{\kappa_i} - \frac{(\sum_i f_i)^2}{\sum_i \kappa_i},$$

a Cauchy–Schwarz gap (Proposition ??) that vanishes if and only if the quartic gradient is proportional to the curvature vector (enhanced symmetry).

1.12.6 The Hamilton–Jacobi equation and inter-channel coupling

On a multi-generator deformation space $X = H^2_{\mathrm{cyc}}(\mathcal{A},\mathcal{A})$ of dimension $r \geq 2$, the algebraicity theorem lifts from the Riccati ODE to a *Hamilton–Jacobi PDE*. Write $U = \sum_{s\geq 3} \mathrm{Sh}_s$ for the nonlinear shadow. The MC equation on X is equivalent to

$$2\,E(U) \;+\; \tfrac{1}{2}\|\nabla U\|_H^2 \;=\; R, \tag{1.12.3}$$

where x_i are coordinates on the deformation space $X = H^2_{\mathrm{cyc}}(\mathcal{A},\mathcal{A})$ (one coordinate per primary generator of $\mathcal{A}$), $E = \sum_i x_i \partial_i$ is the Euler operator (counting degree), $\|\nabla U\|_H^2 = \sum_i P_i\,(\partial_i U)^2$ is the kinetic energy in the *inverse-Hessian metric* $H_{ij} = \mathrm{diag}(P_1,\ldots,P_r)$ (where $P_i = 1/\kappa_i$ is the reciprocal of the channel curvature), and R is a polynomial of degree ≤ 4 encoding the cubic and quartic OPE inputs (Remark ??). On a 1-D subline, $E(U) = rU$ reduces (1.12.3) to the Riccati ODE $2tU' + (P/2)(U')^2 = R(t)$. On the full multi-dimensional space, (1.12.3) is a classical Hamilton–Jacobi equation with Hamiltonian

$$\mathcal{H}(x,p) \;=\; \tfrac{1}{2}\sum_i P_i\,p_i^2 \;+\; V(x),$$

where $p_i = \partial U/\partial x_i$ are shadow momenta and V is the OPE potential. The shadow obstruction tower is the Hamilton principal function: its graph $\{(x,\nabla U(x))\}$ is a Lagrangian submanifold of $T^*(X)$. The genus expansion $\sum_{g\geq 1} F_g(\mathcal{A})\,\hbar^g$ formally resembles a WKB series with U as classical action, but the quantised Hamiltonian needed to make this precise has not been constructed (Remark ??).

The Hamilton–Jacobi structure explains the *inter-channel coupling* on sublines (Proposition ??). On a subline $L \subset X$, the restriction $U|_L$ satisfies the 1-D Riccati equation *plus* a correction from the transverse kinetic energy $(P_\perp/2)\,(\partial_\perp U)^2|_L$: the "transverse momentum" $p_\perp = (\partial_\perp U)|_L$ need not vanish. For $\mathcal{W}_3$ on the T-line ($x_W = 0$), conformal weight forces $p_W|_{x_W=0} = 0$: autonomy. On the W-line ($x_T = 0$), the cubic shadow $\mathrm{Sh}_3 = 2x_T^3 + 3x_T x_W^2$ has nonzero normal derivative $(\partial_{x_T}\mathrm{Sh}_3)|_{x_T=0} = 3x_W^2$: the transverse kinetic energy is nonzero, and the coupling correction first enters at degree 6:

$$\delta_6^{W\text{-line}} \;=\; \frac{576(5c+38)}{c^3(5c+22)^3}.$$

The autonomous W-line recursion misses this. The coupling correction vanishes on any direction where the propagator variance $\delta_{\text{mix}} = 0$.

For the shadow generating function on each primary line, the pole-purity theorem (Proposition ??) gives $S_r = \varepsilon_r P_r(c)/[c^{r-3}(5c+22)^{\lfloor (r-2)/2 \rfloor}]$ for $r \geq 4$, where $\varepsilon_r \in \{+1, -1\}$ is a sign and $P_r \in \mathbb{Z}[c]$ a polynomial of degree $\lfloor (r-4)/2 \rfloor$, with no factors beyond c and $5c+22$ in the denominator at any degree, a direct consequence of the Riccati algebraicity and the Kac-determinant structure of the quasi-primary spectrum.

1.12.7 The all-degree master equation

The degree-r projection of the MC equation is the *all-degree master equation*

$$\nabla_H(\mathrm{Sh}_r) + o^{(r)} = 0 \qquad \text{for all } r \geq 3,$$

where ∇_H is the covariant derivative along the Hodge class direction in $\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}(\mathcal{A})$ (it shifts genus by $+1$ via the non-separating clutching map ξ_{ns}), Sh_r is the shadow at degree r (the degree-r component of $\Theta_{\mathcal{A}}$ in the shadow algebra $\mathcal{A}^{\mathrm{sh}}$), and $o^{(r)}$ is the obstruction class in $H^*(\mathrm{Def}^{\mathrm{mod},(r+1)}_{\mathrm{cyc}}(\mathcal{A}))$. The *genus loop* $\Lambda_P\colon \mathrm{Sym}^r \to \mathrm{Sym}^{r-2}$ is the trace contraction operator $\Lambda_P(f)(x_1, \dots, x_{r-2}) = \sum_{i,j} P^{ij} f(x_1, \dots, x_{r-2}, e_i, e_j)$ (where P^{ij} is the inverse Hessian and $\{e_i\}$ a basis of generators): it governs descent from degree r to degree $r-2$, and its kernel consists of the primitive shadows.

The *primitive logarithmic modular kernel* $\mathfrak{K}_{\mathcal{A}}$ (Definition ??) is the restriction of the modular homological vector field $Q^{\mathrm{mod}}_{\mathcal{A}}$ to primitive corollas and rigid strata. By cofree-coderivation correspondence, $\Theta_{\mathcal{A}} = \mathrm{FT}^{\log}_{\mathrm{mod}}(\mathfrak{K}_{\mathcal{A}})$, and the fundamental equivalence

$$(Q^{\mathrm{mod}}_{\mathcal{A}})^2 = 0 \qquad \Longleftrightarrow \qquad d\,\mathfrak{K}_{\mathcal{A}} + \mathfrak{K}_{\mathcal{A}} \star \mathfrak{K}_{\mathcal{A}} + \hbar\,\Delta_{\mathrm{ns}}(\mathfrak{K}_{\mathcal{A}}) = 0$$

(Theorem ??), where $\star$ is the primitive pre-Lie product from log-FM residues and Δ_{ns} is the non-separating loop operator. The three terms replace the genus-by-genus obstruction programme by a single equation on primitive data.

1.12.8 The Feynman transform: commutative and associative

The convolution algebra (1.12.1) admits two modular operadic inputs:

(i) *Commutative.* $\mathrm{Com}(g, n) = \Bbbk$ for all stable (g, n). FCom is the modular cooperad of unlabelled stable graphs. The bar complex is an FCom-algebra (Theorem ??), and $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}})$ is the full genus tower.

(ii) *Associative.* $\mathrm{Ass}(g, n) = C_*(\overline{\mathcal{M}}^{\mathrm{rib}}_{g,n})$, the ribbon (fatgraph) modular operad (Definition ??). $F\mathrm{Ass}$ is generated by ribbon graphs, stable graphs with cyclic half-edge orderings, equivalently graphs embedded in oriented surfaces. The ordered bar complex is an $F\mathrm{Ass}$-algebra (Chapter ??), and the E_1 MC element $\Theta^{E_1}_{\mathcal{A}} \in \mathrm{MC}(\mathfrak{g}^{E_1}_{\mathcal{A}})$ is the ordered genus tower. The Σ_n-coinvariant map $F\mathrm{Ass} \to \mathrm{FCom}$ (Theorem ??) sends $\Theta^{E_1}_{\mathcal{A}} \mapsto \Theta_{\mathcal{A}}$: the commutative theory is the unordered projection of the associative theory.

The second case is the algebraic form of the 't Hooft large-N expansion. The perturbative free energy

$$\log Z \;=\; \sum_{g=0}^{\infty} N^{2-2g}\, F_g(\lambda) \;=\; \sum_{\Gamma \text{ ribbon}} \frac{N^{2-2g(\Gamma)}}{|\operatorname{Aut}(\Gamma)|}\, W_\Gamma(\lambda)$$

is a sum over ribbon graphs weighted by $\chi(\Gamma) = 2 - 2g(\Gamma)$, where $\lambda = g_{\mathrm{YM}}^2 N$. The MC equation $D\Theta^{E_1} + \frac{1}{2}[\Theta^{E_1}, \Theta^{E_1}] = 0$ encodes the entire ordered shadow package of this genus expansion: its genus-0, degree-2 projection recovers the CYBE for the spectral R-matrix, its degree-3 affine benchmark surface recovers the KZ associator, and its higher-genus degree-2 projections define the formal ordered degree-2 shadow series.

The five facets of Principle 1.9.1 have ordered counterparts (Remark 1.9.2).

1.12.9 The formality principle

The convolution dg Lie algebra $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ is a *strict model* ($\mathrm{Conv}_{\mathrm{str}}$) of an underlying L_∞-algebra $\mathrm{Conv}_\infty(\mathrm{FCom}, \mathrm{End}_{\bar{B}(\mathcal{A})})$. The difference: $\mathrm{Conv}_{\mathrm{str}}$ has a binary Lie bracket $[-,-]$ and a differential D, while Conv_∞ carries brackets ℓ_k for all $k \geq 1$ satisfying the full L_∞ relations ($\ell_1^{(0)} = D$, $\ell_2^{(0)} = [-,-]$, $\ell_k^{(g)}$ for $k \geq 3$ the transferred higher brackets). The two are related by a quasi-isomorphism of L_∞-algebras (Chapter ??), and their MC moduli coincide.

Theorem 1.12.4 (Rectification bridge)**.** *The three rectification principles are:*

(i) Operadic*: the homotopy chiral algebra $\mathcal{A}^{Ch_\infty}$ rectifies to a strict chiral algebra $\mathcal{A}$ via bar-cobar (Theorem* ??*).*

(ii) Convolution*: the L_∞-algebra* $\mathrm{Conv}_\infty(\mathrm{FCom}, \mathrm{End}_{\bar{B}(\mathcal{A})})$ *and the dg Lie algebra* $\mathrm{Conv}_{\mathrm{str}}(\bar{B}(\mathcal{A}), \mathcal{A})$ *carry quasi-isomorphic MC moduli.*

(iii) Reduction*: on the proved generic DS lanes, the derived DS functor* $\mathbf{R}\mathrm{DS}_f$ *and the classical functor $H^0_{Q_{\mathrm{DS}}}$ agree. This is established by Kazhdan filtration formality on the abelian-$\mathfrak{n}_+$ locus (Proposition* ??*) and, in particular, in the principal case (Theorem* ??*).*

Rectification, the passage from homotopy-coherent to strict structures, underlies all five main theorems:

(a) Bar-cobar preserves quasi-isomorphisms because it is a quantum L_∞ functor: it respects the full homotopy package, not merely the strict dg structure. This gives Theorem B.

(b) The transferred A_∞ and L_∞ structures on bar cohomology (homotopy transfer theorem, Chapter ??) yield the shadow obstructions: at genus 0, $o_{r+1}|_{g=0} = [m_{r+1}]$ (Proposition 1.14.4). Formality ($m_n = 0$ for $n \geq 3$) is the Gaussian archetype.

(c) On the proved generic DS lanes, Drinfeld–Sokolov reduction commutes with bar through the derived/classical comparison above: the principal case is unconditional, and additional non-principal cases require the verified DS–KD comparison hypotheses. In those regimes the universal MC class of the full current-plus-ghost system descends to that of the $\mathcal{W}$-algebra (Theorem ??, Corollary ??); the bare affine tower alone does not capture the ghost-created quartic and higher shadows. The geometric content is the Slodowy-slice localization principle (Remark ??): DS reduction is Hamiltonian reduction from $J_\infty(\mathfrak{g}^*)$ to $J_\infty(S_f)$ (the arc space of the Slodowy transverse slice), and the master commutative square (Theorem ??) encodes the compatibility of bar complexes with this geometric localization.

(d) The dg Lie algebra is a faithful chart on the formal moduli problem $\mathcal{M}_{\mathcal{A}}^{\mathrm{mod}}$; every strict-model formula is a projection of $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathrm{Conv}_\infty)$. The space of models is contractible (Proposition **??**).

1.12.10 The modular Koszul datum and the modular envelope adjunction

The canonical input is a *cyclically admissible Lie conformal algebra* L (Definition **??**): a Lie conformal algebra (a $\mathbb{C}[\partial]$-module with a λ-bracket $\{a_\lambda b\} = \sum_{n\geq 0} \lambda^n c_n(a,b)/n!$ satisfying sesquilinearity, skew-symmetry, and the Jacobi identity) equipped with conformal grading, exhaustive filtration, bounded OPE poles, and a non-degenerate invariant pairing.

From L, the *modular Koszul datum* $\Pi_X(L)$ (Construction **??**) assembles six pieces:

$$\Pi_X(L) = \big(\mathrm{Fact}_X(L),\ \bar{B}_X(L),\ \Theta_L,\ \mathcal{L}_L,\ (V_L^{\mathrm{br}}, T_L^{\mathrm{br}}),\ \mathfrak{R}_4^{\mathrm{mod}}(L)\big),$$

where: $\mathrm{Fact}_X(L)$ is the factorization algebra (the enveloping chiral algebra of L); $\bar{B}_X(L)$ is its bar coalgebra; Θ_L is the universal MC class; $\mathcal{L}_L$ is the Lagrangian (the complementarity polarization from Theorem C); $(V_L^{\mathrm{br}}, T_L^{\mathrm{br}})$ is the *branch-line data* (V^{br} is the spectral branch vector space from the Koszul dual Hilbert series, T^{br} its transfer matrix); and $\mathfrak{R}_4^{\mathrm{mod}}(L)$ is the quartic resonance class.

The modular factorization envelope U_X^{mod} (Theorem **??**) is the left adjoint of the modular primitive-current functor $\mathrm{Prim}^{\mathrm{mod}}$:

$$U_X^{\mathrm{mod}} \dashv \mathrm{Prim}^{\mathrm{mod}} \colon \mathrm{Alg}_{\mathrm{aug}}(\mathrm{Fact}^{\mathrm{mod}}(X)) \longrightarrow \mathrm{LCA}^{\mathrm{cyc}},$$

where $\mathrm{Fact}^{\mathrm{mod}}(X)$ is the category of modular factorization algebras (factorization algebras with genus data) and $\mathrm{LCA}^{\mathrm{cyc}}$ is the category of cyclically admissible Lie conformal algebras. For independent sums $L = L_1 \oplus L_2$ with vanishing mixed OPE, all shadows separate: κ additive, T^{br} direct sum, Δ multiplicative, $\mathfrak{R}_4^{\mathrm{mod}}$ additive (Proposition **??**). Cubic gauge triviality (Theorem **??**): if $H^1(F^3\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}/F^4\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}, d_2) = 0$, the cubic MC term is gauge-trivial and the quartic class $[\Theta_4'] \in H^2(F^4\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}/F^5\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}, d_2)$ is canonical.

1.13 The convolution architecture

Every algebraic structure in this monograph is a Maurer–Cartan element in a convolution dg Lie algebra. The bar complex plays the role of the Steinberg variety; $\Theta_{\mathcal{A}}$ plays the role of the fundamental class; its projections recover all convolution products.

1.13.1 The convolution ladder

At three levels of generality, the bar-type construction is a convolution dg Lie algebra:

$$\begin{array}{rll} \text{algebra}: & \mathrm{Hom}\big(B(\mathsf{Ass}),\ \mathrm{End}_A\big) & A_\infty\text{-structures on } A \\ \text{operad}: & \mathrm{Hom}_\Sigma\big(B(\mathcal{P}),\ \mathrm{End}_V\big) & \mathcal{P}_\infty\text{-structures on } V \\ \text{modular}: & \mathrm{Hom}_\Sigma\big(F\mathcal{O},\ \mathrm{End}_{\bar{B}(\mathcal{A})}\big) & \text{consistent genus towers} \end{array}$$

The first line is the topological Hochschild cochain complex; the second is the deformation complex (Ginzburg–Kapranov [?GK94]); the third, with $\mathcal{O} = \mathsf{Com}$, governs this book.

1.13.2 The modular operad and Feynman transform

A *modular operad* $\mathcal{O}$ (Getzler–Kapranov [?GeK98]) has operations indexed by connected stable graphs Γ of arbitrary genus, with structure maps from separating and non-separating gluing of boundary divisors of $\overline{\mathcal{M}}_{g,n}$. Trees are the genus-0 case.

The *Feynman transform* replaces trees by stable graphs:

$$F\mathcal{O}(g,n) = \bigoplus_{\Gamma \in \mathsf{Gr}^{\mathrm{st}}_{g,n}} \frac{\hbar^{g(\Gamma)}}{|\operatorname{Aut}(\Gamma)|} \bigotimes_{v \in V(\Gamma)} \mathcal{O}\big(g(v), \operatorname{In}(v)\big) \otimes \det\big(E(\Gamma)\big).$$

The differential contracts one edge; $d^2 = 0$ is the codimension-2 boundary cancellation on the moduli space. At genus 0: the operadic bar construction.

When $\mathcal{O} = \mathsf{Com}$ (the commutative modular operad), each vertex factor is one-dimensional, and the Feynman transform becomes the stable graph complex $F\mathsf{Com}$, a chain complex spanned by stable graphs with edge-contraction differential. The bar complex $\bar{B}_X(\mathcal{A})$ is an $F\mathsf{Com}$-algebra, and the modular convolution dg Lie algebra

$$\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}} := \operatorname{Hom}_{\Sigma}\big(F\mathsf{Com},\ \operatorname{End}_{\bar{B}(\mathcal{A})}\big)$$

is the ambient dg Lie algebra of $\Theta_{\mathcal{A}}$.

The *modular bar construction*:

$$\bar{B}^{(g)}_X(\mathcal{A}) = \bigoplus_{\Gamma \in \mathsf{Gr}^{\mathrm{st}}_{g,n}} \frac{1}{|\operatorname{Aut}(\Gamma)|} W_\Gamma \otimes \bigotimes_{v \in V(\Gamma)} \Phi^{\mathcal{A}}_v .$$

Replacing Com by Ass equips each vertex with a cyclic ordering, producing the ribbon graph complex $F\mathsf{Ass}$. This is the algebraic form of the 't Hooft $1/N$ expansion.

1.13.3 Three formative preprints

Three preprints require foundational upgrades:

(A) *Malikov–Schechtman* [?MS24]: the primitive local object is a *homotopy chiral algebra* (Ch_∞), not strict. The strict chiral algebra appears after rectification.

(B) *Robert-Nicoud–Wierstra* [?RNW19] + *Vallette* [?Val16]: the universal deformation machine is the filtered convolution $s\mathsf{L}_\infty$-algebra $\hom_\alpha(C, \mathcal{A})$, functorial in either slot separately but *not both simultaneously* (the no-bifunctor obstruction, [?RNW19, Theorem 6.6]).

(C) *Mok* [?Mok25]: the global collision geometry is the log FM compactification $\overline{C}_n(X|D)$ on snc pairs, with tropicalisation $\mathrm{Trop}(\overline{C}_n(C|D))$ recovering planted forests as combinatorial shadows.

Nine of the eleven identification theorems connecting these pillars to the bar construction are proved; the remaining two are structural.

1.13.4 Completion

The strong completion-tower theorem (Theorem ??): the strong filtration axiom $\mu_r(F^{i_1}, \dots, F^{i_r}) \subset F^{i_1+\cdots+i_r}$ forces a degree cutoff (Lemma ??) making continuity and Mittag–Leffler automatic, so finite-stage bar-cobar passes to inverse limits. The completion closure $\mathrm{CompCl}(\mathcal{F}_{\mathrm{ft}})$ carries a quasi-inverse bar-cobar homotopy equivalence (Corollary ??), stable under MC twisting (Theorem ??).

The MC4 splitting: MC4$^+$ (positive towers, including $\mathcal{W}_{1+\infty}$, affine Yangians, and RTT, solved by weight stabilisation) vs. MC4^0 (resonant towers, including Virasoro and non-quadratic $\mathcal{W}_N$, reduced to finite resonance by Theorem ??). The resonance rank $\rho(\mathcal{A})$ classifies completion difficulty. The resonance completion theorem (Theorem ??): the positive-energy axiom gives finite-dimensional weight spaces, so weight-compatible SDR and finiteness of the weight-zero resonance space force $\rho(\mathcal{A}) < \infty$.

1.14 The modular Koszul object

A *modular Koszul chiral algebra* (Definition ??) is a factorization algebra on $\mathrm{Ran}(X)$ whose duality data extends over the modular operad, with curvature classes compatible under clutching, Verdier duality, and trace. Theorems A–D are projections of $\Theta_{\mathcal{A}}$; the modular Koszul structure is the ambient datum from which they are extracted. The full package

$$C_{\mathcal{A}} = (\Theta_{\mathcal{A}}, \kappa(\mathcal{A}), \Delta_{\mathcal{A}}, \Pi_{\mathcal{A}}, \mathcal{H}_{\mathcal{A}})$$

is a hierarchy of increasing refinement (Definition ??). Theorem ?? packages this hierarchy into explicit H-, M-, and S-level data.

The genus tower $\{\bar{B}^{(g)}(\mathcal{A})\}_{g\geq 0}$ is a Maurer–Cartan deformation of the genus-0 bar differential inside a cyclic L_∞ algebra (Definition ??). The proved families include free fields (Proposition ??), affine Kac–Moody at all genera (Theorem ??, Corollary ??), Virasoro (Theorem ??, Corollary ??), and principal $\mathcal{W}$-algebras (Theorem ??, Corollary ??).

Scalar saturation, the identity $\Theta^{\min}_{\mathcal{A}} = \kappa(\mathcal{A}) \cdot \eta \otimes \Lambda$ (Corollary ??), holds whenever $\dim H^2_{\mathrm{cyc}}(\mathcal{A}, \mathcal{A}) = 1$. Theorem ?? extends it to all non-critical levels for algebraic families with rational OPE coefficients; Conjecture ?? addresses the residual non-algebraic-family case.

Definition 1.14.1 (Modular homotopy theory on a smooth curve)**.** A *modular homotopy theory* for a chiral algebra $\mathcal{A}$ on a smooth projective curve X consists of the following data:

(i) *Bar-cobar adjunction.* An adjunction $\bar{B}_X \dashv \Omega_X$ on $D^{\mathrm{co}}(\mathrm{Fact}(X))$, the coderived category of factorization coalgebras on $\mathrm{Ran}(X)$, with $\bar{B}_X(\mathcal{A})$ nilpotent at genus 0 and curved at higher genus.

(ii) *Modular functoriality.* Functoriality of the bar family $\{\bar{B}^{(g)}(\mathcal{A})\}_{g\geq 0}$ over the modular operad $\{\overline{\mathcal{M}}_{g,n}\}$, intertwined with Verdier duality: $\mathbb{D}_{\mathrm{Ran}}\bar{B}^{(g)}_X(\mathcal{A}) \simeq (\mathcal{A}^!)^{(g)}_\infty$ (factorization *algebra*, not coalgebra).

(iii) *Universal Maurer–Cartan class.* A class $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}(\mathcal{A}) \mathbin{\widehat{\otimes}} R\Gamma(\overline{\mathcal{M}}_{g,\bullet}, \mathbb{Q}))$ assembling the genus tower into a single integrability datum, compatible with clutching, trace, and duality.

(iv) *Complementary Lagrangians.* A $(-(3g-3))$-shifted symplectic form on the moduli of genus-g bar families, with $Q_g(\mathcal{A})$ and $Q_g(\mathcal{A}^!)$ as complementary Lagrangians.

(v) *Chain-level modular functor.* The Feynman transform assembles the genus-graded bar complex $\bar{B}^{\mathrm{full}}(\mathcal{A}) = \bigoplus_g \bar{B}^{(g)}(\mathcal{A})$ into a homotopy-coherent modular functor (Theorem ??): to each surface $\Sigma_{g,n}$ a cochain complex, to each degeneration a chain map, to each consistency relation a chain homotopy. Passing to H^0 at integrable level recovers conformal blocks. The scalar genus tower $F_g = \kappa(\mathcal{A}) \cdot \lambda_g^{\mathrm{FP}}$ (uniform-weight) (Theorem D) is the first Chern class of the determinant line bundle of this functor, the trace of the construction, not the construction itself.

Theorems A–D are the first four assertions; item (v) is the structure they build toward, proved in Theorem ??. The formal axiomatic treatment is Definition ??; the concrete dg model is Definition ??.

Proposition 1.14.2 (Classification by modular homotopy type;). *Two chiral algebras $\mathcal{A}$ and $\mathcal{A}'$ on X have equivalent modular homotopy types, i.e., the formal moduli problems $\mathcal{M}^{\mathrm{mod}}_{\mathcal{A}} \simeq \mathcal{M}^{\mathrm{mod}}_{\mathcal{A}'}$, if and only if their modular bar constructions are connected by a zigzag of quasi-isomorphisms of modular-operad coalgebras. The modular homotopy type is the finest invariant visible to the bar construction, coarser than isomorphism: it remembers the deformation-obstruction pattern and forgets multiplication tables.*

Proof. By the Lurie–Pridham correspondence between L_∞-algebras and formal moduli problems (cf. [?Pridham17, Theorem 2.0.2]), an equivalence $\mathcal{M}^{\mathrm{mod}}_{\mathcal{A}} \simeq \mathcal{M}^{\mathrm{mod}}_{\mathcal{A}'}$ is equivalent to a quasi-isomorphism $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}[\mathcal{A}] \xrightarrow{\sim} \mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}[\mathcal{A}']$ of L_∞-algebras. By the convolution identification $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}} = \mathrm{Conv}_\infty(C^{\log \mathrm{FM}}_{\mathrm{mod}}, \mathrm{End}_{\mathrm{Ch}_\infty}(\mathcal{A}))$, a quasi-isomorphism of convolution algebras is induced by a quasi-isomorphism of the modular bar constructions $\bar{B}^{\mathrm{mod}}_X(\mathcal{A}) \xrightarrow{\sim} \bar{B}^{\mathrm{mod}}_X(\mathcal{A}')$ as modular-operad coalgebras (since the cooperad slot is common). The converse holds by the representability of twisting morphisms (Theorem ?? in the completed regime). □

Example 1.14.3 (Modular homotopy type distinguishes families). The Heisenberg algebra $\mathcal{H}_k$ and the affine algebra $\widehat{\mathfrak{sl}}_{2,k}$ have the same underlying graded vector space $\bigoplus_{n\geq 1} \Bbbk \cdot \partial^{n-1}\alpha$ (respectively $\bigoplus_{n\geq 1} \Bbbk^3 \cdot \partial^{n-1}J^a$) in the same conformal weight range. At genus 0, the Heisenberg bar complex has $d_B = 0$ (the bracket is abelian); the $\widehat{\mathfrak{sl}}_2$ bar complex has $d_B(sJ^a|sJ^b) = f^{ab}{}_c\, sJ^c$ (the Lie bracket). The A_∞-minimal model of the Heisenberg has $m_n = 0$ for all $n \geq 2$; the minimal model of $\widehat{\mathfrak{sl}}_2$ has $m_2 =$ the Lie bracket and $m_3 =$ the Jacobiator homotopy F_3 on $\overline{\mathcal{M}}_{0,4} \cong \mathbb{P}^1$.

The modular homotopy types are therefore inequivalent: $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}[\mathcal{H}_k]$ has $\ell_n = 0$ for $n \geq 3$ (Gaussian archetype, shadow depth 2), while $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}[\widehat{\mathfrak{sl}}_{2,k}]$ has $\ell_3 \neq 0$ (Lie/tree archetype, shadow depth 3). The cubic shadow $\mathfrak{C}(\widehat{\mathfrak{sl}}_{2,k})$, the degree-3 component of $\Theta_{\mathcal{A}}$, is the graph sum over trivalent trees, encoding the Lie bracket's contribution to the modular bar construction. No quasi-isomorphism of modular bar coalgebras can kill it, because it represents a nontrivial class in $H^*(\mathrm{Def}^{(3)}_{\mathrm{cyc}}(\mathcal{A})|_{g=0})$ (Proposition 1.14.4).

Shadow depth is therefore a modular-homotopy-type invariant. More precisely: the L_∞-minimal model of $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ has $\ell^{\min}_r = 0$ for $r <$ shadow depth and $\ell^{\min}_r \neq 0$ at $r =$ shadow depth; this cannot change under quasi-isomorphism.

Proposition 1.14.4 (Genus-0 shadow obstructions = A_∞ Massey products;). *Let $\mathcal{A}$ be a chiral algebra on X with bar complex $\bar{B}(\mathcal{A})$, and let $(H^*(\bar{B}(\mathcal{A})), m_2, m_3, m_4, \ldots)$ be the A_∞-minimal model obtained by homotopy transfer (Proposition ??). Then the genus-0 shadow obstructions and the A_∞-Massey products coincide:*

$$o_{r+1}(\mathcal{A})\big|_{g=0} = [m_{r+1}] \in H^*\big(\mathrm{Def}^{(r+1)}_{\mathrm{cyc}}(\mathcal{A})\big|_{g=0}\big) \qquad \textit{for all } r \geq 2. \tag{1.14.1}$$

In particular:

(i) *$m_n = 0$ for all $n \geq 3$ if and only if the genus-0 shadow obstruction tower terminates at degree 2, the Gaussian archetype.*

(ii) *The shadow depth of $\mathcal{A}$ at genus 0 equals the A_∞-depth of the bar cohomology: the smallest r such that $m_r \neq 0$ equals the smallest degree at which the shadow tower has a nontrivial obstruction.*

Proof. The A_∞-products m_n on $H^*(\bar{B}(\mathcal{A}))$ are the Taylor coefficients of the transferred twisting morphism $\tau^{\mathrm{tr}} \colon H^*(\bar{B}(\mathcal{A})) \to \mathcal{A}$ through the deformation retract $(\bar{B}(\mathcal{A}), H^*(\bar{B}(\mathcal{A})), i, p, h)$. The genus-0 shadow obstructions $o_{r+1}|_{g=0}$ are the obstructions to extending the truncated MC element $\Theta_{\mathcal{A}}^{\leq r}|_{g=0}$ by one degree in the genus-0 sector of $\mathrm{Def}_{\mathrm{cyc}}(\mathcal{A})$. But the genus-0 sector of the modular convolution algebra is the *operadic* convolution algebra $\mathrm{Hom}(\bar{B}_{\mathrm{op}}(\mathrm{Com}), \mathrm{End}_{\mathcal{A}})$, and the MC equation there is exactly the A_∞-structure equation. The identification (1.14.1) follows from the standard comparison between the operadic bar-cobar obstruction tower and the A_∞-Massey product tower (cf. [?LPWZ09, Theorem 3.4.1]). □

Remark 1.14.5 (The Sullivan dictionary). The identification of Proposition 1.14.4 is the genus-0 case of a structural parallel between Sullivan's rational homotopy theory and the modular homotopy theory of Definition 1.14.1:

Rational (Sullivan)	**Modular (this monograph)**
topological space X	chiral algebra $\mathcal{A}$ on X
de Rham algebra $\Omega^*(X)$	modular convolution $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$
Sullivan minimal model	L_∞-minimal model of $\mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}}$
Massey products m_n	shadow obstructions o_{r+1} (Prop. 1.14.4)
formality: $m_n = 0$ ($n \geq 3$)	Gaussian: shadow depth 2
Kähler $\Rightarrow$ formal	Heisenberg $\Rightarrow$ Gaussian
non-Kähler, non-formal	Virasoro: infinite shadow depth
Massey cascade from self-map	self-referential OPE: $a \in a_{(k)}a$

The modular upgrade replaces genus-0 graphs (trees) by all-genera graphs, chiral Hochschild cochains by modular-cyclic cochains, and A_∞-obstructions by modular L_∞-obstructions. The formal structure (controlling L_∞-algebra, Maurer–Cartan moduli, obstruction tower, formality criterion) is unchanged.

The dictionary extends to arithmetic content. For lattice vertex algebras, the shadow obstruction tower detects the Hecke eigenform decomposition of the theta function: each degree resolves one Hecke eigenspace, and the depth counts them (Theorem ??). The depth decomposition $d = 1 + d_{\mathrm{arith}} + d_{\mathrm{alg}}$ (Theorem ??) separates the *arithmetic* invariant d_{arith} (Hecke eigenform count, visible through L-functions and critical lines) from the *homotopy-theoretic* invariant d_{alg} (formal non-formality depth, visible through Massey products and A_∞ structure). For lattice algebras: $d_{\mathrm{alg}} = 0$ (the shadow obstruction tower is purely arithmetic). For Virasoro minimal models: d_{arith} is finite but $d_{\mathrm{alg}} = \infty$ (the shadow obstruction tower detects the *algebra itself*, the self-referential OPE $T \in T_{(1)}T$, not its representation theory). The shadow obstruction tower is the universal invariant that encodes both the Langlands programme (arithmetic) and the Sullivan programme (homotopy theory) in a single Maurer–Cartan element $\Theta_{\mathcal{A}}$.

The arithmetic content extends beyond the shadow obstruction tower. The connected sewing amplitudes $a_{\mathcal{A}}(N)$ define a *sewing inner product* on Dirichlet polynomials (§??), whose reproducing kernel is the *sewing kernel* $K_{\mathcal{A}}(s,t) = S_{\mathcal{A}}(s+t) = \zeta(s+t+1)\sum_i(\zeta(s+t) - H_{w_i-1}(s+t))$. For Heisenberg, $K_{\mathcal{H}} = \zeta \cdot \zeta$: a pure Euler product. For Virasoro, $K_{\mathrm{Vir}} = \zeta(u+1)(\zeta(u)-1)$: the weight-1 mode is removed. The surface moment matrix $\mathscr{M}_{ij} = K_{\mathcal{A}}(\alpha/2+i,\ \alpha/2+j)$ is a Gram matrix V^TDV with positive weights, hence $\mathscr{M} \succ 0$ (Theorem ??). The shadow obstruction tower provides interacting corrections to this kernel. When all spectral atoms λ_j of the shadow spectral measure are negative (which holds for all unitary theories with $c > 0$), the interacting Gram matrix is unconditionally positive definite (Theorem ??(i)); the shadow resonance locus is confined to non-unitary $c < 0$ (Theorem ??).

The Euler–Koszul class $\mathrm{ek}(\mathcal{A}) = \max_i(w_i - 1)$ and shadow depth $\kappa_d(\mathcal{A})$ are independent invariants (Theorem **??**): the first reads the character, the second the OPE.

The *Sewing–Hecke reciprocity* theorem (Theorem **??**) shows that the two-variable L-object $L_{\mathcal{A}}(s,u) := \int_{\mathcal{F}}^{\mathrm{reg}} F_{\mathcal{A}}^{\mathrm{conn}} E^*(\tau,s)\, y^{u-2}\, dx\, dy$ *collapses*: $L_{\mathcal{A}}(s,u) = \Gamma(v)(2\pi)^{-v} S_{\mathcal{A}}(v)$ with $v = s + u - 1$. The zero-side Rankin–Selberg variable s and the prime-side Dirichlet variable u are locked together by the Mellin transform of the sewing free energy. The Sewing–Selberg formula ($u = 0$) and the Dirichlet–sewing lift ($s = 1$) are two faces of the same Mellin transform. The quartic resonance class $\mathfrak{Q}(\mathcal{A})$ is the Schur complement of the 3×3 Hankel determinant of the mixed (s,u)-transform, stripping quadratic and cubic contamination: for Virasoro, $\Sigma_2 = 10/[c(5c+22)] = Q^{\mathrm{ct}}_{\mathrm{Vir}}$ exactly (Chapter **??**).

Construction 1.14.6 (Arithmetic shadow chain)**.** The shadow obstruction tower connects modular Koszul duality to arithmetic L-functions via a five-step chain:

(i) *Sewing → partition function*: the HS-sewing criterion (Theorem **??**) gives convergent genus-g amplitudes $Z_g(\mathcal{A}, q)$ for Koszul algebras.

(ii) *Partition function → eta product*: for lattice VOAs, $Z_1(V_\Lambda, q) = 1/\eta(\tau)^{\mathrm{rank}(\Lambda)}$, encoding the modular characteristic $\kappa = \mathrm{rank}(\Lambda)$.

(iii) *Eta product → zeta data*: the Euler product structure of η decomposes Z_1 into prime-indexed factors; the sewing Dirichlet series $S_{\mathcal{A}}(u) = \sum_{n\geq 1} a_n n^{-u}$ extracts the arithmetic content.

(iv) *Zeta data → bar complex*: the Euler–Koszul class $\mathrm{ek}(\mathcal{A}) \in K_0(\mathrm{EK})$ classifies the relationship between sewing data and bar cohomology generating functions.

(v) *Bar complex → L-function*: the shadow generating function $G_{\mathcal{A}}(t) = \sum_n \dim H^n(B(\mathcal{A}))\, t^n$ and the sewing Dirichlet series $S_{\mathcal{A}}(u)$ are related by a Mellin-type integral transform (Approach A, proved for lattice VOAs).

The Euler–Koszul three-tier classification (Definition **??**) partitions chiral algebras into: exact ($\mathrm{ek} = 0$, lattice VOAs), finitely defective (ek finite, affine), and non-EK (ek infinite, Virasoro). For lattice VOAs, this chain is complete (Stages 1–3 proved). For non-lattice algebras, Stage 3 (Hecke decomposition) encounters the non-multiplicativity obstruction: the sewing Dirichlet series is not a Hecke eigenform in general. The shadow-L correspondence $d(\mathcal{A}) = 1 + \#L$-function factors holds for all tested lattice VOAs but fails for $\beta\gamma$ (shadow depth 4, but 1 $U(1)$ critical line).

1.14.1 The arithmetic packet connection and the Miura defect

The depth decomposition $d = 1 + d_{\mathrm{arith}} + d_{\mathrm{alg}}$ acquires a geometric incarnation in the *arithmetic packet connection* $\nabla^{\mathrm{arith}}_{\mathcal{A}} = d - \bigoplus_\chi d \log \Lambda_\chi(s) \cdot (\mathrm{id} + N_\chi)\, ds$ (Definition **??**), a flat meromorphic connection on the Hecke module $M_{\mathcal{A}} = \bigoplus_\chi M_\chi$. Its semisimple part (the *arithmetic skeleton*) controls the singular divisor; its nilpotent part (the *algebraic defect module*) contributes only unipotent monodromy. The *frontier defect form* $\Omega_{\mathcal{A}} = d \log \Lambda_{\mathrm{Eis}} - d \log \varphi$ measures the discrepancy with automorphic scattering (Definition **??**).

For principal $\mathcal{W}_N$, the *finite Miura defect theorem* (Theorem **??**) decomposes the vacuum character as $\chi_{\mathcal{W}_N}(q) = \chi_{\mathcal{H}}(q)^{N-1} \prod_{m=1}^{N-1}(1 - q^m)^{N-m}$: all new genus-1 arithmetic content is carried by a finite polynomial. The Heisenberg core is *arithmetically inert*: its packet connection is diagonal with universal zeta divisors, and its frontier defect form cancels (Proposition **??**). Three closure conjectures form a strict hierarchy: quartic closure, Beilinson functional, clutching closure, the last using the full nonlinear modular boundary geometry (Conjectures **??**–**??**).

1.14.2 The holographic frontier: relative deformation and ribbon structure

The holographic datum $\mathcal{H}(T) = (\mathcal{A}, \mathcal{A}^!, C, r(z), \Theta_{\mathcal{A}}, \nabla^{\mathrm{hol}})$ is a *boundary* object. Lifting it to genuine 3d data is controlled by the *relative holographic deformation complex* $\mathfrak{h}_T^{\Theta} = \mathrm{fib}(\mathfrak{g}_T^{\mathrm{SC}} \to \mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}})_{\alpha_T}$ (Definition ??): its MC elements are exactly lifts of the fixed boundary datum, and the graded obstruction tower lives in $H^2(\mathrm{gr}_r^F\, \mathfrak{h}_T^{\Theta})$ (Proposition ??). The *relative quartic obstruction* $\mathfrak{R}_4^{\mathrm{rel}}(T)$ is the first invariant detecting failure of planarity. The conjectural *Loop–Connes principle* $\beta^*_{\mathrm{der}}(\Delta_{\mathrm{open}}) = \Lambda_P$ (Conjecture ??): loops in the bulk are traces on the boundary.

For literal 't Hooft counting, bare modular graph sums are insufficient (Lemma ??): cyclic trace structure enables ribbon thickening (Theorem ??), and matrix realisation gives the exact weight $N^{\chi(\Sigma_\Gamma)}$ (Theorem ??). The three-level language: plain modular E_1 = genus expansion; cyclic traced modular E_1 = 't Hooft expansion; ribbonised modular Swiss-cheese = holographic open/closed 't Hooft language (Chapter ??).

1.14.3 Completion kinematics versus modular dynamics

The package $C_{\mathcal{A}}$ encodes the *modular dynamics* of $\mathcal{A}$: the OPE-dependent deformation data $D_{\mathcal{A}}$, $\kappa(\mathcal{A})$, $\Theta_{\mathcal{A}}$, and the dg-shifted Yangian kernel $r_{\mathcal{A}}(z)$. But the completed bar-cobar theory also carries a second, independent axis that we call *completion kinematics*: invariants determined solely by the vacuum character $\chi_{\mathcal{A}}(q)$, without reference to OPE coefficients or curvature.

The kinematic data are extracted from the *primitive cumulant quotient* $Q(\mathcal{A}) = \bar{\mathcal{A}}/\partial\bar{\mathcal{A}}$ (Definition ??) and its generating series

$$G_{\mathcal{A}}(t) = \frac{1-t}{t}\big(\chi_{\mathcal{A}}(t) - 1\big),$$

which is canonical: the non-derivative sector is recovered directly from the character. The *completion Hilbert series* $H_{\mathcal{A}}(t) = G_{\mathcal{A}}/(1 - G_{\mathcal{A}})$ counts reduced-weight bar words, and the *completion entropy* $h_K(\mathcal{A}) = \log(\rho_{\mathcal{A}}^{-1})$ (where $G_{\mathcal{A}}(\rho_{\mathcal{A}}) = 1$) is the exponential growth rate of finite bar windows (Definitions ??–??).

The two axes are independent. The Virasoro involution $c \mapsto 26 - c$ changes the dynamic shadow $\mathrm{Vir}_c \leftrightarrow \mathrm{Vir}_{26-c}$ while preserving the kinematic data: both sides have the same $G_{\mathcal{A}}(t)$ and the same entropy. The shadow depth classification (G/L/C/M) measures *dynamic* complexity; the completion entropy h_K measures *kinematic* complexity. Both are needed.

The *primitive defect series* $\Delta_{\mathcal{A}}(t) = G_{\mathcal{A}}(t) - G_{\mathcal{A}}^{\mathrm{naive}}(t)$ (Definition ??) detects nonquadraticity: it is zero for Heisenberg, affine Kac–Moody, and $\beta\gamma$ (all quadratic or free-like), and nonzero starting at t^3 for Virasoro and $\mathcal{W}_N$ ($N \geq 3$). The nonzero defect means the completed dual coalgebra is forced to enlarge its primitive sector beyond the declared generators; the correct dual is the *cumulant coalgebra* $\mathrm{Cum}_c(\mathcal{A}) = \hat{T}^c(sQ(\mathcal{A})^\vee)$, not the tensor coalgebra on the original generators alone (Conjecture ??).

1.15 The standard families and their shadow archetypes

Each standard family determines a *shadow archetype*: termination pattern and qualitative shape of modular invariants. Completion kinematics (§1.14.3) assigns a primitive spectrum $G_{\mathcal{A}}(t)$, entropy $h_K(\mathcal{A})$, and defect $\Delta_{\mathcal{A}}(t)$. The dynamic (shadow depth) and kinematic (completion entropy) classifications are orthogonal: finite shadow depth with zero defect (Heisenberg, Kac–Moody, $\beta\gamma$) coexists with infinite shadow depth and nonzero defect (Virasoro, $\mathcal{W}_N$), and entropy varies continuously within each class.

1.15.1 The Heisenberg atom: Gaussian archetype, termination at degree 2

The Heisenberg algebra $\mathcal{H}_k$ has OPE $\alpha(z)\alpha(w) \sim k/(z-w)^2$: a double pole, no simple pole, no Lie bracket. All five theorems are verified by residue extraction on $\overline{C}_n(X)$.

Bar complex. The bar differential vanishes identically ($d_{\text{bracket}} = 0$); only $d_{\text{curvature}}$ acts, producing scalars. The bar cohomology is $H^p(\bar{B}(\mathcal{H}_k)) = \mathbb{C}$ in bar degree 1, concentrated in the top conformal weight stratum. The Koszul dual is $\mathcal{H}_k^! = \mathrm{Sym}^{\mathrm{ch}}(V^*)$, a curved chiral coalgebra with curvature $-k\omega_g$ (Chapter **??**). Note: $\mathcal{H}_k^! \neq \mathcal{H}_{-k}$ as chiral algebras; the Heisenberg algebra is *not* Koszul self-dual.

Genus tower. The period correction at genus g couples to the Hodge bundle via $\kappa(\mathcal{H}_k) = k$. The obstruction class is $\mathrm{obs}_g = k \cdot \lambda_g$. The genus expansion of the free energy reproduces the Dedekind eta function: $\sum_{g\geq 1} F_g \hbar^{2g} = k(\hat{A}(i\hbar) - 1)$, with the chiral partition function $Z_{\text{ch}} = 1/\eta(\tau)$ yielding $F_1 = k/24$ at genus 1 (Chapter **??**).

Shadow obstruction tower. The Heisenberg shadow obstruction tower terminates at degree 2: $o_3(\mathcal{H}_k) = 0$ (by abelian rigidity), so the cubic shadow vanishes, the quartic shadow vanishes, and $\Theta_{\mathcal{H}_k} = \kappa(\mathcal{H}_k) \cdot \eta \otimes \Lambda$ (the tower terminates at degree 2). The shadow metric is $Q_L = (2k)^2$ (constant); the critical discriminant $\Delta = 0$ (Theorem **??**).

Completion kinematics. The vacuum character $\chi_{\mathcal{H}_k}(q) = \prod_{n\geq 1}(1-q^n)^{-1}$ gives $G_{\mathcal{H}_k}(t) = t$ (one primitive of reduced weight 1). Entropy $h_K(\mathcal{H}_k) = 0$; defect $\Delta = 0$. The Heisenberg algebra is kinematically trivial: the simplest point in the completion phase diagram.

Complementarity. $Q_g(\mathcal{H}_k) \oplus Q_g(\mathcal{H}_k^!) \cong H^*(\overline{\mathcal{M}}_g, \mathbb{C})$: the Koszul pair $(\mathcal{H}_k, \mathcal{H}_k^!)$ fills complementary Lagrangians. The Koszul conductor is $K(\mathcal{H}_k) = c(\mathcal{H}_k) + c(\mathcal{H}_k^!) = 1 + 1 = 2$.

1.15.2 Affine Kac–Moody: Lie/tree archetype, termination at degree 3

The affine Kac–Moody algebra $\widehat{\mathfrak{g}}_k$ has OPE $J^a(z)J^b(w) \sim k\,\delta^{ab}/(z-w)^2 + f^{ab}{}_c J^c(w)/(z-w)$. The simple pole introduces a nonzero bracket component d_{bracket}, and $d^2_{\text{bracket}} \neq 0$ (proved for $\mathfrak{sl}_2$ at all 2048 sign conventions). The full bar differential $d_{\bar{B}} = d_{\text{bracket}} + d_{\text{curvature}}$ satisfies $d^2_{\bar{B}} = 0$ by the Borcherds identity.

Koszul dual and Feigin–Frenkel. The Koszul dual $(\widehat{\mathfrak{g}}_k)^! = (H^*(\bar{B}(\widehat{\mathfrak{g}}_k)))^\vee$ is the chiral Chevalley–Eilenberg algebra $\mathrm{CE}^{\mathrm{ch}}(\widehat{\mathfrak{g}}_{k'})$ at level $k' = -k - 2h^\vee$, where $h^\vee$ is the dual Coxeter number. This is *not* the same object as $\widehat{\mathfrak{g}}_{k'}$ itself: the Koszul dual and the Feigin–Frenkel level-reflected algebra share the same modular characteristic κ, but they are categorically distinct. The Sugawara construction, which provides the stress-energy tensor $T(z) = \frac{1}{2(k+h^\vee)} \sum_a :J^a J^a:(z)$, is undefined at the critical level $k = -h^\vee$ (not "c diverges"; rather, the Sugawara vector is not in the completion of $U(\widehat{\mathfrak{g}}_{-h^\vee})$).

Shadow obstruction tower: degree 3 termination. The modular characteristic is $\kappa(\widehat{\mathfrak{g}}_k) = \dim(\mathfrak{g}) \cdot (k+h^\vee)/(2h^\vee)$, with $\kappa(k) + \kappa(k') = 0$ (anti-symmetry for the affine family). The cubic shadow $\mathfrak{C}(\widehat{\mathfrak{g}}_k)$ is nonzero (it detects the Lie bracket), but $o_4 = 0$: the tower terminates at degree 3. The shadow metric $Q_L = (2\kappa + 3\alpha t)^2$ is a perfect square; the critical discriminant $\Delta = 0$ (Jacobi identity forces the quartic to be entirely derived from the cubic self-bracket).

Completion kinematics. The vacuum character gives $G_{\widehat{\mathfrak{g}}_k}(t) = (\dim\mathfrak{g})\, t/(1-t)$: $\dim\mathfrak{g}$ primitives at each reduced weight. Koszul radius $\rho = 1/(\dim\mathfrak{g} + 1)$, entropy $h_K = \log(\dim\mathfrak{g}+1)$. For $\mathfrak{sl}_2$: $h_K \approx 1.10$, exceeding Virasoro ($h_K \approx 0.57$) despite simpler shadow depth. Defect $\Delta = 0$: the generators J^a span the full primitive sector.

1.15.3 The $\beta\gamma$ system: contact/quartic archetype, termination at degree 4

The class-C witness is the standard conformal-weight family $\beta\gamma_\lambda$: one has $S_2 = 6\lambda^2 - 6\lambda + 1$, $S_3 = 0$, $S_4 = -5/12$, and $S_r = 0$ for $r \geq 5$, so the tower terminates at degree 4. This is the contact archetype, the first system where the quartic shadow is nonzero but the quintic obstruction vanishes. The single-line dichotomy (Theorem ??) shows that this value is impossible on a single primary line, where $r_{\max} \in \{2, 3, \infty\}$. Accordingly, the internal weight-changing line has zero shadow tower by rank-one abelian rigidity (Corollary ??), while the T-line is class M; neither one-dimensional slice is the class-C witness by itself.

Completion kinematics. Two generators of weights 1 and 0 give $G_{\beta\gamma}(t) = t/(1-t)$. Koszul radius $\rho = 1/2$, entropy $h_K = \log 2 \approx 0.693$, defect $\Delta = 0$. Despite greater shadow depth (degree 4 vs 3 for Kac–Moody), the $\beta\gamma$ system is kinematically simpler than any non-abelian affine algebra.

1.15.4 Virasoro and $\mathcal{W}_N$: mixed archetype, infinite tower

The Virasoro algebra has OPE $T(z)\,T(w) \sim c/2(z-w)^4 + 2T(w)/(z-w)^2 + \partial T(w)/(z-w)$, a quartic pole, the highest in the standard landscape. The Koszul dual is Vir_{26-c} (chiral Koszul self-dual at $c = 13$, not $c = 26$).

The shadow obstruction tower is infinite: $o_5(\mathrm{Vir}_c) \neq 0$, and the quintic obstruction is forced. The structural mechanism is *direct self-referentiality* in the OPE: the generator T appears in its own self-OPE via $T_{(1)}T = 2T$. This direct self-loop, absent for Heisenberg ($J \notin J_{(k)}J$), affine ($E^\alpha \notin E^\alpha_{(k)}E^\alpha$), and $\beta\gamma$ ($\beta \notin \beta_{(k)}\beta$), sustains an infinite A_∞ cascade that prevents the shadow obstruction tower from terminating (Proposition ??). This is the mixed archetype: both cubic and quartic shadows are present, and the tower never terminates. The quartic contact invariant $Q^{\mathrm{contact}}_{\mathrm{Vir}} = 10/[c(5c+22)]$ is the explicit measure of the quartic interaction. The poles at $c = 0$ and $c = -22/5$ are the minimal model values $c_{2,5}$ and $c_{2,3}$. The single-line dichotomy (Theorem ??) identifies the structural mechanism: the Virasoro critical discriminant $\Delta_{\mathrm{Vir}} = 40/(5c+22)$ is nonzero for all non-resonant c, and the universal factorization $S_r = \Delta \cdot R_r$ with generically nonvanishing quotients R_r forces all degrees $r \geq 4$ to be populated.

The $\mathcal{W}_N$ algebras share the infinite-tower behavior for $N \geq 3$. The $\mathcal{W}_3$ algebra is the first case where the filtered-curved distinction matters: $\mathcal{W}_3$ carries a truly filtered (not merely curved) A_∞ structure, because the composite field $\Lambda =: (TT) : -\frac{3}{10}\partial^2 T$ introduces a quartic pole that is not a scalar multiple of the identity.

Completion kinematics. Kinematics detects structure invisible to shadow depth. For generic Virasoro, $G_{\mathrm{Vir}}(t) = \frac{1-t}{t}\left(\prod_{n\geq 2}(1-t^n)^{-1} - 1\right)$ gives primitive multiplicities $g_r = 1, 0, 1, 0, 2, 0, 3, 1, \ldots$ Defect $\Delta_{\mathrm{Vir}}(t) = t^3 + 2t^5 + 3t^7 + \cdots \neq 0$: infinitely many primitive cumulants beyond T. Entropy $h_K(\mathrm{Vir}) \approx 0.567$.

For the $\mathcal{W}_N$ tower, the entropy increases monotonically (Proposition ??):

$$h_K(\mathcal{W}_2) \approx 0.567 \;<\; h_K(\mathcal{W}_3) \approx 0.772 \;<\; h_K(\mathcal{W}_4) \approx 0.835 \;<\; \cdots \;<\; h_K(\mathcal{W}_\infty) \approx 0.872.$$

The $\mathcal{W}_3$ jump from 0.567 to 0.772 is the kinematic signature of the composite field Λ, which forces the corrected Koszul recursion $h_q = 2h_{q-1} + h_{q-2} + h_{q-3}$ in place of the naive $h_q = 2h_{q-1} + h_{q-2}$. At the limit $N = \infty$, the character $\chi_{\mathcal{W}_\infty}(q) = \prod_{n\geq 2}(1-q^n)^{-(n-1)}$ admits a MacMahon factorization (Proposition ??), connecting the $\mathcal{W}_\infty$ completion frontier to the combinatorics of plane partitions.

The same-family shadow Vir_{26-c} coexists with the nonzero defect because it is a *dynamic* shadow, not the full kinematic dual. Kinematically, the completed dual coalgebra has infinitely many primitive generators; dynamically, the scalar trace $\kappa(\mathrm{Vir}_c)\cdot\lambda_g$ already reproduces the partner family Vir_{26-c}. The full H-level kinematic identification is the residual MC4 task (§?? of the concordance).

1.15.5 The Yangian: E_1-chiral, R-matrix inversion

The Yangian $Y(\mathfrak{g})$ is an E_1-chiral algebra (Definition 1.23.3), not E_∞-chiral. Its bar complex uses *ordered* configuration spaces $\mathrm{Conf}_n^{\mathrm{ord}}(X)$ rather than unordered ones.

The Yangian is generated by currents $T^{(1)}(z), T^{(2)}(z), \dots$ with spectral parameter dependence. Its OPE is governed by the rational R-matrix

$$R(u) \;=\; 1 \;+\; \frac{\Omega}{u}, \qquad \Omega \;=\; \sum_a I^a \otimes I_a \;\in\; \mathfrak{g}\otimes\mathfrak{g},$$

where Ω is the Casimir tensor and u is the spectral parameter. The RTT relation

$$R_{12}(u-v)\,T_1(u)\,T_2(v) \;=\; T_2(v)\,T_1(u)\,R_{12}(u-v)$$

encodes the exchange algebra. Verdier duality acts by R-matrix inversion $R(u)\mapsto R(u)^{-1}$ rather than by type exchange $\mathcal{A}\mapsto\mathcal{A}^!$ (Theorem ??). The Drinfeld–Kohno bridge (Theorem ??) relates the chiral Koszul duality of affine Kac–Moody algebras at the top of the DK square (1.19.1) to the quantum group R-matrix inversion at the bottom, with the Kazhdan–Lusztig equivalences as vertical arrows.

The modular dg-shifted Yangian $\mathcal{Y}_T^{\mathrm{dg,mod}}$ (Definition ??) is the ambient pronilpotent completion of the convolution dg Lie algebra for the Yangian. A conjectural modular R-matrix $R_T^{\mathrm{mod}}(z;\hbar)$ would be a Maurer–Cartan element in this completed algebra.

Under Drinfeld–Sokolov reduction, the transferred coproduct is strictly primitive at all degrees (Principle ??): a ghost-number obstruction kills every HPL tree correction to the coproduct, while the A_∞ product tower and the r-matrix survive. This separates the dg-shifted Yangian into two regimes: for affine Kac–Moody (class **L**), both the product and coproduct channels are active; for $\mathcal{W}$-algebras (class **M**), the coproduct channel is trivial and all dynamics resides in the r-matrix twist. The ghost-number obstruction extends to all principal DS reductions: the graviton is composite (it arises from the BRST reduction of a Lie-type current); composites do not split.

The binary collision residue $r(z)=\mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}})$ is one object with seven constructions: it is simultaneously the bar-cobar twisting morphism (Theorem ??), the line-operator MC element of [?DNP25], the PVA classical r-matrix of [?KhanZeng25], the leading pole of the commuting sphere Hamiltonians of [?GZ26], the Drinfeld Yangian r-matrix [?Drinfeld85], the Sklyanin Poisson bracket [?STS83], and the Gaudin Hamiltonian [?FFR94]. The identification is Chapter ??. Three distinct invariants control the collision structure: the generator OPE pole order $p_{\max}$, the collision depth $k_{\max}=p_{\max}-1$, and the shadow depth $r_{\max}$ (Chapter ??). These need not coincide: the $\beta\gamma$ system has $p_{\max}=1$, $k_{\max}=0$, $r_{\max}=4$. The collision depth determines the differential operator order of the commuting Hamiltonians (Theorem ??).

At genus 1, the rational propagator $1/z$ is replaced by the Weierstrass zeta function $\zeta(z,\tau)$ on the torus E_τ, and all seven faces acquire elliptic counterparts: the KZB connection, the Belavin–Drinfeld elliptic r-matrix, and the elliptic Gaudin model (Chapter ??). For class M, the higher OPE poles of the stress tensor produce elliptic differential operators involving $\wp(z,\tau)$ and $\wp'(z,\tau)$; these have no counterpart in the existing integrable systems literature.

1.15.6 Lattice vertex algebras: braided, curvature-braiding orthogonal

The lattice vertex algebra V_Λ associated to an even lattice Λ carries an E_1 sublattice structure visible through screening operators. Curvature-braiding orthogonality holds: the modular characteristic $\kappa(V_\Lambda) = \mathrm{rank}(\Lambda)$ is independent of the 2-cocycle that determines the braiding (Theorem ??). The curvature sees only the rank of the lattice; the braiding sees only the cocycle. These two pieces of data are orthogonal in a precise sense: they live in complementary summands of the deformation complex.

Unbounded finite depth and arithmetic content. While generic lattice VOAs have shadow depth 2 (Gaussian), the vertex-operator OPE $e^{i\lambda\cdot X}_{(0)} e^{i\mu\cdot X} \sim \varepsilon(\lambda,\mu)\, e^{i(\lambda+\mu)\cdot X}$ is *nonlinear* (it creates new fields), though not self-referential ($e^{i\lambda\cdot X} \notin e^{i\lambda\cdot X}_{(k)} e^{i\lambda\cdot X}$ since 2λ is never a root of an even lattice). The shadow depth of V_Λ is *finite but unbounded*: it is determined by the modular-form content of the theta function Θ_Λ (Theorem ??). For E_8 ($r{=}8$, $\Theta = E_4$): depth 3. For the Leech lattice ($r{=}24$, $\Theta = E_4^3 - 720\Delta$): depth 4, the first example involving the Ramanujan cusp form Δ_{12} and its L-function. Even unimodular lattices of ranks $48, 72, 96, \ldots$ achieve depths $5, 6, 7, \ldots$: each independent cusp form in $S_{r/2}(\Gamma)$ contributes one critical line of the constrained Epstein zeta $\varepsilon_s^r(V_\Lambda)$ and raises the depth by 1. The shadow depth is a *Galois-theoretic invariant*: it counts the number of independent automorphic L-functions encoded in the lattice's Hecke eigenform decomposition (Chapter ??). All lattice VOAs are exact Euler–Koszul (ek = 0), since every generator has weight 1. Their Dirichlet–sewing lift is a pure Euler product: $S_{V_\Lambda} = \mathrm{rank}(\Lambda)\cdot\zeta(u)\,\zeta(u{+}1)$.

1.16 Strictness as a shadow: the homotopy upgrade

The convolution dg Lie algebra $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ of (1.12.1) is a *strict model*: differential and bracket defined on the nose, not up to homotopy. It is only the first visible layer of a richer homotopy-coherent package. Write

$$\begin{aligned} &\mathrm{Conv}_{\mathrm{str}}(\mathrm{FCom},\ \mathrm{End}_{\bar{B}(\mathcal{A})}) \quad \text{(the strict dg Lie convolution model)},\\ &\mathrm{Conv}_{\infty}(\mathrm{FCom},\ \mathrm{End}_{\bar{B}(\mathcal{A})}) \quad \text{(the homotopy-invariant } L_\infty \text{ convolution)}.\end{aligned}$$

A quasi-isomorphism of L_∞ algebras relates them (Chapter ??), so their MC moduli coincide and every MC element of $\mathrm{Conv}_{\mathrm{str}}$ lifts canonically to Conv_∞. The packages are not interchangeable: Conv_∞ remembers higher homotopies invisible to $\mathrm{Conv}_{\mathrm{str}}$, essential for transfer, formality, and gauge equivalence.

Bar-cobar (Theorem A) preserves quasi-isomorphisms because it is a quantum L_∞ functor, respecting the full homotopy package. Homotopy transfer (Chapter ??) produces transferred A_∞ and L_∞ structures on bar cohomology carrying information lost by passage to a strict model. Every strict-level formula is the leading term of a homotopy-coherent datum, as $\kappa(\mathcal{A})$ is the leading term of $\Theta^{\leq r}_{\mathcal{A}}$.

The same principle applies at the modular level. The full modular deformation object is

$$\mathrm{Def}^{\mathrm{mod}}_{\infty}(\mathcal{A}) \ := \ \mathrm{Conv}_\infty\big(\mathrm{FCom},\ \mathrm{End}_{\bar{B}(\mathcal{A})}\big) \mathbin{\widehat{\otimes}} R\Gamma\big(\overline{\mathcal{M}}_{g,\bullet},\ \mathbb{Q}\big),$$

an L_∞ algebra whose MC elements are complete genus-expansion data compatible with clutching, trace, and duality up to coherent homotopy. The strict model $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ is a chart on this larger formal space. Theorem B (bar-cobar inversion) and Theorem C (complementarity) are proved at the strict level and stated at the homotopy level; the passage between them is guaranteed by the contractibility of the space of models (Proposition ??).

Remark 1.16.1 (Structural consequences of the homotopy upgrade). Four structural consequences:

(i) Every strict-model formula (convolution dg Lie bracket, curved bar differential, finite obstruction class, spectral discriminant) is a projection of $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathrm{Conv}_\infty)$. The strict model computes the correct answer because it is a faithful chart on the larger formal space, not because the L_∞ data are dispensable.

(ii) Genus, clutching, and gluing enter at the chain level (in the bar and cobar constructions, in the boundary stratification of configuration spaces, and in the completed deformation package) rather than as cohomological decorations applied post hoc.

(iii) The standard families (Heisenberg, $\beta\gamma$, affine Kac–Moody, Virasoro, $\mathcal{W}_N$, Yangians, lattice and elliptic) serve as diagnostic benchmarks: each activates a different stratum of the homotopy machinery, and their shadow archetypes (Gaussian, Lie/tree, contact, mixed) are the finite-order witnesses of the transferred L_∞ operations.

(iv) The shadow obstruction tower is a faithful finite-order image of $\Theta_{\mathcal{A}}$: each truncation $\Theta_{\mathcal{A}}^{\leq r}$ captures new modular invariants (κ at r=2, $\mathfrak{C}$ at r=3, $\mathfrak{Q}$ at r=4) that would be invisible at any lower order.

1.17 The three pillars

Three recent preprints force foundational upgrades at three levels of the theory.

1.17.1 Pillar A: homotopy chiral algebras (MS24)

The primitive local object is not a strict chiral algebra but a *homotopy chiral algebra* (Ch_∞) in the sense of Malikov–Schechtman [?MS24]. A Ch_∞-algebra is a $\mathcal{D}_X$-module A equipped with an A_∞ structure: multilinear operations $m_k\colon A^{\otimes k} \to A((\lambda_1))\cdots((\lambda_{k-1}))$ (formal Laurent series in spectral parameters $\lambda_j = z_j - z_k$) satisfying the Stasheff relations in each spectral variable. Here m_1 is the differential, m_2 the OPE product, and m_k ($k \geq 3$) the failure of strict associativity. The strict chiral algebra $\mathcal{A}$ appears after *rectification*: it is H^0 of a Ch_∞ structure, projecting to the Poisson vertex algebra (PVA) shadow at the classical level.

The identification theorem (Theorem **??**): $B(\mathcal{A})$ carries a natural Ch_∞-algebra structure, upgrading the bar construction from a dg coalgebra to a homotopy chiral algebra.

1.17.2 Pillar B: filtered convolution sL_∞ algebras (RNW19+Val16)

The universal deformation machine is the *filtered convolution sL_∞-algebra* $\hom_\alpha(C, A)$ (Robert-Nicoud–Wierstra [?RNW19], Vallette [?Val16]), with underlying space $\mathrm{Hom}(C, A)$ (§1.11.3), L_∞ brackets ℓ_k ($k \geq 1$) where ℓ_2 is the convolution Lie bracket and ℓ_k ($k \geq 3$) arise from the cooperadic decomposition $\Delta_C^{(k)}$. The deformation space of a twisting morphism $\alpha\colon C \to A$ is the MC ∞-groupoid $\mathrm{MC}_\bullet(\hom_\alpha)$; the dg Lie algebra is its strict model.

The functor $\hom_\alpha$ extends to ∞-morphisms in either slot separately but *not* both simultaneously (RNW19, Theorem 6.6). The MC3 categorical lift must therefore proceed one slot at a time.

The identification theorem (Corollary **??**): $\Theta_{\mathcal{A}} \in \mathrm{MC}(\hom_\alpha) \cong \mathrm{Tw}_\alpha$; the universal MC element is a twisting morphism.

1.17.3 Pillar C: logarithmic Fulton–MacPherson (Mok25)

The global collision geometry is the logarithmic Fulton–MacPherson compactification $\mathrm{FM}_n(X|D)$ on a simple normal crossing pair (X, D) in the sense of Mok [?Mok25]. The classical FM compactification is the special case $D = \emptyset$.

Mok's geometric theorem indexes boundary strata by planted forests and gives their codimensions. The tropicalization statement (Theorem ??) is therefore geometric at the level of strata:

$$\mathrm{Trop}(\mathrm{FM}_n(X|D)) = \mathsf{For}_{\mathrm{plant}}(n).$$

The identification with the dg planted-forest coefficient algebra is conditional on determinant-line signs and the signed edge-contraction differential. The semistable degeneration formula supplies the geometric input for the clutching laws; the chain-level residue-pushforward transport is the conditional manuscript package.

1.17.4 Eleven identification theorems

The three pillars yield eleven identification theorems, nine proved and two structural:

(1) $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}} = \mathrm{hom}_\alpha(\mathrm{Com}^!_{\mathrm{ch}}, \mathcal{P}^{\mathrm{ch}})$: the convolution algebra is the twisting hom (Theorem ??).
(2) $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathrm{hom}_\alpha) \cong \mathrm{Tw}_\alpha$: the universal MC element is a twisting morphism (Corollary ??).
(3) $\mathrm{Trop}(\mathrm{FM}_n(X|D)) = \mathsf{For}_{\mathrm{plant}}(n)$ at the level of Mok boundary strata and codimensions; the dg coefficient algebra $\mathfrak{G}^{\mathrm{mod}}_{\mathrm{pf}}$ is the signed chain model, conditional on determinant-line orientations (Theorem ??).
(4) $B(\mathcal{A})$ is a Ch_∞-algebra (Theorem ??).
(5) $\bar{B}_\kappa \dashv \Omega_\kappa$ is a Quillen equivalence (Theorem ??).
(6) $\mathcal{A}^{\mathrm{sh}}$ is a homotopy invariant (Theorem ??).
(7) The one-slot obstruction constrains MC3: the categorical lift must proceed one slot at a time (RNW19, Theorem 6.6).
(8) $F_n = o_n$: the secondary Borcherds operations of [?MS24] at degree $n \geq 3$ coincide with the shadow obstruction tower obstruction classes (Proposition ??).
(9) The quartic clutching law has a log-geometric derivation via Mok's degeneration formula [?Mok25, Corollary 5.3.4] (Construction ??).
(10) The log clutching package uses Mok's degeneration formula [?Mok25, Corollary 5.3.4] as geometric input; the chain-level residue-pushforward transport is conditional.
(11) The ambient $D^2 = 0$ (Theorem ??) is conditional on Mok's relative log-FM geometry and the signed residue-pushforward construction.

1.17.5 Chain-level synthesis

The three pillars produce a single chain-level object on the fixed smooth-curve core and a conditional relative logarithmic enhancement. For a cover $\mathfrak{U}$ of X, set $A^{\mathrm{ch}}_\infty := \check{C}^\bullet(\mathfrak{U}; \mathcal{A})$ (the Ch_∞-input) and $C^{\log\mathrm{FM}}_{\mathrm{mod}}(n) := C_\bullet(\mathrm{FM}_n(X|D))$ (the geometric coefficient system). It becomes a cooperadic chain system after choosing the signed residue-pushforward, orientation-line, and homotopy-coherent cocomposition package. The logarithmic FM convolution algebra

$$\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}} \;:=\; \mathrm{Conv}_\infty\big(C^{\log \mathrm{FM}}_{\mathrm{mod}},\; \mathrm{End}_{\mathrm{Ch}_\infty}(\mathcal{A}^{\mathrm{ch}}_\infty)\big)$$

has five-component differential $D^{\log}_{\mathrm{mod}} = d_{\check{C}} + d_{\mathrm{Ch}_\infty} + d_{\mathrm{coll}} + d_{\mathrm{sew}} + d_{\mathrm{loop}}$, where $d_{\check{C}}$ is the Čech differential on the cover $\mathfrak{U}$, d_{Ch_∞} is the A_∞ differential from the homotopy chiral structure, d_{coll} is the collision differential (residue extraction along the pairwise diagonal divisors of the log-FM compactification), d_{sew} is pushforward along codimension-one separating divisors, and d_{loop} is pushforward along non-separating divisors (Construction ??). The fixed smooth-curve part of $D^2 = 0$ is the ordinary boundary-of-boundary identity. The ambient relative log-FM part is conditional on transporting Mok's codimension-two faces through signed Gysin residues and proper pushforwards.

The *graphwise cocomposition* (Construction ??)

$$\Delta^{\log}_{\Gamma} \;:=\; (\nu_\Gamma)_* \circ i^!_\Gamma$$

decomposes the cooperadic structure over rigid stable graphs: here $D^{\log}_\Gamma$ is the log-FM boundary divisor indexed by the graph Γ (the stratum where points collide according to the combinatorial type of Γ), $i^!_\Gamma$ is the conditional Gysin residue along this divisor, and $\nu_\Gamma\colon D^{\log}_\Gamma \to \prod_{v\in V(\Gamma)} \overline{C}_{\mathrm{val}(v)}(X)$ is the normalisation map that separates the colliding clusters. Once the signed chain package is fixed, this yields Taylor coefficients $\ell^{(g)}_k = \sum_\Gamma |\mathrm{Aut}(\Gamma)|^{-1}\,\ell_\Gamma$ of the modular L_∞-structure.

The universal MC element is the graph sum

$$\Theta_{\mathcal{A}} = \sum_{\Gamma} \frac{W^{\log \mathrm{FM}}_\Gamma}{|\mathrm{Aut}(\Gamma)|} \otimes \Phi^{\mathcal{A}}_\Gamma,$$

summed over connected stable graphs Γ, where:

- $W^{\log \mathrm{FM}}_\Gamma \in C_*(\overline{\mathcal{M}}_\Gamma)$ is the *graph weight*, the fundamental chain of the boundary stratum $\overline{\mathcal{M}}_\Gamma = \prod_v \overline{\mathcal{M}}_{g(v),\mathrm{val}(v)}$ indexed by Γ;
- $\Phi^{\mathcal{A}}_\Gamma = \bigotimes_{v\in V(\Gamma)} \Phi^{\mathcal{A}}_v \circ \bigotimes_{e\in E(\Gamma)} P_e$ is the *algebra amplitude*, the tensor product of transferred chiral operations $\Phi^{\mathcal{A}}_v$ at each vertex, contracted along each internal edge e by the propagator P_e (the homotopy retraction data of §1.11.4).

This is the universal modular twisting morphism: $\mathrm{MC}_\bullet(\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}) \simeq \mathrm{Tw}_{\alpha^{\mathrm{mod}}}$ (Corollary ??). Each coefficient of $\Theta_{\mathcal{A}}$ acquires the refined Mok degree

$$\mu = (g, r;\, b, s, \ell, c, p)$$

(Definition ??): g is loop genus, r is external arity, b counts bar/Hamiltonian contractions, s separating stable nodes, ℓ non-separating stable nodes, c relative Mok crossings, and p pure planted-forest depth. The full log codimension is $s + \ell + c + p$. The coarse tridegree (g, r, d) is a projection; the computational projection used here is $d = c + p$. With the signed residue-pushforward package fixed, this filtration is bar–cobar compatible:

$$\mathrm{gr}_F\, \Omega \bar{B}^{\mathrm{mod},\log}(\mathcal{A}) \;\cong\; \Omega\big(\mathrm{gr}_F\, \bar{B}^{\mathrm{mod},\log}(\mathcal{A})\big).$$

The weight filtration stratifies the MC equation by internal-edge count. At weight 1, the MC element restricts to two types: $\Theta^{[1]}_{\mathcal{A}}|_{(0,4)} = [\delta_s] \otimes \Phi_s + [\delta_t] \otimes \Phi_t + [\delta_u] \otimes \Phi_u$ (the Malikov–Schechtman Jacobiator F_3, where $[\delta_s], [\delta_t], [\delta_u]$ are fundamental classes of the three boundary divisors of $\overline{\mathcal{M}}_{0,4} \cong \mathbb{P}^1$ corresponding to the s-, t-, u-channel degenerations); and $\Theta^{[1]}_{\mathcal{A}}|_{(1,1)} = [\delta_{\mathrm{irr}}] \otimes \Delta$ (the BV operator, where $[\delta_{\mathrm{irr}}]$ is the fundamental class of the irreducible boundary divisor of $\overline{\mathcal{M}}_{1,1}$). At weight 2, the rigid planted-forest corrections enter, the first genuinely logarithmic layer (Proposition **??**).

The modular tangent complex $T^{\mathrm{mod}}_{\Theta_{\mathcal{A}}} := (\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}, d_{\Theta_{\mathcal{A}}})$ with L_∞-twisted differential $d_{\Theta_{\mathcal{A}}}(x) = \sum_{n,g} (\hbar^g/n!)\, \ell^{(g)}_{n+1}(\Theta^{\otimes n}_{\mathcal{A}}, x)$ recovers all invariants as characteristic projections: $\kappa = \mathrm{tr}(\Theta_{\mathcal{A}})_{2,0}$, $\Delta_{\mathcal{A}} = \mathrm{sdet}(1 - t\, H_{d_\Theta})$, $\mathfrak{R}^{\mathrm{mod}}_4 = \mathrm{pr}_4(\Theta_{\mathcal{A}})$ (Construction **??**). The genus filtration on L_{mod} induces a spectral sequence whose differentials are the obstruction maps Ob_g, distinct from the PBW spectral sequence (Construction **??**). The full chain-level architecture (weight filtration, clutching law, genus-two shell decomposition, curved extension for non-quadratic algebras) is in §**??**.

1.18 The total space

The nilpotence-periodicity correspondence becomes geometric in the total space of the universal configuration fibration:

$$\pi : \overline{C}_n(C_g) \longrightarrow \overline{\mathcal{M}}_g.$$

Its fiber over $[\Sigma_g]$ is $\overline{C}_n(\Sigma_g)$. The boundary has two types of divisors:

(a) *Collision divisors D_I* (vertical): marked points collide on a fixed smooth curve. These produce the bar differential.

(b) *Degeneration divisors δ_Γ* (horizontal): the curve acquires a node. These produce the period corrections that restore $\mathrm{d}g g^{\,2} = 0$.

At genus 0, only type (a) exists and the Arnold relation alone gives $d_0^2 = 0$. At genus $g \geq 1$, both coexist.

1.18.1 The Hodge bundle and the propagator

The Hodge bundle $\mathbb{E} = \pi_* \omega_\pi$ has fiber $H^0(\Sigma_g, \Omega^1)$. The propagator depends on a choice of symplectic basis for $H_1(\Sigma_g, \mathbb{Z})$; the period corrections live in $\mathbb{E} \boxtimes \overline{\mathbb{E}}$. For uniform-weight algebras, the scalar trace yields

$$\mathrm{obs}_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot c_g(\mathbb{E}) = \kappa(\mathcal{A}) \cdot \lambda_g \ \in\ H^*(\overline{\mathcal{M}}_g) \qquad \text{(uniform-weight, all } g\text{; arbitrary families, } g = 1\text{)}.$$

1.18.2 The family index

The family $\bar{B}_g(\mathcal{A}) \to \overline{\mathcal{M}}_g$ defines a virtual vector bundle. Grothendieck–Riemann–Roch applied to $\pi : \overline{C}_g \to \overline{\mathcal{M}}_g$: the fiber Chern character contributes $\kappa(\mathcal{A})$; the Todd class produces the $\hat{A}$-series. The generating function is $\sum_{g\geq 1} F_g(\mathcal{A}) \cdot \hbar^{2g} = \kappa(\mathcal{A}) \cdot (\hat{A}(i\hbar) - 1)$ (Theorems 1.10.1 and **??**). For a free fermion, this is the spin partition function; for the Virasoro algebra, the one-loop string amplitude.

1.18.3 The Lagrangian decomposition

Verdier duality $\sigma = \mathbb{D}_{\mathrm{Ran}}$ acts on $\mathbf{C}_g(\mathcal{A}) = R\Gamma(\overline{\mathcal{M}}_g, \mathcal{Z}_{\mathcal{A}})$. The composition

$$\mathbf{C}_g(\mathcal{A}) \otimes \mathbf{C}_g(\mathcal{A}) \xrightarrow{\mathrm{id}\otimes\sigma} \mathbf{C}_g(\mathcal{A}) \otimes \mathbf{C}_g(\mathcal{A}^!) \xrightarrow{\mathrm{eval}} R\Gamma(\overline{\mathcal{M}}_g, \omega_{\overline{\mathcal{M}}_g})$$

is a $(-(3g-3))$-shifted symplectic form (Pantev–Toën–Vaquié–Vezzosi [?PTVV13]). The ± 1 eigenspaces of σ are complementary Lagrangians. Traced through π and projected to moduli, this produces Theorem C.

1.19 Architecture

Remark 1.19.1 (Five geometric ingredients). Modular Koszul duality rests on five geometric ingredients. The first four belong to the commutative (E_∞) face; the fifth to the associative (E_1) face.

(i) *Three-point collision.* The Arnold relation (1.2.1) is factorization coherence: fusing three insertions does not depend on the order of pairwise collisions. The genus-0 seed.

(ii) *Verdier duality on* $\mathrm{Ran}(X)$. $\mathbb{D}_{\mathrm{Ran}}\bar{B}_X(\mathcal{A}) \simeq \mathcal{A}^!_\infty$ (Theorem **??**): the Koszul dual algebra as theorem, not definition-by-analogy.

(iii) *Genus-1 curvature.* $d^2_{\mathrm{fib}} = \kappa(\mathcal{A}) \cdot \omega_1 \cdot \mathrm{id}$ (Theorem **??**). The natural home is Positselski's coderived category: quantum corrections are a change of ambient category, not extra terms.

(iv) *Clutching of stable curves.* The genus tower is compatible with boundary clutching via the modular operad (Theorem **??**). The combinatorics is stable graphs, not trees, and $\Theta_{\mathcal{A}}$ (Theorem **??**, proved intrinsically by Theorem **??**) is compatible with clutching, trace, and Verdier duality.

(v) *Associative structure.* The Yangian $Y(\mathfrak{g})$ is E_1-chiral (Definition 1.23.3): its bar complex uses ordered configuration spaces, and Verdier duality acts by R-matrix inversion $R \mapsto R^{-1}$ (Theorem **??**). The sign flip $\hbar \mapsto -\hbar$ replaces the Feigin–Frenkel involution.

In the Heisenberg algebra, only pieces (i)–(iv) are active, and they already force the full commutative/modular package: Arnold ensures $d^2 = 0$; Verdier produces $\mathcal{H}_k^! = \mathrm{Sym}^{\mathrm{ch}}(V^*)$; the genus-1 period integral creates curvature $\kappa = k$; clutching assembles the genus tower.

In the Yangian, piece (v) is required. Ordered configurations, R-matrix inversion on the module category, and the derived Drinfeld–Kohno bridge (Theorem **??**) are a genuinely braided determination of the same configuration-space mechanism.

Remark 1.19.2 (The DK square). The coexistence of the commutative and braided determinations is witnessed by the derived Drinfeld–Kohno square (Theorem **??**):

$$\begin{tikzcd} O_k^{\mathrm{int}}(\widehat{\mathfrak{g}}) \arrow[r, "\Phi", "\sim"'] \arrow[d, "\Phi_{\mathrm{KL}}"', "\sim"] & O_{k'}(\widehat{\mathfrak{g}}) \arrow[d, "\Phi_{\mathrm{KL}}", "\sim"'] \\ C(U_q(\mathfrak{g})) \arrow[r, "\sim", "q\mapsto q^{-1}"'] & C(U_{q^{-1}}(\mathfrak{g})) \end{tikzcd} \tag{1.19.1}$$

Top: level inversion $k \mapsto k' = -k - 2h^\vee$ (Frame A). Bottom: R-matrix inversion $q \mapsto q^{-1}$ (Frame B). Verticals: Kazhdan–Lusztig equivalences. Commutativity: bar-cobar duality geometrically implements the quantum group involution. The duality constraint on κ ($\kappa + \kappa' = 0$ for KM/free fields; $\kappa + \kappa' =$

$\varrho(\mathfrak{g}) \cdot K(\mathfrak{g})$ for $\mathcal{W}$-algebras, where $\varrho(\mathfrak{g}) = \sum_i 1/(m_i + 1))$ is the scalar shadow of the top arrow; $q \mapsto q^{-1}$ is the quantum-group shadow. The four theorems and the DK square are four projections of one geometric structure on $\mathrm{Ran}(X) \times \overline{\mathcal{M}}_g$.

Remark 1.19.3 (Master functorial correspondence). Every row is a theorem.

Geometry	**Algebra**	**Physics**
Verdier duality $\mathbb{D}_{\mathrm{Ran}}$	Koszul duality $\mathcal{A} \leftrightarrow \mathcal{A}^!$	CPT
Residue at D_{ij}	Bar differential d_{res}	OPE fusion
FM boundary stratum	Tree contribution to m_k	Feynman diagram
Arnold relation	$d^2 = 0$ (genus 0)	Factorization coherence
Period integral on $\overline{\mathcal{M}}_g$	Obstruction class obs_g	One-loop determinant
Clutching $\partial\overline{\mathcal{M}}_g$	Modular operad	String sewing
DS reduction	Koszul functor = Ham. red. to $J_\infty(S_f)$	Gauge fixing = localization to Slodowy slice
Log form η_{ij}	Propagator	Free-field propagator
Forgetful $\overline{\mathcal{M}}_{g,n+1} \to \overline{\mathcal{M}}_{g,n}$	Forgetting a marked point	Integrating out a field
Branch locus of $P_{\mathcal{A}^!}(x)$	Discriminant $\Delta_{\mathcal{A}}(x)$	Spectral curve
MC in $\bar{B}(\widehat{\mathfrak{g}}_k)$	L_∞ flatness	CS equations of motion
$\fint_{\overline{\mathcal{M}}_{0,n}} (\bar{B}^{\mathrm{geom}})^{(0)}_n$	BRST cohomology pairing	Tree-level string amplitude

The first four rows are proved in Part I (Chapters ??–??); the next three in Part II (Chapters ??–??). The physics column is made precise in Chapters ?? and ??.

Part II computes bar complexes for all major families (Table ??). The shared discriminant $\Delta(x) = 1 - 2x - 3x^2$ (Theorem ??) and the universal generating function $\sum_{g\geq 1} F_g x^{2g} = \kappa \cdot (\hat{A}$-genus) (uniform-weight) (Theorem ??) are consequences of the proved scalar and spectral layers.

1.20 Beyond the five theorems: three concentric rings

The results form three concentric rings of decreasing logical certainty.

Ring 1: the proved modular Koszul core. Theorems A/B/C/D/H and their E_1 counterparts (Remark 1.9.2), Master Conjectures 1 and 2 (at finite order), and the DK ladder through step 3 on the evaluation-generated core at all simple Lie types. This ring is closed: every claim is proved in the text, and the outlook (the Frontier part, Chapter ??) records its boundary precisely.

Ring 2: the shadow obstruction tower. The primary mathematical object beyond the linear regime. Between the proved scalar tower $\kappa(\mathcal{A})\lambda_g$ and the full universal class $\Theta_{\mathcal{A}}$ sits a constructive finite-order layer whose visible invariants form a hierarchy:

$$\underbrace{\kappa(\mathcal{A})}_{\text{scalar}} \longrightarrow \underbrace{\Delta_{\mathcal{A}}}_{\text{spectral}} \longrightarrow \underbrace{[\mathfrak{C}_{\mathcal{A}}],\ \mathfrak{o}^{(4)}_{\mathcal{A}},\ [\mathfrak{Q}_{\mathcal{A}}]}_{\text{nonlinear shadows}} \longrightarrow \underbrace{\Theta_{\mathcal{A}}}_{\text{universal class}} .$$

The scalar characteristic $\kappa(\mathcal{A})$ is proved. The spectral discriminant $\Delta_{\mathcal{A}}$ is proved. The cubic and quartic shadow classes $[\mathfrak{C}_{\mathcal{A}}]$ and $[\mathfrak{Q}_{\mathcal{A}}]$ are the first genuinely nonlinear modular invariants, computable on any family where the cyclic deformation package is available through the relevant order (Chapter ??). The branch-line reductions (Appendix ??) extract exact finite-dimensional quotients of this layer, including genus-2 transparency and the universal genus-3 constant 7 on the shared $\widehat{\mathfrak{sl}}_2$/Virasoro/$\beta\gamma$ spectral sheet.

The unifying principle is the modular L_∞ convolution algebra $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$, whose L_∞ structure comes from the Feynman transform of the modular operad; $\mathrm{Conv}_{\mathrm{str}}$ is its strict model. The shadow obstruction tower consists of finite-order projections of $\Theta_{\mathcal{A}}$: κ (degree 2), $\mathfrak{C}$ (degree 3), $\mathfrak{Q}$ (degree 4), each a projection of the next. The weight filtration on $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$ controls extension; each truncation $\Theta^{\leq r}_{\mathcal{A}}$ is constructive without requiring the full all-genera modular envelope. The all-degree master equation $D\Theta + \frac{1}{2}[\Theta, \Theta] = 0$ is proved by the bar-intrinsic construction (Theorem ??): $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_0$ is automatically Maurer–Cartan because $D^2_{\mathcal{A}} = 0$.

Ring 3: the physics-facing frontier. Three axes extend the proved core toward mathematical physics.

(a) *The $\mathcal{W}$-algebra axis.* Completed Koszulity is ubiquitous: every affine $\mathcal{W}$-algebra at generic level is completed Koszul. Strict Koszulity is exceptional: the subregular family $\mathcal{W}^{(2)}_n$ has unbounded canonical homotopy degree, and the classical W_3 sector defines the first finite-degree modular higher-spin package. MC4 is proved: the strong completion-tower theorem (Theorem ??) gives a completion-closed bar-cobar homotopy equivalence on CompCl(Fft), with automatic continuity via the degree cutoff (Lemma ??) and coefficient-stability criterion (Theorem ??). The DS-HPL transfer theorem (Volume II, Theorem thm:ds-hpl-transfer) closes the local nonlinear bridge from affine to $\mathcal{W}$-algebra data: homological perturbation through the BRST SDR transfers the full dg-shifted Yangian triple, with the gravitational coproduct proved strictly primitive at all degrees (Principle ??).

(b) *The Yangian/RTT axis.* Shifted Yangian duality decomposes into a local quadratic RTT kernel, a weightwise stabilized completed envelope, and a boundary quotient whose dual is an orthogonal coideal. For the Yangian MC3 programme, the all-types proved core is narrower: categorical prefundamental Clebsch–Gordan closure is proved for all simple types (Theorem ??), and the DK comparison on the evaluation-generated core is proved separately (Corollary ??). For type A, shifted-prefundamental generation and pro-Weyl recovery are also proved, while the compact/completed extension packet remains open.

(c) *The holographic/celestial axis.* The proved modular core supplies a protected transform calculus: dual transform, cotangent reciprocity, Weyl sewing, and a metaplectic determinant-line anomaly. The full bulk/boundary/line Koszul triangle and the celestial boundary transfer programme are developed in Volume II.

1.20.1 Koszulness and holographic reconstruction

The bar-cobar adjunction $B \dashv \Omega$ encodes boundary data into a bar coalgebra and recovers it via the cobar construction. On the Koszul locus, Theorem B guarantees exact recovery. Chapter ?? proves the equivalence: *chiral Koszulness is equivalent to exact holographic reconstruction*. The twelve characterisations K1–K12 of Koszulness (Theorem ??), comprising eight genuinely independent bidirectional equivalences, one Barr–Beck–Lurie restatement of the bar-cobar counit quasi-isomorphism, one one-way chiral Hochschild consequence on the Koszul locus, one perfectness-conditional Lagrangian criterion (unconditional for the standard landscape), and one one-directional $\mathcal{D}$-module purity implication (the converse is proved for affine Kac–Moody at non-critical level; Virasoro and principal $\mathcal{W}_N$ open), provide a comprehensive suite of conditions for the holographic code to admit exact recovery. The bifunctor decomposition theorem is a further proved consequence outside the twelve numbered items. The Lagrangian isotropy of Theorem C provides a symplectic code structure (Verdier-isotropic summands with non-degenerate cross-pairing), not an orthogonal one. Shadow depth classifies redundancy channels: class G has no channels, class L has one, class M has infinitely many (convergent).

1.20.2 Entanglement entropy from the shadow obstruction tower

At the scalar level, the modular characteristic $\kappa(\mathcal{A})$ (equal to $c(\mathcal{A})/2$ for the Virasoro algebra; the ratio κ/c is family-dependent in general) determines the Calabrese–Cardy entropy of a single interval: $S_{\mathrm{EE}} = \frac{c}{3}\log(L/\varepsilon)$, recovered as the genus-1 shadow of $\Theta_{\mathcal{A}}$ on the replica orbifold (Theorem ??; Chapter ??). For Virasoro, complementarity sharpens this: the sum $S_{\mathrm{EE}}(\mathrm{Vir}_c) + S_{\mathrm{EE}}(\mathrm{Vir}_{26-c}) = \frac{26}{3}\log(L/\varepsilon)$ is a shadow of Theorem C with Koszul conductor $K = 26$, saturated at the self-dual point $c = 13$. Beyond the scalar level, the four shadow-depth classes G/L/C/M produce distinct entanglement complexity: class G (Heisenberg) has Gaussian entanglement with no subleading corrections; class L (Kac–Moody) acquires a single logarithmic correction from the cubic shadow; class C ($\beta\gamma$) adds a quartic contact channel; and class M (Virasoro, $\mathcal{W}_N$) produces an infinite tower of Rényi corrections controlled by the shadow growth rate $\rho(\mathcal{A})$. The full Knill–Laflamme condition for the holographic code requires the physical inner product and is conjectured at higher genus; at genus 1 it is automatic.

1.21 The MC frontier

The five Master Conjectures MC1–MC5 organize the structural results. MC1 through MC4 are proved; MC5 is partially proved, with the analytic HS-sewing package established at all genera while the genuswise BV/BRST/bar identification remains conjectural. MC3 holds for all simple types on the evaluation-generated core (Theorem ??, Corollary ??). The residual DK-4/5 problem (extension beyond evaluation modules) is downstream of MC3. Proof status and refinements are in Chapter ??.

MC	Statement	Status	Key input
MC1	PBW concentration	**proved**	all standard families
MC2	Universal MC class $\Theta_{\mathcal{A}}$	**proved**	bar-intrinsic (Thm ??)
MC3	Thick generation	**proved**	CG all types + eval-core DK (Cor. ??); DK-4/5 downstream
MC4	$\mathcal{W}_\infty$ closure	**proved**	completion towers + weight cutoff
MC5	Genus tower (analytic sewing)	**analytic part proved**	HS-sewing (Thm ??); genuswise BV/BRST/bar identification conjec

MC1 (PBW concentration): Bar cohomology of every standard-landscape chiral algebra concentrates in PBW degrees, proved for all standard families by explicit computation.
MC2 (Universal MC class): $\Theta_{\mathcal{A}} \in \mathrm{MC}(\mathrm{Def}_{\mathrm{cyc}}(\mathcal{A})\widehat{\otimes}\mathfrak{G}^{\mathrm{mod}})$ exists for every modular Koszul chiral algebra by the bar-intrinsic construction (Theorem ??): $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_0$ is MC because $D^2_{\mathcal{A}} = 0$. The shadow obstruction tower $\Theta^{\leq r}_{\mathcal{A}}$ consists of its finite-order projections. When $\dim H^2_{\mathrm{cyc}} = 1$, the MC element collapses to $\Theta_{\mathcal{A}} = \kappa(\mathcal{A}) \cdot \eta \otimes \Lambda$ (Corollary ??).
MC3 (Thick generation): Proved for all simple types on the evaluation-generated core via multiplicity-free ℓ-weights [?ChariMoura06] (Theorem ??), with DK comparison proved separately (Corollary ??, Corollary ??). For type A, shifted-prefundamental generation and pro-Weyl recovery are also proved (Theorem ??). DK-4/5 (extension beyond evaluation modules) is downstream.

MC4 ($\mathcal{W}_\infty$ closure and completed bar-cobar): The completed bar-cobar problem asks whether finite-stage Koszul duality passes to inverse limits. Two complementary mechanisms cover the entire standard landscape:

Mechanism 1: strong filtration. If the algebra $\mathcal{A}$ carries a filtration satisfying the strong axiom $\mu_r(F^{i_1}\mathcal{A}, \ldots, F^{i_r}\mathcal{A}) \subset F^{i_1+\cdots+i_r}\mathcal{A}$, the degree cutoff (Lemma ??) makes continuity and Mittag–Leffler automatic. The completion closure CompCl(Fft) then carries a quasi-inverse bar-cobar homotopy equivalence (Corollary ??), stable under MC twisting (Theorem ??), with completed twisting representability (Theorem ??). This applies to: $V_k(\mathfrak{g})$ (non-critical), Vir_c (all c), $\mathcal{W}^k(\mathfrak{g}, f_{\mathrm{prin}})$ (non-critical), lattice algebras (Proposition ??).

Mechanism 2: weight cutoff. The algebra $\mathcal{W}_\infty = \varprojlim \mathcal{W}_N$ does not satisfy the strong filtration axiom. Instead, the weight-cutoff criterion (Proposition ??) exploits that bar chains of conformal weight w involve only generators of spin $\leq w$: the inverse system stabilizes levelwise, and the completed duality follows by conformal-weight induction. $\mathcal{W}_N$ rigidity (Theorem ??) resolves 21 conjectures and verifies the Yangian evaluation core at 249 test points.

Structural splitting. The coefficient-stability criterion (Theorem ??) reduces convergence to finite matrix stabilization; the uniform PBW bridge (Theorem ??) connects MC1 to MC4. The problem splits into MC4$^+$ (positive towers, solved by weight stabilization; Theorem ??) and MC4° (resonant towers, reduced to a finite-dimensional problem by Theorem ??). The remaining example-specific task is coefficient stabilization on finite windows and H-level target identification.

MC5 (Genus tower / BV-BRST comparison): the algebraic genus recursion and the analytic sewing package are **proved**, but the full genuswise BV/BRST/bar identification remains open. The algebraic bar differential at genus g is determined inductively from lower genera (Theorem ??). Graph amplitudes on Σ_g converge without UV renormalization in dimension 2 (Proposition ??). The analytic and algebraic differentials agree on the algebraic core (Theorem ??). HS-sewing holds for the standard landscape (Theorem ??): polynomial OPE growth and subexponential sector growth imply convergent mode sums for all $0 < q < 1$. For the Heisenberg algebra, the genus-g partition function is the Fredholm determinant of the one-particle Bergman sewing operator (Theorem ??). At genus 0, the algebraic BRST/bar comparison is proved, while the top-degree amplitude pairing is only conditional (Theorem ??, Corollary ??).

The proved core has a sharp boundary: the *four-test interface* (Principle ??; Remark ??). $D^2_{\mathcal{A}} = 0$, the Hodge-class identification $\mathrm{obs}_g = \kappa \cdot \lambda_g$ (uniform-weight lane), complementarity, and sewing are four independent tests forming the complete algebraic-geometric interface with $\overline{\mathcal{M}}_{g,n}$. What lies outside (multi-weight universality at $g \geq 2$, BV/BRST = bar at higher genus, non-perturbative completion) is in §??.

1.22 Traverse of the volume

Each chapter builds on the previous one; the five theorems emerge as milestones.

Part ??: The Bar Complex *(proved core; Thms A–D+H)*

Fourier seed (Ch. ??)	Bar construction as transform; product-formula GF; inversion as Fourier inversion
Algebraic foundations (Ch. ??)	Classical Koszul duality; twisting morphisms; operadic bar-cobar; HTT (A_∞/L_∞ transfer)

Configuration spaces (Ch. ??)	FM compactification $\overline{C}_n(X)$; Arnold relations; residue calculus; cooperadic boundary strata
Geometric bar complex (Ch. ??)	$\bar{B}_X(\mathcal{A})$ via residue extraction; **Theorem A** (adjunction + Verdier intertwining); $d^2 = 0$
Geometric cobar complex (Ch. ??)	Ω_X as left adjoint; $d^2_{\mathrm{cobar}} = 0$ via Verdier duality
Bar-cobar adjunction (Ch. ??)	**Theorem B** (inversion on Koszul locus); curved A_∞; coderived continuation
Non-abelian Poincaré duality (Ch. ??)	$\mathcal{A}^!$ as Verdier dual of $\bar{B}(\mathcal{A})$; resolves circularity in defining Koszul dual
Higher genus (Ch. ??)	Genus tower; curvature $d^2_{\mathrm{fib}} = \kappa\omega_g$; **Theorems C+D** (complementarity, modular characteristic); Lagrangian upgrade
Koszul pairs + deformation theory (Chs. ??–??)	Chiral Koszul pairs; chiral modules; **Theorem H** (ChirHoch* polynomial, Koszul-functorial)
Quantum corrections (Ch. ??)	Higher-genus Arnold deformations; filtered-curved hierarchy

Part ??: The Characteristic Datum *(shadow obstruction tower; E_1 wing; E_n Koszul duality)*

Nonlinear modular shadows (Ch. ??)	Shadow obstruction tower $\kappa \to \Delta \to \mathfrak{C} \to \mathfrak{Q} \to \cdots$; clutching law; archetypes G/L/C/M; branch-line reductions
E_1 modular Koszul duality (Ch. ??)	Ribbon modular operad; FAss; E_1 convolution; E_1 shadow tower; Thms A^{E_1}–H^{E_1}
Ordered associative chiral KD (Ch. ??)	Diagonal bicomodule; chiral Hochschild-coHochschild dictionary; open trace formalism
E_n Koszul duality (Ch. ??)	Higher-dimensional bar complexes; Totaro relations; Ayala–Francis

Part ??: The Standard Landscape *(complete portrait of each family)*

Lattice VOAs (Ch. ??)	$\kappa(V_\Lambda) = \mathrm{rank}(\Lambda)$; curvature-braiding orthogonality; screening operators
Free fields (Ch. ??)	Heisenberg, free fermion; Eisenstein genus expansion; periodic table of the volume
$\beta\gamma$ system (Ch. ??)	First nonabelian computation; $\mu_{\beta\gamma} = 0$ (rank-one rigidity); shadow terminates at degree 4
Kac–Moody (Ch. ??)	$\widehat{\mathfrak{g}}_k$ bar complex; Feigin–Frenkel = Koszul duality; PBW at all genera
$\mathcal{W}$-algebras (Ch. ??)	DS reduction as Koszul functor; $\mathcal{W}_3$ composites; curved A_∞ from higher poles
Deformation quantization (Chs. ??–??)	Chiral Kontsevich formality; $P_\infty \to E_\infty$; star products; MC deformations
Yangians (Ch. ??)	E_1-chiral bar on ordered configurations; R-matrix inversion; DK bridge; dg-shifted Yangian
Genus expansions (Ch. ??)	Explicit expansions: modular forms, Eisenstein, Mumford, bosonization (Remark ??: bosonization ≠ Koszul duality)
Computations + combinatorial frontier (Chs. ??–??)	Bar tables through high degree; OEIS; growth rates

Part ??: Physics Bridges *(connections to physics and adjacent mathematics)*

Feynman diagrams (Ch. ??)	Perturbative incarnation: Feynman graphs on $\overline{C}_n(X)$; higher operations from nested collapses

BV-BRST (Ch. ??)	Genus-0 bar/BRST quasi-iso; BV/bar dictionary; QME/MC comparison
Open/closed realization (§??)	Derived center = universal bulk; open-sector primitive; four-stage architecture
HT boundary conditions (Ch. ??)	3d holomorphic-topological twists; open-sector language (modules, lines, boundaries)
Yang–Mills boundary (Chs. ??–??)	Boundary BRST; central formality; instanton completion; screening; mass-gap reduction
Derived Langlands (Ch. ??)	Critical-level bar complex; opers; Kazhdan–Lusztig from bar-cobar
Arithmetic shadows (Ch. ??)	Shadow L-function; depth decomposition $d = 1 + d_{\mathrm{arith}} + d_{\mathrm{alg}}$; Eisenstein spectral content
Part ??: The Seven Faces of the Collision Residue *(archive-only)*	
Holographic datum master (Ch. ??)	One object $r(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{\mathcal{A}})$ in seven frameworks; falsification checklist
Genus-1 seven faces (Ch. ??)	Elliptic r-matrix; KZB connection; Belavin; elliptic Gaudin
Part ??: The Frontier *(archive-only)*	
Concordance (Ch. ??)	Five theorems; three pillars; three rings; Koszulness programme; open frontiers
Volume II	Swiss-cheese on $\overline{C}_k(\mathbb{C}) \times \mathrm{Conf}_k(\mathbb{R})$; curved at $g \geq 1$; recognition; homotopy-Koszulity of $\mathrm{SC}^{\mathrm{ch,top}}$; bulk/boundary/line triangle

1.23 Dictionary: chiral algebra types and Koszul dualities

Collision geometry on a curve forces three levels of algebraic structure: commutative (E_∞), associative (E_1), semiclassical (P_∞), each carrying its own Koszul duality.

Definition 1.23.1 (E_∞-chiral algebra = vertex algebra)**.** Throughout, $k = \mathbb{C}$ and X is a smooth algebraic curve. The ambient ∞-category is $\mathcal{D}$-mod(X), equipped with the chiral tensor product $\otimes^{\mathrm{ch}}$ [?BD04, §3.4]. Duality means Verdier duality $\mathbb{D}$ of holonomic $\mathcal{D}$-modules (Remark 1.23.2).

An *E_∞-chiral algebra* on X is a commutative algebra in $(\mathcal{D}\text{-mod}(X), \otimes^{\mathrm{ch}})$ (cf. [?HA, §5.1]). Equivalent formulations: (a) $\mathcal{D}_X$-module with chiral Lie bracket [?BD04, §3.3]; (b) vertex algebra [?FBZ04, §3.7] for $X = \mathbb{A}^1$; (c) factorization algebra on $\mathrm{Ran}(X)$ (Definition ??).

Remark 1.23.2 (Verdier versus linear duality)**.** $\mathbb{D}(\mathcal{M})$ is the Verdier dual $\mathcal{D}$-module, not the naïve linear dual $\mathrm{Hom}_k(\mathcal{M}, k)$. The Koszul dual $\mathcal{A}^!$ is constructed via Verdier duality on $\overline{C}_n(X)$ (Theorem ??).

Definition 1.23.3 (E_1-chiral algebra = nonlocal vertex algebra)**.** An *E_1-chiral algebra* on X is an associative (but not necessarily commutative) algebra over the associative chiral operad $\mathsf{Ass}^{\mathrm{ch}}$ (Definition 1.23.4): a $\mathcal{D}_X$-module $\mathcal{A}$ with binary chiral operation $\mu\colon \mathcal{A} \otimes_{\mathrm{ch}} \mathcal{A} \to \Delta_! \mathcal{A}$ satisfying $\mu \circ (\mu \otimes \mathrm{id}) = \mu \circ (\mathrm{id} \otimes \mu)$ but not necessarily skew-symmetry. In vertex algebra language, a nonlocal vertex algebra (Li [?Li96]; cf. [?HA, §4.1]; Beilinson–Drinfeld [?BD04, Theorem 3.4.9]; Francis–Gaitsgory [?FG12]).

Definition 1.23.4 (The associative chiral operad)**.** The *associative chiral operad* $\mathsf{Ass}^{\mathrm{ch}}$ on X is the non-Σ chiral operad with $\mathsf{Ass}^{\mathrm{ch}}(n) = \Delta_!^{(n)}\, \omega_X$, supported on the small diagonal without Σ_n-action. Quadratic presentation: $\mathsf{Ass}^{\mathrm{ch}} = \mathrm{Free}^{\mathrm{ch}}(\mu)/(\mu \circ_1 \mu = \mu \circ_2 \mu)$.

Proposition 1.23.5 ($\mathsf{Ass}^{\mathrm{ch}}$ self-duality;)**.** $(\mathsf{Ass}^{\mathrm{ch}})^! \cong \mathsf{Ass}^{\mathrm{ch}} \otimes \mathrm{sgn}$. *$E_1$–$E_1$ Koszul duality stays within the same operadic type.*

Proof. Classically $\mathrm{Ass}^! \cong \mathrm{Ass} \otimes \mathrm{sgn}$ [?LV12, Theorem 7.4.1]. The chiral lift preserves Koszul duality: $(\mathcal{P}^{\mathrm{ch}})^! \cong (\mathcal{P}^!)^{\mathrm{ch}}$ since the orthogonal complement commutes with the chiral tensor product. □

Definition 1.23.6 (P_∞-chiral algebra = chiral Poisson)**.** A *P_∞-chiral algebra* is an E_∞-chiral algebra with a compatible L_∞ structure via Leibniz: commutative chiral product and Lie bracket coexist.

Remark 1.23.7 (E_n terminology). The subscripts E_∞, E_1, P_∞ refer to the factorization level inside the BD chiral tensor category. Bare E_n and E_n refer to little-disks algebras [?HA, §5.1]. An E_1-chiral algebra on a curve is not a topological E_1-algebra; the suffix "-chiral" is essential.

The hierarchy E_∞-chiral ⊂ P_∞-chiral ⊂ E_1-chiral. Deformation quantization passes from P_∞ to E_1. Bar-cobar duality operates at each level: E_∞ computes factorization homology; E_1 connects to Yangians and quantum groups (Chapter **??**).

Remark 1.23.8 (Operadic hierarchy and configuration spaces). The hierarchy is organized by the geometry of insertion points:

Algebra type	**Operad**	**Configuration space**	**Koszul exchange**
Heisenberg, KM, Vir, $\mathcal{W}$	E_∞ (comm.)	$C_n(X)$ unordered	type exchange $\mathcal{A} \leftrightarrow \mathcal{A}^!$
Lattice VOA	E_∞ with E_1 sublattice	$C_n(X)$ with monodromy	type exchange + screening
Yangian $Y(\mathfrak{g})$	E_1 (assoc.)	$\mathrm{Conf}_n^{\mathrm{ord}}(X)$ ordered	R-matrix inversion
Elliptic QG	modular deformation	Conf on E_τ	elliptic R-inversion

The progression from E_∞ to E_1 is not "more generality"; it is a change in the geometry of collision, from symmetric to ordered, with correspondingly different Koszul exchange mechanisms.

Remark 1.23.9 (Four Koszul duality mechanisms and three bar complexes).

(i) *Francis–Gaitsgory* [?FG12]: $\mathsf{Lie}^{\mathrm{ch}}$-alg $\simeq$ $\mathsf{Com}^{\mathrm{ch}}$-coalg via Chevalley/Primitives, under pro-nilpotence. Uses the FG bar $\bar{B}^{\mathrm{FG}}(\mathcal{A})$ (only the zeroth product $a_{(0)}b$).

(ii) *Symmetric bar-cobar resolution*: $\Omega(\bar{B}(\mathcal{A})) \to \mathcal{A}$ is a quasi-isomorphism on the Koszul locus, or in completed/coderived regimes (Appendix **??**). Uses the symmetric bar $\bar{B}^{\Sigma}(\mathcal{A}) = \bar{B}(\mathcal{A})$ (all OPE products, Σ_n-coinvariants).

(iii) *Quadratic/Priddy*: $\mathcal{A} \mapsto \mathcal{A}^! = T(V^*)/R^\perp$, with $\bar{B}(\mathcal{A})^\vee \simeq \mathcal{A}^!$ when E_2-degeneration holds (the bar coalgebra, dualized, recovers the Koszul dual algebra). Uses $\bar{B}^{\Sigma}(\mathcal{A})^\vee$. Gui–Li–Zeng [?GLZ22] develop this quadratic duality in the chiral setting independently, establishing a bar-cobar adjunction for quadratically presented chiral algebras.

(iv) *Ordered bar-cobar (E_1-chiral Koszul duality)*: the ordered bar $\bar{B}^{\mathrm{ord}}(\mathcal{A})$ retains the linear ordering on configuration spaces; no Σ_n quotient. Its Koszul dual is the dg-shifted Yangian $\mathcal{A}^!_{\mathrm{line}}$. There is a natural map $\bar{B}^{\mathrm{ord}}(\mathcal{A}) \to \bar{B}^{\Sigma}(\mathcal{A})$ (take coinvariants), twisted by the R-matrix whenever the OPE has poles (Chapter **??**).

Every theorem specifies which mechanism is invoked. Mechanisms (i)–(iii) are the three E_∞ variants; mechanism (iv) is the E_1 counterpart. A filtration on $\bar{B}^{\Sigma}(\mathcal{A})$ whose associated graded recovers $\bar{B}^{\mathrm{FG}}(\mathcal{A})$ (retain only zeroth poles) relates (i) to (ii); the coinvariant descent $\bar{B}^{\mathrm{ord}} \to \bar{B}^{\Sigma}$ relates (iv) to (ii) (Remark **??**). At higher genus, all four fuse via the Feynman transform of the modular operad (FCom for (i)–(iii), FAss for (iv)).

Definition 1.23.10 (Anomaly ratio)**.** For $c(\mathcal{A}) \neq 0$: $\varrho(\mathcal{A}) := \kappa(\mathcal{A})/c(\mathcal{A})$. For $\mathcal{W}$-algebras, $\varrho(\mathfrak{g}) = \sum_{i=1}^{r} 1/(m_i + 1)$, independent of k.

Definition 1.23.11 (Koszul conductor)**.** $K(\mathcal{A}) := c(\mathcal{A}) + c(\mathcal{A}^!)$. Values: $2 \dim \mathfrak{g}$ for $\widehat{\mathfrak{g}}_k$; 26 for Virasoro; 100 for $\mathcal{W}_3$.

Definition 1.23.12 (Koszul spectrum)**.** $\mathrm{KSpec}(\mathcal{A}) := (c(\mathcal{A}), \kappa(\mathcal{A}), \varrho(\mathcal{A}))$. Additive under tensor products in c and κ.

Remark 1.23.13 (Existence regimes). Three regimes, not one universal theorem: (i) *strict*, where $\bar{B}_X$ and inversion hold on the Koszul locus; (ii) *completed*, under inverse-limit hypotheses (Appendix **??**); (iii) *comparison* (Appendix **??**). The essential distinction: $\bar{B}_X(\mathcal{A})$ may be constructed beyond where $\Omega_X \bar{B}_X(\mathcal{A}) \simeq \mathcal{A}$ is proved.

1.24 The master invariant table

The standard landscape of the Standard Landscape part is summarized in a single table recording κ, c, $K = c + c'$, Δ, and the shadow archetype for each family.

Family	c	κ	K	Δ	**Archetype**
Heisenberg $\mathcal{H}_k$	1	k	2	0	Gaussian, @2
Free fermion $\mathcal{F}$	$\frac{1}{2}$	$\frac{1}{4}$	0	0	Gaussian, @2
$\widehat{\mathfrak{sl}}_2$ at level k	$\frac{3k}{k+2}$	$\frac{3(k+2)}{4}$	6	$(1-3x)(1+x)$	Lie/tree, @3
$\beta\gamma$	2	1	0	$(1-3x)(1+x)$	contact, @4
Virasoro c	c	$\frac{c}{2}$	26	$(1-3x)(1+x)$	mixed, ∞
$\mathcal{W}_3$	$c(k)$	DS	100	quartic	mixed, ∞
V_Λ (lattice)	$\mathrm{rk}(\Lambda)$	$\mathrm{rk}(\Lambda)$	2rk	--	braided
$Y(\mathfrak{sl}_2)$	--	--	--	--	E_1, R-matrix

The shared discriminant $\Delta(x) = (1-3x)(1+x)$ across $\widehat{\mathfrak{sl}}_2$, Virasoro, and $\beta\gamma$ reflects Drinfeld–Sokolov reduction as a Koszul functor: quantum Hamiltonian reduction preserves the discriminant. The zeros at $x = 1/3$ and $x = -1$ mark bar-acyclic/bar-active transitions, invariants of the planted-forest tropicalization (Theorem **??**). These three algebras have different shadow depths ($r_{\max} = 3, \infty, 4$) but identical genus-0 spectral geometry: the discriminant depends only on $\Theta_{\mathcal{A}}^{\leq 2}$, a DS invariant (Definition **??**).

1.25 Conventions

Convention 1.25.1 (Semantic levels: homotopy, model, shadow)**. H-level (homotopy/native).** The statement in the stable ∞-category of factorization (co)algebras on $\mathrm{Ran}(X)$, or in the associated formal moduli / shifted-symplectic language. Defined up to contractible ambiguity.

M-level (model). An explicit dg / L_∞ / filtered complex. The geometric bar complex $\bar{B}_X(\mathcal{A}) = (T^c(s^{-1}\bar{\mathcal{A}}), d)$ built from configuration space integrals on $\overline{C}_n(X)$ is the canonical model.

S-level (shadow). The cohomological, Ext, numerical, or generating-function consequence: bar cohomology dimensions, Hilbert series, $\kappa(\mathcal{A})$, free energies $F_g(\mathcal{A})$.

Notation.

- $\simeq$: equivalence in the homotopy/∞-categorical sense (H-level).
- $\cong$: strict isomorphism of chosen dg models (M-level).
- $=$: definition or literal equality only.

The proved results are proved at M-level and stated at H-level.

Model independence. Any two admissible dg presentations are connected by a contractible space of quasi-isomorphisms (Proposition **??**).

Convention 1.25.2 (Regime tags)**.** Four regimes, in order of increasing generality. The three bar differentials are: d_0 (the genus-0 bar differential), d_{fib} (the fibered bar differential, genus-g but uncorrected for period monodromy), and dg$g$ (the full genus-g corrected bar differential, incorporating period corrections via the Hodge bundle).

(i) **Quadratic.** $d_0^2 = 0$; Theorems A–D hold without qualification. Heisenberg, free fermions, lattice VOAs.
(ii) **Curved-central.** $d_{\text{fib}}^2 = \kappa \cdot \omega_g$; dg$g^2 = 0$. $\widehat{\mathfrak{g}}_k$ at non-critical level, Virasoro with $c \neq 0$.
(iii) **Filtered-complete.** Complete filtration with $\mathrm{gr}^F \mathcal{A}$ quadratic Koszul. $\mathcal{W}$-algebras via DS reduction, deformation quantizations.
(iv) **Programmatic.** Stratum II (Remark 1.1); stated as conjectures. Derived DK, $\mathbb{E}_n$ for $n \geq 2$.

The four regimes correspond to increasing monodromy of the categorical logarithm: single-valued, scalar curvature, filtered convergence, full structure.

Convention 1.25.3 (Standing assumptions)**.** (SA1) Ground field $\mathbb{C}$. All vector spaces, algebras, and categories are over $\mathbb{C}$.
(SA2) Cohomological grading: $|d| = +1$. See Appendix **??**, §**??**.
(SA3) Formal power series in completed tensor products. Configuration space integrals converge absolutely on FM compactifications (Theorem **??**).
(SA4) Smooth projective geometrically connected curves over $\mathbb{C}$.
(SA5) Completed chiral algebras at genus $g \geq 1$ (Appendix **??**).

1.25.1 Ambient category

All $\mathcal{D}$-modules are holonomic. The cobar construction is the Verdier dual of the bar complex (Definition **??**), so $d_{\text{cobar}}^2 = 0$ follows from $d_{\text{bar}}^2 = 0$ (Corollary **??**). Grading is cohomological ($|d| = +1$); see Appendix **??**. All chiral algebras are augmented ($\varepsilon\colon \mathcal{A} \to \omega_X$, $\bar{\mathcal{A}} = \ker(\varepsilon)$); the bar complex is always reduced. Forms $\eta_{ij} \in \Omega^1(\log D)$ and residue maps are intrinsic to the FM compactification.

Bar complex notation. Several variants of the bar complex appear; here is the dictionary:

- $\bar{B}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$: the *algebraic* bar complex (underlying coalgebra, no curve dependence);
- $\bar{B}_X(\mathcal{A})$: the *geometric* bar complex on X (incorporating configuration-space forms on $\overline{C}_n(X)$);
- $\bar{B}_X^{(g)}(\mathcal{A})$: the *genus-$g$ bar complex* (the component of $\bar{B}_X(\mathcal{A})$ over $\overline{\mathcal{M}}_g$);
- $\Omega_X(C)$: the *cobar* of a factorization coalgebra C (the right adjoint of $\bar{B}_X$).

1.25.2 Critical pitfalls

The following distinctions are the most common sources of error:

Four objects. $\mathcal{A}$ is the algebra. $\bar{B}(\mathcal{A})$ is the bar coalgebra. $\mathcal{A}^{¡} = H^*(\bar{B}(\mathcal{A}))$ is the Koszul dual coalgebra. $\mathcal{A}^! = (\mathcal{A}^{¡})^\vee$ is the Koszul dual algebra. The cobar $\Omega(\bar{B}(\mathcal{A}))$ recovers $\mathcal{A}$ (bar-cobar *inversion*, not duality); $\mathcal{A}^!$ is obtained by Verdier/linear duality, not cobar.

Grading. Cohomological ($|d| = +1$) throughout. The bar complex uses desuspension: s^{-1} lowers degree by 1.

Classical vs. chiral Koszul duality. $\mathrm{Com}^! = \mathrm{Lie}$ (not coLie): the Koszul dual *operad* of Com involves an operadic suspension $\mathfrak{S}$ that shifts degrees and twists by the sign representation ($\mathfrak{S}\mathcal{P}(n) = \mathrm{sgn}_n \otimes \mathcal{P}(n)[n-1]$); omitting the suspension yields coLie instead of Lie, which changes the sign of the bracket in all convolution computations (Appendix **??**, Pitfall 9). The Heisenberg algebra is *not* self-dual. Chiral Koszulness is not the same as classical Koszulness.

Curved A_∞. $m_1^2(a) = [m_0, a]$ (commutator, with a minus sign). The bar differential satisfies $d^2 = 0$ always; curvature shows as $m_1^2 \neq 0$ in the A_∞ algebra, not in the bar complex.

Sugawara. The Sugawara construction is *undefined* at critical level $k = -h^\vee$; the Sugawara vector does not lie in the appropriate completion. This is not "c diverges"; the construction does not exist.

Virasoro self-duality. The Virasoro algebra is chiral Koszul self-dual at $c = 13$ ($\mathrm{Vir}_c^! = \mathrm{Vir}_{26-c}$), not $c = 26$. Quadratic self-duality (uncurved bar, $\kappa = 0$) holds at $c = 0$.

FM compactification. The Fulton–MacPherson compactification is the blowup along diagonals, not the complement of diagonals.

Prime form. The prime form $E(z_1, z_2)$ is a section of $K^{-1/2} \boxtimes K^{-1/2}$, not $K^{+1/2}$.

QME. The quantum master equation is $\hbar\Delta S + \frac{1}{2}\{S, S\} = 0$ (factor $\frac{1}{2}$, not 1).

$d^2_{\mathrm{bracket}} \neq 0$. The bracket component of the bar differential does not square to zero on its own: this is proved for all 2048 sign conventions for $\mathfrak{sl}_2$. The full Borcherds differential gives $d^2 = 0$.